Depth and learning bounds for critical Bayesian neural networks

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Abstract

Can a Bayesian neural network learn with polynomial dependence on its learner depth, without restricting the prior to internally stable networks? We prove such bounds for the original critical Gaussian prior on tanh networks, conditioned on absolute output margin greater than $1/2$ at every Boolean input and exact noiseless training fit. The bounds hold at every width capable of implementing the target. Every parity has polynomial sample complexity once learner depth is a sufficiently large multiple of $\log(d+L+2)$, independently of parity degree. More generally the bound is polynomial in Fourier variation. A bounded-weight tanh teacher of depth D gives $R^{O(D+1)}$ samples, where R includes dimension, teacher resources, and learner depth, when $L \geq C(D+1)\log R$. Boolean threshold teachers need no bound on their original coefficient magnitudes. The exponent still grows with teacher depth: joint polynomial dependence on arbitrary teacher width and depth remains open.

The proof gives an intrinsic theorem at every finite width N. If a_N is the target's normalized negative log-mass under the raw Gaussian prior, the sample bound is a universal polynomial in dimension, depth, a_N, and logarithmic accuracy factors, times $\exp(C\sqrt{L(a_N - V_*)_+})$, where $V_* \in (1,2)$ is a scalar baseline. Forward conditional entropy extracts approximate covariance paths from finite-width events. Preserving a small input-anchor covariance block then gives finite tanh representations and VC bounds. A separate finite-width construction tracks the target signal and residual variances to bound its actual prior mass. Ordinary VC bounds cover the remaining small widths.

We also obtain bounds with only polynomial dependence on sequential length for several sequential target classes, a quasipolynomial bound for general formulas, sharp excess-rate estimates, and complementary parity lower bounds at smaller depths.

1 Model and meaning of the result

Let $\mathcal{X} = \{-1,1\}^d$, $d \geq 1$, and $L \geq 2$. A width-N network has $L-1$ hidden layers and computes

$$h^0(x) = \bar{x} = (1,x), \qquad h^s(x) = \tanh(W^s h^{s-1}(x)) \quad (1 \leq s < L), \qquad z_w(x) = v^T h^{L-1}(x).$$

There are no biases after the augmented input. Under the raw critical Gaussian prior $\mu_{N,1}$, entries are independent centered Gaussians with variances $1/d$ in W^1, $1/N$ in internal matrices, and $1/N^2$ in v. We also write $\mu_N = \mu_{N,1}$ for this raw weight law. For a Boolean function f, write

$$A_{N,f} = \{w : f(x)z_w(x) > 1/2 \text{ for every } x \in \mathcal{X}\}, \qquad J_L(f) = \lim_{N\to\infty} -\frac{1}{N}\log \mu_{N,1}(A_{N,f}).$$

The limit exists at each fixed d, L. Its proof, including the strict margin convention, appears in Appendix A.

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For a feature law ν on $[-1,1]^{\mathcal{X}}$, let $Q(\nu) = \mathbb{E}_\nu hh^T$. Let M_Q be the law of coordinatewise tanh applied to a centered Gaussian with covariance Q, and put $Q_0(x,y) = (1 + x \cdot y)/d$. The rate is

$$J_L(f) = \inf\Big\{ D(\nu_1 \| \mathsf{M}_{Q_0}) + \sum_{s=2}^{L-1} D(\nu_s \| \mathsf{M}_{Q(\nu_{s-1})}) + \tfrac{1}{2} z^T Q(\nu_{L-1})^+ z : \\ f(x)z(x) \geq 1/2 \quad (x \in \mathcal{X}) \Big\}.$$

The last term is infinite outside the covariance range. The matrices $Q(\nu)$ are uncentered second moments, not necessarily centered covariances. We use $H = L - 1$ for the number of hidden layers and r for a terminal readout cost when it is not otherwise specified. All logarithms are natural, and $u_+ = \max\{u, 0\}$.

1.1 Learning at every realizable width

Write $A_N = \bigcup_g A_{N,g}$ and $\pi_N(g) = \mu_N(A_{N,g})/\mu_N(A_N)$. After a labeled sample S, let $p_N(f(x) \mid x)$ denote the posterior probability of the correct label at x, with conditioning on S implicit; we also abbreviate it as $p_N(x)$ when the target is fixed. Thus

$$p_N(f(x) \mid x) = \frac{\sum_{g:\, g|_S = f|_S,\, g(x) = f(x)} \mu_N(A_{N,g})}{\sum_{g:\, g|_S = f|_S} \mu_N(A_{N,g})}.$$

The common global-margin normalization cancels in this ratio.

A width is *realizable* if a depth-L, width-N tanh network of the specified architecture implements the target's signs on the cube with no zero output. Rescaling its readout gives strict output margin $1/2$; continuity then makes $A_{N,f}$ have positive Gaussian probability. Our target-based learning theorems hold at every such width, without an additional width threshold. In particular, they hold at every width above the minimum implementing width at the specified learner depth.

Training inputs are independent draws from any distribution $\mathcal{D}$ on the cube and have noiseless labels $f(x)$. The learner conditions the original Gaussian prior on $|z_w(x)| > 1/2$ at every input and on exact training signs. No internal stability event is added. Every sample bound below ensures, with training probability at least $1 - \delta$, that the correct-label posterior probability exceeds $2/3$ outside a set of $\mathcal{D}$-mass at most ε. The conclusions apply to the full posterior, including mixtures.

For the normalized cube Fourier expansion, put

$$\chi_S(x) = \prod_{i \in S} x_i, \qquad \widehat{f}(S) = 2^{-d} \sum_{x \in \mathcal{X}} f(x)\chi_S(x).$$

For Fourier variation $A_F(f) = \sum_{S \subseteq [d]} |\widehat{f}(S)|$, put $R_F = (d + L + 2)A_F(f)$. Theorem 2.2 gives the sufficient sample bound

$$n \geq \frac{C}{\varepsilon} \left\{ R_F^c (1 + \log(1/\varepsilon))^c + \log(1/\delta) \right\} \qquad \text{if } L \geq C_0 \log R_F.$$

Every signed parity has $A_F = 1$. This therefore gives polynomial samples for all parities, including full parity, at every realizable width and at sufficiently large logarithmic learner depth.

For a depth-D, width-C_T tanh teacher with parameter magnitudes bounded by $B \geq 1$ and global signed output margin at least $1/2$, put $R = d + C_T + B + D + L + 2$. Corollary 2.5 gives

$$n \geq \frac{C}{\varepsilon} \left\{ R^{c(D+1)} (1 + \log(1/\varepsilon))^c + \log(1/\delta) \right\} \qquad \text{if } L \geq C_0 (D+1) \log R.$$

The constants c, C, C_0 are universal. Thus learner depth enters polynomially for each fixed teacher depth. The exponent still depends on teacher depth, and joint polynomial dependence on arbitrary teacher width and depth remains open.

For Boolean threshold teachers, the corresponding bound does not depend on the magnitudes or precision of their real coefficients. A teacher of width C_T and depth h gives $R^{O(h)}$ samples with $R = d + C_T + h + L + 2$, at $L \geq Ch \log R$; see Corollary 2.4. In particular every Boolean halfspace has a polynomial sample bound at every width $N \geq 1$ under the logarithmic learner-depth condition.

Several target classes permit milder dependence on their original sequential depth. Decision trees and two-state branching programs have polynomial Fourier variation. Programs over fixed solvable finite groups, including permutation programs of width at most four, have sample bounds polynomial in their sequential length, with an exponent depending only on the group. A fixed-rank first-layer bottleneck gives polynomial dependence on teacher product norm even when the remaining teacher depth is arbitrary. For general formulas the bound is quasipolynomial in dimension and formula size at polynomial learner depth. The precise resource and depth conditions, all at every realizable width, appear in Section 2.

1.2 An intrinsic bound and its limiting-rate counterpart

There is a universal scalar baseline $V_* \in (1, 2)$, defined below as the limiting scalar cost of producing the required output margin from a vanishing Gaussian state. At a realizable width put

$$a_N(f) = -\frac{1}{N} \log \mu_{N,1}(A_{N,f}).$$

Theorem 27.1 proves the following sufficient sample bound at that exact width:

$$\frac{C}{\varepsilon}\left\{[(d + L + a_N(f) + 2)(1 + \log(1/\varepsilon))]^c \exp\left(C\sqrt{L(a_N(f) - V_*)_+}\right) + \log(1/\delta)\right\}.$$

It depends on width only through the actual raw target cost. The finite-width constructions in Sections 28 and 30 bound this cost explicitly for the preceding target classes once width exceeds a quantified threshold. Ordinary network VC bounds handle all smaller realizable widths. This is how the target-based conclusions remove that threshold.

A limiting-rate version is also useful. If $J_L(f) \leq a$, then Theorem 22.3 gives, at all sufficiently large widths,

$$n \geq \frac{C}{\varepsilon}\left\{[(d + 1)(L + 1)(a + 2)]^c \exp\left(C\sqrt{L(a - V_*)_+}\right) + \log(1/\delta)\right\}.$$

This rate-based statement alone does not quantify its width threshold. The finite-width constructions bound the cost of explicit target representations; they do not assert convergence within $1/L$ to the unknown optimal rate $J_L(f)$ for every target.

1.3 Why the finite representation is possible

For a path with layer entropies D_s and action at most $V_* + \eta$, Lemma 12.1 gives

$$S := \sum_s \sqrt{D_s} \leq C[\log(L + 1) + \sqrt{L\eta}].$$

Theorem 15.1 constructs a same-depth finite tanh network with the same Boolean function and width at most $\mathrm{poly}(d, L, A_0)e^{CS}$, where $A_0 = \max\{1, \text{total action}\}$. A greedy procedure selects

inputs whose evaluations account for all significant conditional entropies and for the readout. Their covariance block is then preserved exactly at every layer. Leverage-based importance sampling and a positive definite correction approximate each remaining input jointly with these anchors. The correction is absorbed into the next incoming matrix, so it adds no nonlinear layer.

This construction avoids requiring a low-rank approximation in the full Gaussian-coupling metric. Such an approximation can be impossible for a high-rank covariance, even when only one final Boolean readout matters. The compressed network's weights may be very large; the classical polynomial VC bound for sigmoid networks still controls its number of Boolean sign patterns. Applying a finite-class consistency bound to all functions of rate at most $J_L(f)$ then gives the rate-based posterior theorem.

For finite width, forward conditional entropy and exchangeability extract approximate covariance paths on a finite set of input probes. A projected readout entropy bound preserves the scalar baseline, and the entropy estimate is stable under the resulting moment errors. Local finite-network representations then give a sample-independent class of controlled VC dimension and a quantitative bound on its exceptional prior mass. Ordinary network VC bounds cover the remaining smaller widths.

1.4 Rate estimates and complementary lower bounds

The rate estimates used in the learning theorem are proved in this paper. Universal flattening gives, for every Boolean function,

$$V_* - \frac{C}{L-1} \le J_L(f) \le V_* + C\frac{(d+1)^2 + [\log(2L)]^2}{L} \qquad (L \ge C(d+1)).$$

A bound polynomial in normalized excess alone would be insufficient: at sufficiently large polynomial depth that quantity is polynomial for every truth table, whereas arbitrary truth tables require exponentially many observations. The exponential of its square root in our capacity bound is consistent with this obstruction.

Controlled Fourier features give

$$J_L(f) \le V_* + C\frac{[\log(d+L+2) + \log A_F(f)]^2}{L}$$

under the stated depth condition. A layerwise circuit simulation and finite-precision teacher compilation give $J_L(f) \le V_* + C(D+1)^2 \log^2 R/L$ for the teachers above. These constructions supply the two learning corollaries. Every sample of n consistently labeled points also has an extension with excess rate $O(\log^2(d+n+L)/L)$ at sufficient depth. This bounds the posterior's rate without asserting that all such extensions are accurate.

The excess-rate scale is sharp for constants, dictators, and all parities: it is $\Theta(\log^2 d/L)$ between sufficiently large logarithmic depth and a sufficiently small fixed power of dimension. The lower bound holds for every Boolean function. A separate comparison proves that, for all sufficiently large d depending on θ, full parity still has limiting Gibbs error $1/2$ with high training probability when

$$2 \le L \le c\theta\frac{\log d}{\log\log d}, \qquad n \le d^{(1-\theta)/(L-1)}, \qquad 0 < \theta < 1.$$

Its confidence-error lower limit is at least $1/4$, including mixtures of minimizers. This polynomial lower bound at smaller depths is consistent with the positive logarithmic-depth result.

Neither rate flattening nor our learning proof makes the function prior uniform. For a sample S, a consistent function's exponential posterior rate is

$$J_L(g) - \min_{h \text{ consistent with } S} J_L(h).$$

Any positive gap is multiplied by width N. In the rate-based statements, dimension, depth, and sample size are fixed before width tends to infinity. The intrinsic theorem and target-based every-width theorems instead apply at each specified realizable width. The analytic proofs have received independent AI-agent mathematical review; they have not been formalized in Lean. Constants are not optimized, and no efficient posterior-sampling algorithm is claimed.

1.5 Reading guide

Section 2 collects the finite-width learning statements before their proofs. Their common argument has two inputs: the entropy-certificate and intrinsic learning results in Sections 26 and 27, and the target-mass constructions in Sections 28– 30. The intrinsic argument uses the approximate entropy budget and anchored compression from Sections 13 and 15.

For the underlying variational analysis, begin with the scalar value in Section 8; Appendix A proves its connection to the actual prior's large-width rate. The Fourier and circuit constructions bound target costs, whereas Section 22 counts competing functions and supplies the separate rate-based learning theorem. The later geometric comparisons and parity lower bounds give limitations and complementary results; they are not additional premises for the every-width conclusions. In particular, the smaller-depth parity theorem is stated in Section 44.

2 Target-based learning bounds at every realizable width

The quantitative target-mass constructions remove the width-limit qualification from several learning bounds. Throughout this section, training inputs are iid from any distribution D on the Boolean cube, labels are noiseless, and the learner is the original critical Gaussian prior conditioned on a global absolute output margin and exact training signs. A conclusion at *every realizable width* means every integer $N \geq 1$ for which $\mu_N(A_{N,f}) > 0$. Equivalently, a depth-L, width-N network of the specified architecture realizes the signs of f with no zero output on the cube: rescaling its readout gives strict margin $1/2$, and continuity makes that event have positive Gaussian probability.

Every sample bound below guarantees, with sample probability at least $1 - \delta$,

$$D\{x : p_N(f(x) \mid x) \leq 2/3\} \leq \varepsilon, \qquad 0 < \varepsilon, \delta < 1.$$

All constants are universal. The mild logarithmic accuracy factor comes from the intrinsic finite-width theorem; no efficient posterior algorithm is asserted.

Lemma 2.1 (From a quantitative target mass to every-width learning)**.** *Let $c_0 > 0$ be a universal constant. Suppose $T \geq \log(d + L + 2)$, $L \geq T$, and the raw target cost obeys*

$$a_N(f) \leq V_* + c_0 T^2/L \qquad (N \geq e^{c_0 T}).$$

Then, at every realizable width, with c, C depending only on c_0, a sufficient sample size is

$$n \geq \frac{C}{\varepsilon}\left\{e^{CT}(1 + \log(1/\varepsilon))^c + \log(1/\delta)\right\}.$$

Proof. Above the displayed width threshold, apply Theorem 27.1. The target cost is at most $2+c_0T$, and its exponential factor is at most e^{CT}. Since $d+L \le e^T$, all remaining resource factors are absorbed in the asserted bound.

Below that threshold, the whole class of depth-L, width-N tanh sign networks has VC dimension polynomial in d, L, N, hence at most e^{CT}. Sauer's lemma on the finite cube gives at most $\exp(e^{CT})$ distinct sign functions after increasing C. A finite-class consistency bound therefore makes every posterior-supported function have test error at most $\varepsilon/3$ with the required sample probability. Averaging and the probability-2/3 criterion give the result. Positive target mass ensures that the conditioning event is nonempty for every sample. □

Theorem 2.2 (Every-width learning from Fourier variation)**.** *Put $A_F(f) = \sum_S |\widehat{f}(S)|$ and $R = (d+L+2)A_F(f)$. If $L \ge C \log R$, then at every realizable width a sufficient sample size is*

$$n \ge \frac{C}{\varepsilon}\left\{R^c(1+\log(1/\varepsilon))^c + \log(1/\delta)\right\}.$$

In particular this is polynomial for every signed parity, independently of its degree. The same conclusion holds for a decision tree with s leaves or a length-t binary-state branching program, with R replaced by $d+L+s+2$ or $d+L+t+2$, respectively. The latter programs may query different coordinates in different states and may repeat queries.

Proof. Corollary 28.5 supplies the preceding lemma with T a universal multiple of $\log R$; enlarge the depth constant accordingly. Every signed parity has $A_F = 1$, while Proposition 23.1 gives $A_F \le s$ for the tree and $A_F \le t+1$ for the binary-state program. Products such as $(d+L+2)(t+1)$ are bounded by a universal power of the stated replacement resource. □

Theorem 2.3 (Every-width learning from a bounded threshold implementation)**.** *Suppose f has a layered Boolean threshold implementation of depth $h \ge 1$, width at most C_T including constant wires, unit gate margins, and gate coefficient Euclidean norms at most $B \ge 1$. Put*

$$T = \log(d+2) + h\log(C_T + B + h + L + 2).$$

If $L \ge CT$, then at every realizable width a sufficient sample size is

$$n \ge \frac{C}{\varepsilon}\left\{e^{CT}(1+\log(1/\varepsilon))^c + \log(1/\delta)\right\}.$$

In particular, the conclusion holds for every $N \ge C_T$.

Proof. The depth hypothesis implies the remaining-depth requirement $L-h \ge C_0T$ in Corollary 30.1, after increasing its universal constant. That corollary gives the raw target-cost hypothesis of Lemma 2.1.

For the final assertion, multiply the coefficients of every circuit gate by one sufficiently large common gain and replace its sign by tanh. The unit margins and finitely many gates ensure that all hidden activations uniformly approximate the corresponding Boolean wires closely enough to preserve every later gate sign. Constant wires are included in this approximation. Thus h hidden layers of width C_T and a linear readout realize the target. Any further hidden layers may propagate its scalar output through a positive multiple of tanh, preserving its sign. Rescale the final readout and pad unused neurons with zero weights. This gives a strict-margin realization at every $N \ge C_T$ and $L \ge h+1$, as ensured by the stated depth condition. □

Corollary 2.4 (Real threshold teachers without a weight bound)**.** *Suppose f has a layered real threshold circuit of depth $h \geq 1$ and width $C_T \geq 1$, with arbitrary coefficients and biases and a fixed tie convention at each gate. Put $R = d + C_T + h + L + 2$. If $L \geq Ch \log R$, then at every realizable width a sufficient sample size is*

$$n \geq \frac{C}{\varepsilon}\left\{R^{ch}(1+\log(1/\varepsilon))^c + \log(1/\delta)\right\}.$$

There is no dependence on the original coefficient magnitudes or precisions. In particular, every Boolean halfspace has a universal polynomial sample bound at every width $N \geq 1$, whenever $L \geq C \log(d + L + 2)$.

Proof. Lemma 31.2 gives a unit-margin implementation of depth $O(h)$, width and gate norms bounded by a universal polynomial in $d + C_T + h + 2$. Apply Theorem 2.3; its parameter is at most $Ch \log R$. For a halfspace, perturb its bias if necessary to obtain strict signs on the finite cube. A width-one tanh network realizes those signs at any depth: use the affine form in the first layer and positive scalar weights in each subsequent hidden layer, then scale the readout. Thus every width is realizable. □

Corollary 2.5 (Bounded-weight tanh teachers)**.** *Suppose a depth-D, width-C_T tanh teacher has parameter magnitudes at most $B \geq 1$ and output margin $1/2$ on the cube. Put $R = d + C_T + B + D + L + 2$. If $L \geq C(D+1)\log R$, then at every realizable width a sufficient sample size is*

$$n \geq \frac{C}{\varepsilon}\left\{R^{c(D+1)}(1+\log(1/\varepsilon))^c + \log(1/\delta)\right\}.$$

Proof. Lemma 21.1 gives a threshold circuit of depth $O(D+1)$, with width and gate coefficient norms bounded by a universal polynomial in $d + C_T + B + D + 2$. Its arithmetic circuits use unweighted majority gates: the required precision does not introduce a polynomial degree depending on D. The preceding theorem therefore has $T \leq C(D+1)\log R$ and proves the claim. □

Corollary 2.6 (Low-rank teachers and solvable-group programs)**.** *The following two classes have every-width bounds.*

1. *Suppose f has a unit-margin tanh representation with product of layer Frobenius norms at most $P \geq 1$, and first-layer rank at most $r \geq 1$. Put $S = 1 + \log(d + L + 2) + r \log(P + r + 2)$. If $L \geq CS$, a sufficient sample size at every realizable width is*

$$\frac{C}{\varepsilon}\left\{(d+L+2)^c(P+r+2)^{cr}(1+\log(1/\varepsilon))^c + \log(1/\delta)\right\}.$$

   The approximate-rank condition in Corollary 24.2 has the same consequence with P replaced by $8P/7$.

2. *Suppose f is a length-t program over a solvable group of order $q \geq 2$ and derived length $\ell \geq 1$. Put $R = d + q + t + L + 2$. If $L \geq C\ell \log R$, a sufficient sample size at every realizable width is*

$$\frac{C}{\varepsilon}\left\{R^{c\ell}(1+\log(1/\varepsilon))^c + \log(1/\delta)\right\}.$$

   This includes a universal polynomial bound in d, t, and L for permutation branching programs of width at most four; the exponent does not depend on the program length.

Proof. The proof of Theorem 24.1 constructs a three-stage threshold circuit. Its width and gate norms are polynomial in P, r and the number of boxes $M \le (CP^2\sqrt{r})^r$. The parameter in Theorem 2.3 is thus at most CS. This proves the first assertion and its approximate-rank variant. For the second, Lemma 23.2 gives depth $O(\ell)$ and universal polynomial width and gate norms in q, t. The same parameter is at most $C\ell \log R$. The permutation corollary uses the bounded order and derived length of S_4, as before. □

Theorem 2.7 (Every-width learning after Boolean layer blocking)**.** *Suppose f has a layered fan-in-two AND/OR/NOT circuit of width C_T and depth $h \ge 1$. For $1 \le b \le h$, put $k = \lceil h/b \rceil$ and*

$$U_b = \log(d+2) + k\{\log(C_T + h + L + 2) + 2^b\}.$$

If $L \ge CU_b$, a sufficient sample size at every realizable width is

$$n \ge \frac{C}{\varepsilon}\left\{e^{CU_b}(1+\log(1/\varepsilon))^c + \log(1/\delta)\right\}.$$

Consequently, for a fan-in-two circuit of size s, depth h, and $S = \max\{16, d+s+2\}$, the conditions

$$L \ge C\log(S+L)\left(1 + \frac{h}{\log\log(S+L)}\right),$$

$$n \ge \frac{C}{\varepsilon}\left\{(S+L)^{C(1+h/\log\log(S+L))}(1+\log(1/\varepsilon))^c + \log(1/\delta)\right\}$$

suffice at every realizable width. For formulas, h may be replaced by $C\log S$ after changing universal constants. At sufficiently large learner depths polynomial in S, this is $\exp\{O(\log^2 S/\log\log S)\}$ apart from statistical factors.

Proof. The block construction in Theorem 25.2 has $2k$ threshold stages, width at most $(C_T+1)2^{C2^b}$, and coefficient norms at most 2^{C2^b} up to universal factors. Its finite-width mass parameter is at most CU_b, so apply Theorem 2.3. For a size-s circuit insert identity gates along skipped wires, then use the same block choice and formula-balancing argument as in Corollary 25.4. This yields $U_b \le C\log(S+L)(1+h/\log\log(S+L))$. □

The parameter U_b records the finite-width construction proved here. We have not asserted that the sharper optimized-prefix parameter T_b or T_* in Section 25 has the same finite-width target-mass estimate. The every-width formula bound does retain the stated quasipolynomial order. None of these conclusions proves joint polynomial sample complexity for arbitrary small deep circuits.

3 Universal critical-rate bounds

The limiting constant is the scalar entropy-control value V_* from Definition 8.1. Its definition involves only one-dimensional Gaussian laws and the activation tanh.

Theorem 3.1 (A common critical rate at large depth)**.** *There are universal constants $C < \infty$ and $V_* \in (1,2)$ such that, for every input dimension $d \ge 1$, depth $L \ge 2$, and Boolean function $f : \{-1,1\}^d \to \{-1,1\}$,*

$$J_L(f) \ge V_* - \frac{C}{L-1}.$$

With $D = d+1$, whenever $L \ge CD$, one also has

$$J_L(f) \le V_* + C\frac{D^2 + [\log(2L)]^2}{L}.$$

Both bounds hold uniformly over all Boolean functions at the same specified depth. In particular, at every fixed input dimension, $J_L(f) \to V_*$ uniformly in f as $L \to \infty$.

Proof. Lemma 8.2 constructs V_* and proves $1 < V_* < 2$. The lower estimate is proved in Theorem 10.2; the upper estimate is proved in Theorem 17.1. Increasing the common universal constant gives the displayed simultaneous bounds. □

For every fixed $\varepsilon > 0$, a depth polynomial in d and $1/\varepsilon$ therefore places all 2^{2^d} Boolean functions inside the same absolute rate interval $[V_* - \varepsilon, V_* + \varepsilon]$. An absolute rate sublevel cannot by itself provide a polynomial-depth bound on the logarithm of the number of possible functions. This statement does not identify the ordering of rates inside the interval; such remaining differences can still determine a width-first posterior.

4 A depth-independent lower bound on every critical rate

At every prior scale $0 < t = t(d, L, N) \le 1$, the global clean-margin event has an exponentially small probability, uniformly in depth. The exponent has a universal positive lower bound as the input dimension grows. This result does not compare the excess rates of different Boolean functions and is not a posterior sample-complexity lower bound.

Throughout this section the hidden activation is tanh, the input is $\bar{x} = (1, x)$ for $x \in \{-1, 1\}^d$, the first-layer variance is t^2/d, the subsequent hidden-weight variance is t^2/N, and the readout variance is t^2/N^2. The bounds hold at each specified (d, L, N), so t may follow any prescribed schedule in these parameters; it need not remain constant as width varies. At fixed d, L, write this schedule as $t(N)$. The critical model of the rest of the paper is the constant schedule $t(N) = 1$. There are $L - 1$ hidden layers, with $L \ge 2$. Write $A_{N,f}$ for the event $f(x)z(x) > 1/2$ at every cube input, and $A_N = \bigcup_f A_{N,f}$ for the global clean-margin event.

Lemma 4.1 (A scalar exponential potential)**.** *For $0 < t \le 1$, $0 \le a \le 1$, and $G \sim \mathcal{N}(0, t^2 q)$, $q \ge 0$,*

$$\mathbb{E} \exp\{(2a/t^2) \tanh^2 G\} \le e^{2aq/t^2}.$$

If (G_x, G_y) is any centered Gaussian pair, $s \in \{-1, 1\}$, and $t^2 v = \mathrm{Var}[(G_x + sG_y)/2]$, then

$$\mathbb{E} \exp\left\{ \frac{2a}{t^2} \left(\frac{\tanh G_x + s \tanh G_y}{2} \right)^2 \right\} \le e^{2av/t^2}.$$

Proof. The case $a = 0$ is immediate. Otherwise put $\lambda = 2a/t^2$ and $F_\lambda(z) = \exp(\lambda \tanh^2 z)$. Writing $u = \tanh^2 z \in [0, 1)$, direct differentiation gives

$$\frac{F_\lambda''(z)}{F_\lambda(z)} = 2\lambda(1 - u)[1 - 3u + 2\lambda u(1 - u)] \le \frac{2\lambda}{t^2}.$$

To check the inequality, multiply its bracket by t^2. The resulting quantity $t^2(1-u)(1-3u)+4au(1-u)^2$ is affine in $t^2 \in [0, 1]$. At the two endpoints it is at most $16/27$ and $1 - 5u^2 + 4u^3 \le 1$, respectively. The Gaussian heat equation and Gronwall's inequality therefore give $\mathbb{E}F_\lambda(\sqrt{r}Z) \le \exp(\lambda r/t^2)$ for $r \ge 0$. All derivatives needed here are bounded for each fixed λ. Taking $r = t^2 q$ proves the first assertion. The elementary identities for the sum and difference of two hyperbolic tangents imply

$$\left| \frac{\tanh a + s \tanh b}{2} \right| \le \tanh \left| \frac{a + sb}{2} \right|.$$

For example, after writing $a = A + D$, $b = A - D$, the half-sum equals $\sinh(2A)/[\cosh(2A) + \cosh(2D)]$, whose absolute value is at most $\tanh|A|$. The half-difference is identical with A, D interchanged. Applying the scalar assertion to $(G_x + sG_y)/2$ proves the claim, without any nonsingularity assumption. □

Theorem 4.2 (Finite-width clean-margin upper bounds for all $\gamma \geq 1$)**.** *For every $d \geq 1$, $N \geq 1$, $L \geq 2$, $0 < t \leq 1$, and every Boolean function f,*

$$\mu_{N,t}(A_{N,f}) \leq \exp\{-(N/t^2)\psi(d)\}, \qquad \psi(d) = \begin{cases} d/8, & d \leq 4, \\ 1 - 2/d, & d \geq 4. \end{cases}$$

The unconditioned Gaussian probability of global clean margin satisfies

$$\mu_{N,t}(A_N) \leq \min\{1,\ 2^{d+1}\exp[-(N/t^2)\psi(d)]\}.$$

In the usual scaling $t^2 = N^{1-\gamma}$, the exponential speed N/t^2 is N^γ, and the assumption is $\gamma \geq 1$. The first bound holds with no factor depending on depth or width outside its exponential. The simpler exponent $1 - 2/d$ is valid in every dimension.

Proof. Fix two inputs x, y, a choice $s \in \{-1, 1\}$, and $a \in (0, 1]$. For the empirical hidden feature covariance Q_j, set

$$v_j = \frac{1}{4}(Q_j(x,x) + Q_j(y,y) + 2sQ_j(x,y)), \qquad 0 \leq j \leq L-1,$$

where $Q_0(x,y) = \bar{x}\cdot\bar{y}/d$. Conditional on the previous layer, the N preactivation pairs are independent centered Gaussians with covariance given by $t^2 Q_{j-1}$. Lemma 4.1 therefore gives

$$\mathbb{E}[\exp((2aN/t^2)v_j) \mid \text{previous layers}] \leq \exp((2aN/t^2)v_{j-1}).$$

Consequently $\mathbb{E}\exp((2aN/t^2)v_{L-1}) \leq \exp((2aN/t^2)v_0)$.

Conditional on the hidden layers, the readout combination $Z = (z(x) + sz(y))/2$ is centered Gaussian with variance $t^2 v_{L-1}/N$. For either specified sign of Z, its probability of exceeding $1/2$ in that sign is at most $\exp[-N/(8t^2 v_{L-1})]$, interpreted as zero when $v_{L-1} = 0$. Since

$$\frac{1}{8v} \geq \sqrt{a} - 2av \qquad (v > 0),$$

the unconditional one-sided probability is at most $\exp[-(N/t^2)(\sqrt{a} - 2av_0)]$.

If f is nonconstant, some adjacent cube vertices have opposite labels. Choose such x, y and $s = -1$. Their input difference has squared norm four, so $v_0 = 1/d$, and $A_{N,f}$ implies a specified sign of Z exceeds $1/2$. If f is constant, use antipodal inputs $y = -x$ and $s = +1$. Their augmented-input half-sum is the bias coordinate, again giving $v_0 = 1/d$. Maximizing $\sqrt{a} - 2a/d$ over $a \in [0, 1]$ gives $\psi(d)$, proving the first assertion.

For the second assertion fix a spanning tree of the cube graph and one antipodal pair. Every nonconstant sign function changes sign on one of the tree's $2^d - 1$ edges; a constant function has equal signs on the fixed antipodal pair. Each case implies $|Z| > 1/2$ for its corresponding difference or sum. The two-sided bound and a union bound over these 2^d pairs give the result. □

Corollary 4.3 (Universal lower critical-rate baseline)**.** *For the critical variational rate functional J_L with output margin $1/2$, every Boolean function satisfies*

$$J_L(f) \geq \psi(d) \geq 1 - \frac{2}{d} \qquad (L \geq 2).$$

Proof. This can be read directly from the finite-width upper bound whenever the corresponding large-deviation formula is available. There is also a direct proof from the variational functional, which avoids any interchange of width and depth limits. Choose the signed pair from the preceding proof, put $b = (e_x + s e_y)/2$, and write $v_j = b^T Q_j b$ for the covariances along any admissible variational path. For any $a \in (0, 1]$, the entropy variational inequality and Lemma 4.1 at $t = 1$ give, for every hidden layer,

$$D(\nu_j \| \mathsf{M}_{Q_{j-1}}) \geq 2a(v_j - v_{j-1}).$$

The output constraints imply $|b^T z| \geq 1/2$. If the readout cost is finite, Cauchy–Schwarz on the range of Q_{L-1} gives

$$\tfrac{1}{2} z^T Q_{L-1}^{+} z \geq \frac{1}{8 v_{L-1}}.$$

The inequality is immediate if this cost is infinite. Summing the entropy inequalities and minimizing over $v = v_{L-1} > 0$ now yields

$$J_L(f) \geq \inf_{v>0} \left(2av + \frac{1}{8v} \right) - 2av_0 = \sqrt{a} - \frac{2a}{d}.$$

Maximizing in a gives $\psi(d)$. □

For completeness, the same argument with an arbitrary required output margin $r > 0$ replaces $\psi(d)$ by

$$\psi_r(d) = \max_{0 \leq u \leq 1} (2ru - 2u^2/d) = \begin{cases} r^2 d/2, & rd \leq 2, \\ 2r - 2/d, & rd \geq 2. \end{cases}$$

This applies to both finite-width bounds and the corresponding critical variational rate. No change to the network architecture is needed.

The lower bound supplies a common baseline for rate comparisons. It does not identify the minimum rate, bound the difference between a target's rate and a competitor's rate, or show that the posterior is diffuse among functions of bounded rate. In particular, even a small positive rate gap is amplified by width in the critical posterior.

5 A quantitative first-layer spectrum bound

The proof starts with a quantitative lower bound on every Fourier eigenvalue of the first tanh feature kernel.

Lemma 5.1 (A quantitative first-layer spectrum bound)**.** *Let $d \geq 1$, $D = d + 1$, $m = 2^d$, and let Q be the $m \times m$ matrix*

$$Q(x, y) = \mathbb{E}_w \tanh(w \cdot \bar{x}) \tanh(w \cdot \bar{y}), \qquad w \sim \mathcal{N}(0, I_{d+1}/d).$$

Then

$$\lambda_{\min}(Q) \geq e^{-13D}.$$

Proof. Write $q = D/d \in [1, 2]$, and let h_n be the coefficient of the normalized probabilists' Hermite polynomial $\mathrm{He}_n / \sqrt{n!}$ in $z \mapsto \tanh(\sqrt{q} z)$. The coefficients vanish for even n. For every odd $n \geq 1$, the Fourier sine identity

$$\tanh(\sqrt{q} z) = \frac{1}{\sqrt{q}} \int_0^\infty \frac{\sin(tz)}{\sinh(\pi t/(2\sqrt{q}))} \, dt$$

and $\mathbb{E}[\mathrm{He}_n(Z)e^{itZ}] = (it)^n e^{-t^2/2}$ give

$$|h_n| = \frac{1}{\sqrt{qn!}} \int_0^\infty \frac{t^n e^{-t^2/2}}{\sinh(\pi t/(2\sqrt{q}))}\, dt.$$

The sine identity follows, for example, by expanding $1/\sinh(\pi t/2) = 2\sum_{j\ge 0} e^{-\pi(j+1/2)t}$ and using the logarithmic derivative of the Euler product for cosh. Fubini is valid in the displayed Hermite integral: near zero use $|\sin(tZ)| \le t|Z|$, and at infinity the reciprocal hyperbolic sine has exponential decay.

Restrict the positive integral to $[\sqrt{n}, \sqrt{n}+1]$. Using $n! \le n^n$, $1/\sinh u \ge 2e^{-u}$, and $1 \le q \le 2$ gives

$$|h_n| \ge \sqrt{2} \exp\left[-\frac{n}{2} - \left(1+\frac{\pi}{2}\right)\sqrt{n} - \frac{1+\pi}{2}\right] \ge e^{-6n}.$$

The last inequality follows from $\sqrt{n} \le n$, $n \ge 1$ and $\pi < 4$.

Mehler's identity now yields the nonnegative power series

$$Q(x,y) = \sum_{n \text{ odd}} h_n^2 \left(\frac{1+\sum_{i=1}^d x_i y_i}{D}\right)^n.$$

The Boolean characters diagonalize this translation-invariant kernel. For a character of degree r, choose the odd integer $n = r$ if r is odd and $n = r+1$ if r is even. Then $1 \le n \le D$. In the multinomial expansion of the nth power, choosing every selected coordinate exactly once, and choosing the constant coordinate once when r is even, contributes $n!/D^n$ to this character's normalized eigenvalue. Every other contribution is nonnegative. Therefore its eigenvalue for Q/m is at least

$$h_n^2 \frac{n!}{D^n} \ge e^{-12n-D} \ge e^{-13D}.$$

Here $n! \ge (n/e)^n$, and $n[1+\log(D/n)] \le D$, prove $n!/D^n \ge e^{-D}$. Multiplying by $m \ge 1$ proves the stated lower bound for Q. □

6 A subexponential lower bound for the first-layer spectrum

Theorem 6.1 (Subexponential spectral floor)**.** *Let $D = d+1 \ge 2$, $m = 2^d$, and*

$$K(x,y) = \mathbb{E}_w \tanh(w\cdot\bar{x})\tanh(w\cdot\bar{y}), \qquad w \sim \mathcal{N}(0, I_D/d).$$

Then

$$\lambda_{\min}(K) \ge p_d := \exp\{-13\sqrt{D\log(4D)}\}.$$

Proof. Set $q = D/d \in [1,2]$. The positive integral formula from Lemma 5.1 gives, for every odd $n \ge 1$,

$$|h_n| = \frac{1}{\sqrt{qn!}} \int_0^\infty \frac{t^n e^{-t^2/2}}{\sinh(\pi t/(2\sqrt{q}))}\, dt,$$

where h_n is the normalized Hermite coefficient of $z \mapsto \tanh(\sqrt{q}z)$. Use the elementary Stirling upper bound $n! \le 3\sqrt{n}(n/e)^n$ and restrict the integral to $[\sqrt{n}, \sqrt{n}+1]$. Since $1/\sinh u \ge 2e^{-u}$,

$$|h_n| \ge \sqrt{2/3}\, n^{-1/4} \exp\left[-(1+\pi/2)\sqrt{n} - (1+\pi)/2\right] \ge e^{-6\sqrt{n}}.$$

For the last inequality, use $\pi < 4$, $\log n \le \sqrt{n}$, and $\sqrt{n} \ge 1$.

The normalized Walsh coefficient of

$$\left(\frac{1+\sum_{i=1}^{d} z_i}{D}\right)^n$$

at a subset $S \subseteq \{1,\dots,d\}$ is the probability $P_n(S)$ that the lazy walk on $\mathbb{F}_2^d$, choosing uniformly between staying put and toggling any of its d coordinates, reaches S after n steps. Fourier inversion gives

$$|mP_n(S) - 1| \le \sum_{j=1}^{d} \binom{d}{j} \left|1 - \frac{2j}{D}\right|^n.$$

For $1 \le j \le D-1$, put $k = \min(j, D-j)$ and bound the last factor by $e^{-2nk/D}$. Pairing $j = k$ with $j = D-k$ and using Pascal's identity bounds the sum by

$$\sum_{k=1}^{\lfloor D/2 \rfloor} \binom{D}{k} e^{-2nk/D} \le (1+e^{-2n/D})^D - 1.$$

At the middle index when D is even this only overcounts, so the inequality remains valid. Choose the smallest odd integer $n \ge (D/2)\log(4D)$. Then $n \le D\log(4D)$ and the displayed sum is at most $e^{1/4} - 1 < 1/2$. Thus $mP_n(S) \ge 1/2$ for every S.

The kernel's Hermite expansion has nonnegative coefficients, so every eigenvalue of K is at least

$$mh_n^2 P_n(S) \ge \frac{1}{2} e^{-12\sqrt{n}} \ge e^{-13\sqrt{D\log(4D)}}.$$

□

Corollary 6.2 (Empirical first-layer kernel)**.** *Let $h_1,\dots,h_N$ be independent first-layer feature vectors and $\widehat{K} = N^{-1}\sum_i h_i h_i^T$. For $0 < \delta < 1$, if*

$$N \ge \frac{8m}{p_d} \log \frac{m}{\delta},$$

then $\widehat{K} \succeq K/2 \succeq (p_d/2) I_m$ with probability at least $1-\delta$.

Proof. The whitened summands $K^{-1/2} h_i h_i^T K^{-1/2}$ have expectation I_m and operator norm at most m/p_d, because $\|h_i\|_2^2 \le m$. The lower-tail matrix Chernoff inequality gives failure probability at most $m\exp[-Np_d/(8m)]$. □

7 Free critical cooling preserves a small target component

Write $F(q) = \mathbb{E}\tanh^2(\sqrt{q}Z)$ for a standard Gaussian Z. For a feature second moment Q, let M_Q denote the law of coordinatewise tanh of a centered Gaussian with covariance Q.

Lemma 7.1 (Uniform free-cooling estimates)**.** *There are universal constants $B < \infty$ and $b > 0$ with the following property. Suppose Q_0 has constant diagonal $q_0 \in [F(1), F(2)]$ and $Q_0 \succeq v_0 f f^T$, where f is Boolean and $v_0 > 0$. Successively apply the unmodified feature law M_Q. For every $0 < \delta \le 1/128$, the first layer whose diagonal q_k is at most δ occurs after at most B/δ transitions. At this layer,*

$$\delta/2 \le q_k \le \delta, \qquad Q_k \succeq b v_0 \delta\, f f^T.$$

All these transitions cost zero in the critical variational functional.

Proof. The diagonal evolves exactly by $q_{j+1} = F(q_j)$. The function F is increasing and $0 < F(q) < q$ for $q > 0$. Taylor expansion with bounded derivatives gives

$$F(q) = q - 2q^2 + O(q^3).$$

It follows by continuity on $(0, F(2)]$ and the limit at zero that there is a universal $c_F > 0$ satisfying $q - F(q) \geq c_F q^2$ throughout this interval. Hence $1/q_{j+1} - 1/q_j \geq c_F$, which bounds the first hitting time of $(0, \delta]$ by $O(1/\delta)$. The elementary bound $F(q) \geq q/(1+3q)$ follows from $\tanh^2 t \geq t^2/(1+t^2)$ and Cauchy–Schwarz. Monotonicity therefore implies that the first value below δ is at least $F(\delta) \geq \delta/(1+3\delta) \geq \delta/2$.

If $Q \succeq v f f^T$ has constant diagonal q, decompose its Gaussian as $\sqrt{v}\, fZ + G$, with G independent and covariance $Q - v f f^T$. The conditional feature mean is $f\psi(Z)$, with the same scalar function at every input because all residual variances equal $q - v$. Conditional expectation, projection onto the first Hermite polynomial, and Gaussian integration by parts give

$$Q' \succeq \mathbb{E}\psi(Z)^2 f f^T \succeq v[1 - F(q)]^2 f f^T.$$

Thus we may track $v_{j+1} = v_j[1 - F(q_j)]^2$. The ratio v_j/q_j is multiplied by

$$R(q_j) = \frac{q_j[1 - F(q_j)]^2}{F(q_j)}.$$

The expansion of F yields $R(q) = 1 + O(q^2)$ at zero, and R is positive on $(0, F(2)]$. Consequently $|\log R(q)| \leq C_R q^2$ there for a universal C_R. Also

$$\sum_{j<k} q_j^2 \leq c_F^{-1} \sum_{j<k} (q_j - q_{j+1}) \leq F(2)/c_F.$$

The product of all ratio factors is therefore at least a universal positive constant. Since $q_0 < 1$ and $q_k \geq \delta/2$, this gives $v_k \geq b v_0 \delta$ for a universal $b > 0$. □

8 Scalar control and target-component constructions

We establish the scalar value and the constructions used in the proof of Theorem 3.1. The critical variational functional is the one defined in the model section, with output margin $1/2$. All logarithms are natural.

Definition 8.1 (Scalar control value)**.** *For $q > 0$, a scalar path of length $m \geq 0$ starts at $q_0 = q$. At step j, choose a probability law η_j on $\mathbb{R}$ with finite entropy relative to $\mathcal{N}(0, q_{j-1})$, and set*

$$q_j = \int \tanh^2 z \, d\eta_j(z).$$

Its total cost is

$$\sum_{j=1}^{m} D(\eta_j \| \mathcal{N}(0, q_{j-1})) + \frac{1}{8q_m}.$$

Let $V(q)$ be the infimum of these costs over all finite paths, including the length-zero path. Define

$$V_* := \lim_{q \downarrow 0} V(q).$$

The next lemma establishes that this limit exists and is finite.

8.1 Properties of the scalar value

Lemma 8.2 (Scalar value and increasing controls)**.** *The function V is nonincreasing, $1 < V_* < 2$, and there is a universal constant C_0 such that*

$$0 \le V_* - V(q) \le C_0 q \qquad (0 < q \le 1/128).$$

For every $0 < p \le 1/128$, a scalar path of cost at most $V(p) + p$ can be chosen with strictly increasing states.

Proof. Monotonicity follows from rescaling controls. If one starts a path at a larger variance, rescale its first preactivation control by the square root of the variance ratio. Relative entropy from the corresponding Gaussian is unchanged, while the next second moment increases because $|\tanh z|$ increases with $|z|$. Repeat this at every step. The final readout cost decreases. Taking infima proves that V is nonincreasing.

Put $F(u) = \mathbb{E}\tanh^2(\sqrt{u}Z)$. The elementary inequality $\tanh^2 z \ge z^2/(1+z^2)$ and Cauchy–Schwarz give $F(u) \ge u/(1+3u)$. Here is an elementary uniform upper bound. For $q < 1/10$, replace its reference Gaussian by $\mathcal{N}(0, (1+6q)q)$, at entropy cost at most $9q^2$. The bound $F(u) \ge u/(1+3u)$ gives

$$q' - q \ge \frac{q^2(3-18q)}{1+3q+18q^2} > 0.$$

The cost divided by this variance increment is at most

$$R_0(q) = \frac{3(1+3q+18q^2)}{1-6q},$$

which is increasing on $[0, 1/10]$. Repeatedly take full steps while the new state remains below $1/10$. Their total cost is bounded by

$$\int_0^{1/10} R_0(q)\, dq = -\frac{69}{200} + \log(5/2).$$

On the last step, continuously reduce the variance inflation to hit $1/10$ exactly; this costs at most $9/100$. The readout then costs $5/4$. The total is at most $199/200 + \log(5/2) < 2$. For starting states at least $1/10$, the length-zero path already costs at most $5/4$. Thus $V(q) < 2$ for all $q > 0$. Since V is nonincreasing, its limit at zero exists and is at most two.

For a quantitative small-state bridge, let $q \le p_0 := 1/128$ and replace its reference Gaussian by $\mathcal{N}(0, (1+16q)q)$. The entropy cost is at most $64q^2$. The first Hermite coefficient and Gaussian integration by parts give

$$F(u) \ge u[1-F(u)]^2 \ge u(1-2u).$$

Consequently this control gives a new state satisfying

$$q + 4q^2 \le q' \le q + 16q^2 \qquad (q \le p_0).$$

The scalar exponential potential from Lemma 4.1, at $t = 1$ and $a = 1$, gives entropy cost at least $2(q_j - q_{j-1})$ at each step. Summing and including the readout shows $V(q) \ge 1 - 2q$, so $V_* \ge 1$. In fact the inequality is strict in the limit. Let

$$g(q) = 2q - \log \mathbb{E}\exp\{2\tanh^2(\sqrt{q}Z)\}.$$

The strict heat-equation inequality gives $g(q) > 0$ for $q > 0$. Consider a path from $q < 1/100$ with cost at most two. If its final state is below $1/10$, its readout alone costs more than $5/4$. Otherwise

consider its first crossing of $1/10$. The preceding state must exceed $1/100$: a transition with $q'/q \geq 10$ has entropy at least $[10 - 1 - \log 10]/2 > 2$. To see this, its preactivation second moment is at least q', since $\tanh^2 z \leq z^2$; Gaussian entropy duality gives the lower bound $[r - 1 - \log r]/2$ at a second-moment ratio $r \geq 1$. The pre-crossing state therefore belongs to $[1/100, 1/10]$. Summing the strengthened step bounds $D \geq 2(q' - q) + g(q)$, and minimizing the terminal $2q_m + 1/(8q_m)$, gives total cost at least $1 - 2q + \min_{[1/100,1/10]} g$. The alternative final-state-below-$1/10$ case satisfies the same bound, since $\min_{[1/100,1/10]} g \leq g(1/100) \leq 1/50 < 1/4$. Paths of cost greater than two satisfy a stronger bound automatically. Letting $q \downarrow 0$ proves $V_* > 1$.

The same Gaussian-growth construction, stopped exactly at any $p \leq p_0$, costs at most $17p$ when started at an arbitrarily small positive state. Indeed all full steps cost at most 16 times their variance increment. For the last step, continuously reduce the variance inflation to hit p exactly; its cost is at most $64p^2 \leq p$. Appending a nearly optimal path from p and letting the starting state tend to zero yields $V_* \leq 17p + V(p)$.

We also need a strict dissipation estimate. There is a universal $c > 0$ such that

$$\log \mathbb{E} \exp\{\tanh^2(\sqrt{q}Z)\} \leq q - cq^2, \qquad 0 \leq q \leq 1. \tag{1}$$

For a direct verification, $G(z) = \exp(\tanh^2 z)$ satisfies $G'' < 2G$ away from zero. The heat equation gives a strict inequality $\log \mathbb{E}G(\sqrt{q}Z) < q$ for $q > 0$. Taylor expansion gives $\log \mathbb{E}G(\sqrt{q}Z) = q - q^2 + O(q^3)$ at zero. Compactness on $[0, 1]$ therefore gives a positive c in (1). Entropy duality now implies

$$D(\eta_j \| \mathcal{N}(0, q_{j-1})) \geq q_j - q_{j-1} + cq_{j-1}^2. \tag{2}$$

An arbitrary finite path can be made strictly increasing without raising its cost. If a step decreases or preserves the state, delete that step. Start the remaining controls at the larger preceding state and rescale each successive control as in the monotonicity argument. Their entropies are unchanged and the final readout cost cannot increase. Repeat until there are no nonincreasing steps; the procedure terminates because each deletion shortens the path. Take a path within p of $V(p)$ and apply this procedure. The resulting increasing path has the asserted cost, proving the remaining claim. □

8.2 A small-variance layer forces the scalar cost

Lemma 8.3 (Comparison of a signed pair with the scalar control problem)**.** *For any admissible critical variational path and any signed input pair, put $b = (e_x + se_y)/2$, $s \in \{-1, 1\}$, and $v_j = b^T Q_j b$. If the output satisfies $|b^T z| \geq 1/2$, the part of the path after layer j, including the readout, costs at least $V(v_j)$. Moreover, a transition from pair variance $v \leq 1$ to v' has entropy cost at least $v' - v + cv^2$, with the constant from (1).*

Proof. It suffices to compare one transition. Lift a finite-entropy feature law through the inverse coordinatewise tanh map to its preactivation law. Under the reference law, the preactivation combination $b^T G$ is $\mathcal{N}(0, v)$. Under the controlled law, denote its distribution by η. Data processing bounds its Gaussian relative entropy by the original layer entropy. The pair contraction inequality gives

$$\int \tanh^2 z \, d\eta(z) \geq v',$$

where v' is the next feature-pair second moment. If the left side is q', the definition of V and its monotonicity imply

$$V(v) \leq D(\eta \| \mathcal{N}(0, v)) + V(q') \leq D(\nu' \| \mathsf{M}_Q) + V(v').$$

The final readout costs at least $1/(8v_{L-1}) \geq V(v_{L-1})$. Telescoping proves the assertion. Zero-variance or singular-covariance cases cause no problem: a finite-cost path reaching the stipulated nonzero pair output must have positive pair variance at every relevant step, and Cauchy–Schwarz is taken on the covariance range. For the final assertion, entropy duality applied to (1) gives

$$D(\eta\|\mathcal{N}(0,v)) \geq \int \tanh^2 z \, d\eta(z) - v + cv^2 \geq v' - v + cv^2.$$

Data processing transfers this bound to the full layer entropy. □

8.3 Holding and growing an arbitrarily weak target component

Suppose a feature second moment has the decomposition

$$Q = vff^T + R, \qquad R \succeq 0, \qquad R(x,x) = r \quad \text{for every } x.$$

Its diagonal is $q = v + r$. Inflate only the target component, taking the next preactivation covariance to be $Q + \epsilon vff^T$. Its entropy cost is at most $\epsilon^2/4$. Indeed, f lies in the range of Q, and $u = vf^TQ^+f \leq 1$. The Gaussian relative entropy on that range is $[\epsilon u - \log(1+\epsilon u)]/2 \leq \epsilon^2/4$; applying tanh cannot increase it. Put

$$\psi_r(z) = \mathbb{E}\tanh(z + \sqrt{r}Z), \qquad v' = \mathbb{E}\psi_r(\sqrt{(1+\epsilon)v}Z)^2, \qquad q' = F(q+\epsilon v).$$

Conditional expectation gives $Q' \succeq v'ff^T$. Conditional Gaussian Poincare, using the unit Lipschitz constant of tanh, also gives

$$r' := q' - v' \leq r. \tag{3}$$

The map $\epsilon \mapsto v'$ is continuous and nondecreasing. At $\epsilon = 0$ it is at most v, while the first Hermite projection gives

$$v' \geq (1+\epsilon)v[1 - F(q+\epsilon v)]^2 \geq (1+\epsilon)v[1 - 2(q+\epsilon v)]. \tag{4}$$

These statements depend only on the tracked scalars v, r, ϵ, not on the remaining entries of R or on the choice of f.

Lemma 8.4 (Cheap holding and prescribed growth)**.** *There are universal constants A, B, C_2 with the following properties. Let $D = d+1$ and $0 < \delta \leq p \leq 1/128$. Starting from the unperturbed first feature law, at zero entropy cost and after at most B/δ hidden layers, one can track*

$$Q \succeq v_0 ff^T, \qquad v_0 = \delta e^{-AD}, \qquad Q(x,x) \leq \delta.$$

One may hold this v_0 exactly fixed for any specified number H of layers, at total entropy cost at most $64H\delta^2$, retaining diagonal residual at most δ. Thereafter one can grow the tracked component exactly to p in at most BD/δ layers, with entropy cost at most

$$C_2 p + C_2 D\delta,$$

and with residual diagonal still at most δ. The number of growth layers is fixed in advance from v_0, δ, p, so it is independent of the preceding holding time and of f.

Proof. The first unperturbed covariance has smallest eigenvalue at least e^{-13D} by Lemma 5.1. Since the cube has $2^d \leq e^D$ points, it dominates $e^{-14D}ff^T$ for every f. Lemma 7.1 shows that, after $O(1/\delta)$ unmodified layers, the diagonal lies in $[\delta/2, \delta]$ and the tracked target component is at

least $b\delta e^{-14D}$. Enlarging A gives the displayed common lower component v_0; reduce the tracked component to precisely that value if necessary.

For holding, use an inflation $\epsilon \in [0, 16\delta]$ chosen to make $v' = v_0$. Such a choice exists by continuity: at zero inflation $v' \le v_0$, and (4) gives $v' \ge v_0$ at 16δ when $q \le \delta \le 1/128$. The residual does not increase, so the same argument applies at every holding layer. Each costs at most $64\delta^2$.

For growth, prescribe the scalar sequence

$$v_{j+1} = \min\{p,\ v_j[1 + 2(\delta + v_j)]\},$$

stopping upon reaching p. Throughout it, $r \le \delta$ and $v_j \le p$. At inflation $\epsilon = 16(\delta + v_j)$, (4) gives $v' \ge v_j[1 + 4(\delta + v_j)]$. Therefore an inflation no larger than this realizes the prescribed next value exactly. Its entropy cost is at most $64(\delta + v_j)^2$. For a full step, with $\Delta v = 2(\delta + v_j)v_j$, this cost is

$$32\Delta v + 32\delta\frac{\Delta v}{v_j}.$$

Here $\Delta v/v_j \le 4p < 1$, so $\Delta v/v_j \le 2\log(v_{j+1}/v_j)$. Summing, bounding the possible last step separately by $256p^2$, and using $\delta\log(p/\delta) \le p/e$, proves a bound $C_2p + C_2D\delta$.

Before v reaches δ, each full step multiplies it by at least $1 + 2\delta$, so at most $AD/\delta + 1$ steps are needed. Afterwards each full step decreases $1/v$ by at least one, so at most $1/\delta + 1$ further full steps and one final step are needed. This proves the stated BD/δ length bound. The prescribed scalar sequence fixes the length independently of the residual or holding time. □

8.4 Stable lifting of scalar controls

Lemma 8.5 (Lifting with a small residual)**.** *Suppose $Q \succeq pff^T$ has constant diagonal $p + r$, with $0 \le r \le \delta$. Take a scalar path of length m starting at p. It can be lifted to a vector feature path with no greater total hidden entropy. Its final second moment dominates*

$$(1 - \delta)^{2m} q_m ff^T.$$

The residual diagonal stays at most δ throughout. If the scalar path has total cost at most three and $2m\delta \le 1/2$, the lifted path, including output $z = f/2$, costs at most the scalar cost plus $12m\delta$.

Proof. For $z \ge 0$ and $r \ge 0$, symmetry of a standard Gaussian gives

$$\psi_r(z) = \mathbb{E}\tanh(z + \sqrt{r}Z) \ge [1 - F(r)]\tanh z \ge (1 - r)\tanh z.$$

Indeed, writing $u = \tanh z$ and $w = \tanh(\sqrt{r}Z)$, the average of the two values corresponding to w and $-w$ is $u(1 - w^2)/(1 - u^2w^2) \ge u(1 - w^2)$. Oddness gives the corresponding absolute-value inequality for all real z.

Suppose the current tracked vector component is v, whereas the nominal scalar state is q. Rescale the nominal scalar preactivation control η by $\sqrt{v/q}$. Its entropy relative to $\mathcal{N}(0, v)$ equals the nominal entropy relative to $\mathcal{N}(0, q)$. Independently add a centered Gaussian with the residual covariance. Data processing bounds the vector feature entropy by that same scalar entropy. Its new tracked component is the second moment of the conditional feature mean. Since $\tanh(cz) \ge c\tanh z$ for $z \ge 0$ and $0 \le c \le 1$, the new component v' obeys

$$\frac{v'}{q'} \ge (1 - \delta)^2 \min\{1, v/q\},$$

where q' is the nominal next scalar state. The conditional-variance argument from (3) applies equally to a non-Gaussian scalar control, so the residual diagonal never increases. Starting from $v = q = p$ proves the component bound by induction.

The lifted readout cost is at most the nominal readout cost divided by $(1-\delta)^{2m} \geq 1 - 2m\delta$. If $2m\delta \leq 1/2$, its increase is at most $4m\delta$ times the nominal readout cost, which is at most three. This proves the final assertion. □

9 Short near-optimal scalar paths and residual spectral control

Near-optimality controls the entropy spent at every prefix of an increasing scalar path. This yields the path-length estimate used in the upper bound, and controls the spectral loss when the path is lifted to vector features. We use V, V_* and the constant $c > 0$ from (2).

Lemma 9.1 (Dyadic bounds for a near-optimal increasing path)**.** *There is a universal C such that, for every $0 < p \leq 1/256$, an increasing scalar path from p with total cost at most $V(p) + p$ has length m and states $q_0 = p, q_1, \ldots, q_m$ satisfying*

$$m \leq C/p, \qquad \sum_{j=0}^{m-1} q_j \leq C \log(2/p).$$

Such a path exists by the increasing-path reduction. The same sum bound holds with the post-step states $q_1, \ldots, q_m$.

Proof. Write D_j for the entropy cost of step j. Every suffix from q_i including its readout costs at least $V(q_i)$, so

$$\sum_{j=1}^{i} D_j \leq V(p) + p - V(q_i) \leq V_* - V(q_i) + p. \tag{5}$$

Consider a dyadic band $[u, 2u)$, where $u = 2^k p < 1/256$. Because states increase, its transitions form one consecutive segment. At most one transition exits the band. Every other transition has post-state below $2u$, and the prefix ending at the last such transition has cost, by (5), at most

$$V_* - V(2u) + p \leq 34u + p \leq 35u.$$

Every step whose pre-state lies in this band has cost at least cu^2, by increasing states and strict dissipation. Thus at most $35/(cu) + 1$ steps start in the band, contributing at most $70/c + 2u$ to the sum of their pre-states.

Choose the last band so that its next lower endpoint lies in $[1/256, 1/128)$. All remaining pre-states are at least $1/256$. Since total cost is less than three, at most $3 \cdot 256^2/c$ steps remain. Summing the dyadic counts gives $m \leq C/p$, and summing the pre-state contributions gives $C \log(2/p)$. Finally, the post-state sum differs from the pre-state sum by $q_m - p \leq 1$. □

Lemma 9.2 (A lower spectral bound for a lifted residual)**.** *Let $f \in \{-1, 1\}^M$, let $R \succeq 0$ have constant diagonal r, and let $G \sim \mathcal{N}(0, R)$ be independent of a scalar random variable ξ. Define vector features*

$$h_x = \tanh(f_x \xi + G_x), \qquad \psi_r(z) = \mathbb{E} \tanh(z + \sqrt{r} Z),$$
$$v' = \mathbb{E}\psi_r(\xi)^2, \qquad q' = \mathbb{E} \tanh^2(\xi + \sqrt{r} Z), \qquad R' = \mathbb{E} h h^\top - v' f f^\top.$$

Then

$$R' \succeq (1 - q')^2 R. \tag{6}$$

This applies to arbitrary scalar control laws with finite second moment, not only Gaussian controls.

Proof. Condition on ξ. The conditional mean of h is $f\psi_r(\xi)$. Put $a(\xi) = \mathbb{E}\operatorname{sech}^2(\xi+\sqrt{r}Z)$. Gaussian integration by parts gives

$$\operatorname{Cov}(h, G \mid \xi) = a(\xi)R.$$

The coefficient is the same at every input because the derivative is even and the residual diagonal is constant. The covariance matrix of $h - \mathbb{E}[h \mid \xi] - a(\xi)G$ is positive semidefinite, hence

$$\operatorname{Cov}(h \mid \xi) \succeq a(\xi)^2 R.$$

This argument also handles singular R by Gaussian approximation, or directly by a linear representation of G. Average over ξ and apply Jensen: $R' = \mathbb{E}\operatorname{Cov}(h \mid \xi) \succeq (\mathbb{E}a(\xi))^2 R = (1-q')^2 R.$ □

Corollary 9.3 (Only polynomial spectral loss through a near-optimal lift)**.** *Lift a scalar path from Lemma 9.1 by the earlier scalar-lifting construction. Suppose the initial residual has constant diagonal at most $\delta \le p$. Then for universal $c_0 > 0, C_0 < \infty$, its final residual satisfies*

$$R_m \succeq c_0 p^{C_0} R_0.$$

Proof. At each step the scalar control is rescaled by the square root of the ratio of the tracked component v to the nominal state q. The tracked component never exceeds the nominal state. To check this, for $z \ge 0$ and $r \ge 0$, averaging the tanh addition formula at opposite Gaussian shifts gives

$$0 \le \psi_r(z) \le \tanh z.$$

If $v \le q$, rescaling by $\sqrt{v/q} \le 1$ can therefore only decrease the absolute scalar feature. Induction from $v = q = p$ proves the claim. The residual diagonal stays at most δ by conditional Gaussian Poincare. Thus the actual total feature variance at step j is at most $q_j + \delta$.

Every nominal step has entropy at most three and reference variance at most one. Gaussian entropy duality gives its preactivation second moment at most $12 + 2\log 2 < 14$. Rescaling does not increase this moment; adding the independent residual raises it by at most one. Consequently for each actual total post-step variance $\widehat{q}_j$,

$$1 - \widehat{q}_j = \mathbb{E}\operatorname{sech}^2(\xi + \sqrt{r}Z) \ge \mathbb{E}e^{-2|\xi+\sqrt{r}Z|} \ge e^{-2\sqrt{15}} =: a_0 > 0.$$

For $0 \le u \le 1 - a_0$, one has $\log(1-u) \ge -u/a_0$. Lemma 9.2 therefore gives

$$R_m \succeq \exp\left[-\frac{2}{a_0}\sum_{j=1}^{m}\widehat{q}_j\right] R_0 \succeq \exp\left[-\frac{2}{a_0}\left(\sum_{j=1}^{m} q_j + m\delta\right)\right] R_0.$$

The dyadic lemma bounds the state sum by $C\log(2/p)$ and $m\delta \le C$, proving the assertion. □

These are variational-path and population-covariance statements. They do not alone give a finite-width lower probability or a sample lower bound. Their role is to avoid an unnecessarily exponential spectral loss when constructing such a finite-width comparison.

10 The inverse-depth lower bound

The lower bound follows from the fact that a low-total-cost path has a small entropy budget at *every* prefix ending at a small variance.

Lemma 10.1 (A last-exit occupation bound)**.** *Let $q_0, \dots, q_H \in (0,1]$, with transition costs $D_j \ge 0$. Suppose there are universal constants $c, C_0 > 0$ such that*

$$D_j \ge q_j - q_{j-1} + cq_{j-1}^2, \qquad \sum_{j=1}^{i} D_j \le C_0 q_i \quad (0 \le i \le H), \qquad \sum_{j=1}^{H} D_j \le 2.$$

Then, for a constant C depending only on c, C_0,

$$H \le C/\epsilon, \qquad \epsilon = \min_{0 \le j \le H} q_j.$$

The states need not be monotone.

Proof. Concavity of $q \mapsto -1/q$ gives

$$\frac{q_j - q_{j-1}}{q_{j-1}^2} \ge \frac{1}{q_{j-1}} - \frac{1}{q_j}.$$

On an interval of ℓ transitions whose pre-states are all at least $u > 0$, division of the dissipation inequality and summation therefore imply

$$c\ell \le \frac{\text{entropy spent on the interval}}{u^2} + \frac{1}{q_{\text{end}}} - \frac{1}{q_{\text{start}}}. \tag{7}$$

Fix $u_* = 1/256$. If $\epsilon \ge u_*$, apply this inequality to the entire path with total entropy at most two; its length is bounded by a universal constant, which is enough. Otherwise choose the least J such that $u_J = 2^J \epsilon \ge u_*$. Then $u_J < 2u_*$. For $0 \le j \le J$, let

$$\tau_j = \max\{i : q_i \le u_j\}.$$

These indices exist and are nondecreasing. The prefix ending at τ_0 ends at the global minimum ϵ, stays above it, and has entropy at most $C_0\epsilon$. By (7), its length is at most $(C_0 + 1)/(c\epsilon)$.

Consider a nonempty block from τ_j to τ_{j+1}. Remove its first transition temporarily. Every pre-state in the remaining interval exceeds u_j, by definition of τ_j, and its endpoint lies in $(u_j, 2u_j]$. The entropy of this interval is at most the total prefix entropy at τ_{j+1}, hence at most $2C_0u_j$. Its length is consequently at most $(2C_0 + 1)/(cu_j)$. Restoring the first transition adds one. If the two last-exit indices coincide, the block is empty.

After τ_J, remove one first transition if present. Every remaining state exceeds $u_J \ge u_*$, and the total remaining entropy is at most two. The same weighted inequality bounds this final tail by a universal constant. Summing the geometric series $\sum_{j<J} u_j^{-1} \le 2/\epsilon$ and absorbing the at most J removed transitions proves the assertion. □

Theorem 10.2 (Uniform inverse-depth lower bound)**.** *There is a universal constant C such that, for every input dimension $d \ge 1$, depth $L \ge 2$, and Boolean function f,*

$$J_L(f) \ge V_* - \frac{C}{L-1}.$$

Proof. For a nonconstant target, choose opposite labels on one cube edge; for a constant target, choose equal labels on an antipodal pair. In each case take the signed pair whose required output magnitude is at least $1/2$. Write $q_0 = 1/d$ and let $q_1, \dots, q_H$, $H = L - 1$, be its successive pair variances. Every finite-cost path satisfying the required nonzero readout has positive pair variances.

All lie in $(0,1]$. By Lemma 8.3, the full layer entropy costs satisfy the strict pair dissipation inequality required in Lemma 10.1.

Let A be the total path cost including the readout. If $A > V_*$, the desired conclusion is immediate. Otherwise the suffix comparison with the scalar control value gives, at every layer i,

$$\sum_{j=1}^{i} D_j \le A - V(q_i) \le V_* - V(q_i) \le C_0 q_i.$$

The last inequality follows from the small-state bridge for $q_i \le 1/128$ and from $0 \le V(q_i) \le V_* < 2$ for larger states; for example one may take $C_0 = 256$. Also $\sum_j D_j \le A \le V_* < 2$. Lemma 10.1 shows that the minimum pair variance ϵ is at most C/H. At a layer attaining it, the scalar suffix comparison gives

$$A \ge V(\epsilon) \ge V_* - C_0\epsilon \ge V_* - \frac{C'}{H}.$$

Take the infimum over admissible variational paths and rename the universal constant. □

11 Separating the scalar baseline from directional entropy

This section gives a structural consequence of a critical variational path whose cost is close to the universal scalar value V_*. It does not yet give a product-norm or sample-complexity bound.

Lemma 11.1 (Last-exit occupation with an additive budget)**.** *Let $q_0, \dots, q_H \in (0,1]$ and $D_1, \dots, D_H \ge 0$ satisfy*

$$D_j \ge q_j - q_{j-1} + cq_{j-1}^2, \qquad \sum_{j=1}^{i} D_j \le \varepsilon + C_0 q_i, \qquad \sum_{j=1}^{H} D_j \le 3,$$

where $c, C_0 > 0$ are fixed and $0 \le \varepsilon \le 1$. Writing $m = \min_{0\le j\le H} q_j$, there is a constant C, depending only on c, C_0, such that

$$H \le C\left(\frac{1}{m} + \frac{\varepsilon}{m^2}\right).$$

Consequently, for $H \ge 1$,

$$m \le \frac{C'}{H} + C'\sqrt{\frac{\varepsilon}{H}}.$$

Proof. For an interval of ℓ transitions whose pre-states are at least $u > 0$, the dissipation inequality and concavity of $q \mapsto -1/q$ give

$$c\ell \le \frac{\text{entropy on the interval}}{u^2} + \frac{1}{q_{\text{end}}} - \frac{1}{q_{\text{start}}}.$$

Fix a universal $u_* \in (0, 1/4)$. If $m \ge u_*$, applying this bound to the entire path gives a universal upper bound for H, which suffices. Otherwise choose the least J with $u_J = 2^J m \ge u_*$. Define the nondecreasing last-exit indices

$$\tau_j = \max\{i : q_i \le u_j\}, \qquad 0 \le j \le J.$$

The prefix ending at τ_0 has entropy at most $\varepsilon + C_0 m$ and all its states are at least m, so its length is at most $C(\varepsilon/m^2 + 1/m)$.

For a nonempty block from τ_j to τ_{j+1}, remove its first transition. All remaining pre-states exceed u_j, and the endpoint lies in $(u_j, 2u_j]$. The entropy of this interval is at most the full prefix entropy at τ_{j+1}, hence at most $\varepsilon + 2C_0u_j$. Its length is therefore at most $C(\varepsilon/u_j^2 + 1/u_j)$. Restoring the removed step adds one. After τ_J, remove one first step if present; the remaining pre-states exceed $u_J \geq u_*$ and the remaining entropy is at most three, so this tail has universally bounded length. Summing the geometric series in u_j^{-1} and u_j^{-2}, and absorbing the at most J removed transitions, proves the first claim. Solving $Hm^2 \leq C(m + \varepsilon)$ proves the second. □

Theorem 11.2 (Near-baseline directional budget)**.** *Consider an admissible critical variational path of depth $L \geq 2$ with hidden second moments $Q_0, \ldots, Q_H$, $H = L - 1$, hidden entropy costs $D_1, \ldots, D_H$, output z, and readout cost*

$$r = \tfrac{1}{2} z^{\mathsf{T}} Q_H^{+} z.$$

Assume its total cost $A = \sum_{j=1}^{H} D_j + r$ satisfies $A \leq V_* + \varepsilon$ with $0 \leq \varepsilon \leq 1$ and $A \leq 3$, and that $f(x)z(x) \geq 1/2$ on the Boolean cube.

Choose a bichromatic cube edge if f is nonconstant, and an antipodal pair if f is constant. In the first case set $b = (e_x - e_y)/2$; in the second set $b = (e_x + e_y)/2$. Write

$$v_j = b^{\mathsf{T}} Q_j b, \qquad a = b^{\mathsf{T}} z, \qquad m = \min_{0 \leq j \leq H} v_j,$$

and choose an index i attaining the minimum. All v_j are positive, $v_0 = 1/d$, and

$$m \leq \frac{C}{H} + C\sqrt{\frac{\varepsilon}{H}}.$$

For each $j > i$, lift the controlled feature law through the inverse coordinatewise tanh *map to a preactivation law ρ_j. Let η_j be its pushforward under $G \mapsto b^{\mathsf{T}} G$, and define the conditional entropy*

$$E_j = D(\rho_j \| N(0, Q_{j-1})) - D(\eta_j \| N(0, v_{j-1})) \geq 0.$$

Then the nonnegative quantities

$$r_\perp = r - \frac{a^2}{2v_H}, \qquad r_{\mathrm{amp}} = \frac{a^2 - 1/4}{2v_H}$$

satisfy

$$\boxed{\sum_{j=1}^{i} D_j + \sum_{j=i+1}^{H} E_j + r_\perp + r_{\mathrm{amp}} \leq \varepsilon + C_0 m.}$$

In particular this budget is at most $C\{\varepsilon + H^{-1} + \sqrt{\varepsilon/H}\}$. More precisely, define the nonnegative scalar Bellman slacks

$$s_j = D(\eta_j \| N(0, v_{j-1})) + V(v_j) - V(v_{j-1}), \qquad s_{\mathrm{out}} = \frac{1}{8v_H} - V(v_H).$$

They satisfy the exact decomposition

$$A - V(m) = \sum_{j=1}^{i} D_j + \sum_{j=i+1}^{H} (E_j + s_j) + r_\perp + r_{\mathrm{amp}} + s_{\mathrm{out}}.$$

Thus all the displayed nonnegative terms together obey the same upper bound $\varepsilon + C_0 m$.

Proof. The chosen pair has $|a| \geq 1/2$ and input variance $v_0 = 1/d$. The signed-pair comparison in Lemma 8.3 gives, for every prefix,

$$\sum_{j=1}^{t} D_j \leq A - V(v_t) \leq \varepsilon + C_0 v_t.$$

Here $V_* - V(v) \leq C_0 v$ extends from small v to $(0, 1]$ by increasing C_0, since $0 \leq V(v) \leq V_* < 2$. The same comparison gives the dissipation inequality, and the full hidden entropy is at most $A \leq 3$. Thus Lemma 11.1 gives the claimed minimum-variance bound. Positivity follows because a zero signed-pair variance propagates under every finite-entropy transition and is incompatible with $|a| \geq 1/2$.

Coordinatewise tanh is a measurable bijection onto its image, so lifting preserves relative entropy. The chain rule under the scalar projection gives $D_j = D_j^{\mathrm{sc}} + E_j$, where $D_j^{\mathrm{sc}} = D(\eta_j \| N(0, v_{j-1}))$; equivalently, E_j is the average KL divergence of the controlled conditional law given $b^{\mathsf{T}} G$ from the corresponding conditional Gaussian law. This identity remains valid for singular covariances by disintegration on their Gaussian support.

The signed-pair contraction inequality gives $\int \tanh^2 s \, d\eta_j(s) \geq v_j$. The scalar Bellman inequality and monotonicity of V therefore imply

$$D_j^{\mathrm{sc}} \geq V(v_{j-1}) - V(v_j).$$

This proves $s_j \geq 0$, while the length-zero scalar path gives $s_{\mathrm{out}} \geq 0$. After summing over $j > i$ and adding $1/(8v_H) \geq V(v_H)$,

$$\sum_{j=i+1}^{H} D_j^{\mathrm{sc}} + \frac{1}{8v_H} \geq V(v_i).$$

The finite readout cost implies $z \in \operatorname{range} Q_H$. Cauchy–Schwarz on that range gives $r_\perp \geq 0$, and $r_{\mathrm{amp}} \geq 0$ follows from $|a| \geq 1/2$. Since

$$r = \frac{1}{8v_H} + r_\perp + r_{\mathrm{amp}},$$

the total path cost is at least

$$\sum_{j=1}^{i} D_j + \sum_{j=i+1}^{H} E_j + r_\perp + r_{\mathrm{amp}} + V(v_i).$$

Subtract $V(v_i)$ and use $A \leq V_* + \varepsilon$ and $v_i = m$ to obtain the boxed inequality. Retaining the scalar slacks in the same telescoping calculation gives the asserted exact decomposition. □

Corollary 11.3 (Exact classification by one anchored kernel section)**.** *Under the theorem's hypotheses set $B = \varepsilon + C_0 m$. If $B < 1/8$, then on the entire Boolean cube*

$$f = \operatorname{sign}\left(\frac{a}{v_H} Q_H b \right).$$

If $B \leq 1/32$, then the signed section $\operatorname{sign}(a) Q_H b$ has uniform margin at least $1/(16A) \geq 1/48$ for f.

Proof. Put $z_\perp = z - (a/v_H) Q_H b$. Orthogonal decomposition in the Q_H^+ seminorm gives

$$\tfrac{1}{2} z_\perp^{\mathsf{T}} Q_H^+ z_\perp = r_\perp \leq B.$$

Since $Q_H(u,u) \le 1$, coordinatewise Cauchy–Schwarz gives $|z_\perp(u)| \le \sqrt{2B}$. This is strictly less than the output margin $1/2$ when $B < 1/8$, proving the sign identity. If $B \le 1/32$, the anchored output has margin at least $1/4$. Finally $a^2/(2v_H) \le A$ and $|a| \ge 1/2$ imply $|a|/v_H \le 4A$. Dividing the anchored output by that factor gives the asserted margin for the unscaled signed section. □

Proposition 11.4 (Zero directional entropy gives a shallow classifier)**.** *Under the theorem's hypotheses, suppose additionally that*

$$\sum_{j=1}^{i} D_j = 0, \qquad \sum_{j=i+1}^{H} E_j = 0, \qquad r_\perp < 1/8.$$

Then f has a depth-two tanh *implementation with unit global margin, width polynomial in d, and product of layer Frobenius norms polynomial in d. The bounds are independent of L.*

Proof. Let $\mathcal{G}$ be the group of signed coordinate permutations fixing the two selected cube inputs individually. It fixes the vector b under its induced permutation of input indices. The raw input covariance is invariant under this group, and the zero-entropy prefix is exactly the uncontrolled feature recursion, so Q_i is invariant.

At a later layer, $E_j = 0$ means that the controlled preactivation law is obtained by changing only the distribution of $S = b^\mathsf{T} G$ and retaining the reference conditional Gaussian law given S. If the reference covariance is $\mathcal{G}$ invariant, so is this conditional construction, because S is fixed by every group element. Coordinatewise tanh preserves the invariance. Induction therefore makes Q_H invariant. The anchored section $Q_H b$ is consequently constant on every $\mathcal{G}$ orbit. The readout residual bound, by the preceding proof, identifies its sign with f.

After an input sign change, an edge pair is $(1,\dots,1)$ and $(-1,1,\dots,1)$. Permuting coordinates $2,\dots,d$ shows that f depends only on x_1 and $\sum_{j=2}^{d} x_j$, giving at most $2d$ orbit classes. These classes have distinct, separated integer codes

$$t(x) = (d+1)x_1 + \sum_{j=2}^{d} x_j.$$

For an antipodal pair the corresponding orbit classes are determined by $t(x) = \sum_{j=1}^{d} x_j$ and number at most $d+1$. In either case the codes have magnitude $O(d)$ and gaps at least one.

Order their distinct values $t_1 < \cdots < t_M$, and let f_j be the desired sign at t_j. With thresholds halfway between consecutive values, the exact step expansion is

$$f(x) = c_0 + \sum_{j=1}^{M-1} c_j \operatorname{sign}(t(x) - \theta_j), \qquad c_j = (f_{j+1} - f_j)/2, \quad c_0 = (f_1 + f_M)/2.$$

All these coefficients have absolute value at most one. Replace each step by $\tanh(A(t-\theta_j))$ with $A = C\log(d+2)$. The sum's uniform error is at most $1/4$ for a sufficiently large universal C. A constant first-layer feature implements c_0 exactly by adjusting its readout coefficient. The resulting width, incoming coefficients, and readout norm are polynomial in d, and doubling the readout gives unit margin. Input sign changes preserve Frobenius norms. This proves the claim. □

The zero-entropy proposition is a rigidity statement. A quantitative version controlling intrinsic product norm by $\mathrm{poly}(d,L)\exp\{C\sqrt{L\varepsilon}\}$ would require stability of the conditional-Gaussian suffix under the small directional budget. That implication is not proved here.

12 The entropy budget above the scalar baseline

The total action includes an unavoidable scalar cost. Removing that baseline gives a stronger bound on the accumulated layerwise distortion than Cauchy–Schwarz applied to the total action. The resulting estimate also gives a dimension-dependent lower bound on the excess rate. Neither conclusion is a posterior sample-complexity theorem.

Lemma 12.1 (Square-root entropy along a near-baseline path)**.** *There is a universal C such that every admissible critical path with $H = L - 1$ hidden transitions, layer entropies $D_1, \dots, D_H$, and total action $A \le V_* + \epsilon$, $\epsilon \ge 0$, satisfies*

$$\sum_{j=1}^{H} \sqrt{D_j} \le C\big[\log(H+2) + \sqrt{H\epsilon}\big].$$

The path need not have monotone variances or constant diagonal.

Proof. Choose the signed pair from the proof of Theorem 10.2: opposite labels on a cube edge for a nonconstant target, or equal labels on an antipodal pair for a constant target. Its successive half-difference or half-sum variances are $q_0, \dots, q_H \in (0, 1]$, with $q_0 = 1/d$. Positivity follows because zero variance could never subsequently produce the required nonzero readout at finite entropy cost. The pair comparison and scalar suffix bound give, for universal $c, C_0 > 0$,

$$D_j \ge q_j - q_{j-1} + c q_{j-1}^2, \qquad S_i := \sum_{j=1}^{i} D_j \le A - V(q_i) \le \epsilon + C_0 q_i. \tag{8}$$

The small-state bridge extends the last inequality to all $q_i \le 1$ by increasing C_0. These are inequalities for the full vector-layer entropies, not just their selected-pair marginals.

If $\epsilon \ge 1/4$, then $S_H \le V_* + \epsilon \le 9\epsilon$, and Cauchy–Schwarz gives the claim. Hence assume $\epsilon < 1/4$, so $S_H < 3$. Put $u_* = 1/4$ and

$$a = \max\{\epsilon, (H+1)^{-1}\}.$$

If $a \ge u_*$, then H is universally bounded and the result is immediate. We may suppose $a < u_*$.

We use an occupation estimate that does not require monotonicity. For any consecutive interval of ℓ transitions with pre-states at least $u > 0$, division of (8) by q_{j-1}^2 gives

$$c\ell \le \frac{E}{u^2} + \frac{1}{q_{\mathrm{end}}} - \frac{1}{q_{\mathrm{start}}}, \tag{9}$$

where E is its entropy expenditure. Indeed

$$\frac{q_j - q_{j-1}}{q_{j-1}^2} \ge \frac{1}{q_{j-1}} - \frac{1}{q_j};$$

the difference is $(q_j - q_{j-1})^2/(q_{j-1}^2 q_j)$. If the endpoint is also at least u, the interval length is at most $(E/u^2 + 1/u)/c$.

Let $b = \min_i q_i$. If $b \ge u_*$, applying the occupation estimate to the whole path bounds H by a universal constant. Otherwise set $u_0 = \max\{a, b\} < u_*$. Let J be the least integer such that $u_J = 2^J u_0 \ge u_*$, and define

$$\tau_j = \max\{i : q_i \le u_j\}, \qquad 0 \le j \le J.$$

These indices exist and are nondecreasing, and $J = O(\log(H+2))$ because $u_0 \ge (H+1)^{-1}$.

For the prefix ending at τ_0, consider two cases. If $b \le a$, its entropy is at most $\epsilon + C_0 a$ and its length at most H. Its square-root entropy sum is therefore at most

$$\sqrt{H(\epsilon + C_0 a)} \le C(1 + \sqrt{H\epsilon}).$$

If $b > a$, its endpoint is b and its entropy is at most $(1 + C_0)b$. Every pre-state is at least b, so the occupation estimate bounds its length by C/b. Cauchy–Schwarz then bounds its square-root entropy sum by a universal constant.

For a nonempty block from τ_j to τ_{j+1}, temporarily remove its first transition. All remaining pre-states exceed u_j, and its endpoint lies in $(u_j, 2u_j]$. Its entropy is at most

$$S_{\tau_{j+1}} \le \epsilon + 2C_0 u_j \le (1 + 2C_0)u_j.$$

The occupation estimate bounds its length by C/u_j; restoring the first transition preserves that form. The entire block has the same entropy upper bound. Cauchy–Schwarz therefore bounds its square-root entropy sum by a universal constant. Empty blocks contribute zero.

After τ_J, remove a first transition if present. Every remaining pre-state and the endpoint exceed $u_J \ge u_*$, while the total entropy is below three. The remaining length and its square-root entropy sum are universally bounded. Restoring the first transition changes only the constant. Summing the initial prefix, the J blocks, and this final suffix proves the lemma. □

Theorem 12.2 (A lower bound on excess critical rate)**.** *There is a universal C such that, for every Boolean target f, input dimension $d \ge 1$, and depth $L \ge 2$,*

$$\log d \le C \log(L + 1) + C\sqrt{L(J_L(f) - V_*)_+}, \qquad t_+ := \max\{t, 0\}.$$

Consequently there are universal $c_0, c_1 > 0$ and d_0 such that

$$J_L(f) \ge V_* + c_1 \frac{\log^2 d}{L} \qquad (d \ge d_0, \quad 2 \le L \le d^{c_0}).$$

Both statements include constant targets.

Proof. Consider an admissible path of cost $A \le V_* + \epsilon$ and the same signed pair as in Lemma 12.1. At transition j, its scalar preactivation coordinate B is Gaussian with variance q_{j-1} under the reference law. Under the controlled law, put $r = \mathbb{E}B^2/q_{j-1}$. Oddness and one-Lipschitzness of tanh imply $q_j \le \mathbb{E}B^2$, for both the half-sum and half-difference conventions. Data processing and Gaussian entropy duality give

$$D_j \ge \tfrac{1}{2}(r - 1 - \log r).$$

If $q_j > q_{j-1}$, then $r \ge q_j/q_{j-1} > 1$. For $r \ge 1$, $r - 1 - \log r \ge \frac{1}{2}\log^2 r$, since $e^s - 1 - s \ge s^2/2$ for $s \ge 0$. Thus in all cases

$$q_j \le q_{j-1} e^{2\sqrt{D_j}}, \qquad q_H \le d^{-1} \exp\left(2 \sum_j \sqrt{D_j}\right).$$

The selected scalar variance is positive, so this argument also covers singular full covariance matrices.

Let b be the signed pair vector including its factor $1/2$. The readout z has $|b^T z| \ge 1/2$ and $b^T Q_H b = q_H$. Its quadratic action and RKHS Cauchy–Schwarz imply

$$\frac{1}{4} \le (z^T Q_H^\dagger z) q_H \le 2A q_H.$$

Consequently

$$\log d \le \log(8A) + 2\sum_j \sqrt{D_j} \le C\log(L+1) + C\sqrt{L\epsilon}.$$

Here the lemma bounds the sum. The remaining logarithm is absorbed using $A \le V_* + \epsilon < 2 + \epsilon$: it is uniformly bounded for $\epsilon \le 1$ and is at most $C + C\sqrt{\epsilon}$ for larger ϵ.

Apply this inequality to arbitrarily near-minimizing paths and let ϵ decrease to $(J_L(f) - V_*)_+$. No attainment of the variational infimum is needed. If $J_L(f) < V_*$, paths of cost below V_* directly give the same conclusion with $\epsilon = 0$.

Finally choose $c_0 > 0$ sufficiently small and d_0 sufficiently large that $C\log(L+1) \le \frac{1}{2}\log d$ whenever $d \ge d_0$ and $L \le d^{c_0}$. The first inequality then forces $J_L(f) > V_*$ and gives the second with $c_1 = 1/(4C^2)$. □

Corollary 12.3 (Matching excess-rate order for parities)**.** *There are universal $c, C, C_0, c_0 > 0$ and d_0, with $c_0 \le 1$, such that every signed parity $f = \pm\chi_S$, $S \subseteq [d]$, satisfies*

$$c\frac{\log^2 d}{L} \le J_L(f) - V_* \le C\frac{\log^2 d}{L}$$

whenever $d \ge d_0$ and

$$C_0\log(d+2) \le L \le d^{c_0}.$$

This includes both constants and signed dictators, and the constants are independent of the parity degree.

Proof. The lower bound is Theorem 12.2. For every signed parity, the Fourier variation is $A(f) = 1$. Theorem 18.1 therefore gives the upper bound $C\log^2(d+L+2)/L$ when $L \ge C'\log(d+L+2)$. Because $c_0 \le 1$, the stated upper depth bound implies $L \le d$. Thus $\log(d+L+2) = O(\log(d+2))$, and increasing the universal lower depth constant C_0 ensures the Fourier theorem's condition. For $d \ge d_0 \ge 2$, this gives the asserted upper bound. □

The matching order does not say that these targets have the same rate or the same leading constant. It leaves their relative ordering, and the resulting posterior preference among sample-consistent functions, undetermined.

The theorem does not imply $J_L(f) \ge V_*$ outside the stated small-polynomial depth range. At other depths the positive part in the first inequality can vanish; the separate lower bound $V_* - O(1/L)$ remains applicable. Nor does the square-root entropy estimate by itself compress an arbitrary controlled path into a polynomial-product finite network. Product-norm compression is a separate unresolved route. Section 15 instead bounds finite width with unrestricted weight magnitudes, which suffices for the VC counting argument used in this paper.

13 The entropy budget for an approximate covariance path

The near-baseline entropy estimate is stable under entrywise errors in the covariance recursion. This statement does not require a finite-width large-deviation approximation.

Lemma 13.1 (Approximate covariance paths)**.** *Let $H \ge 1$, let X be a nonempty finite set of inputs, and let $Q_0, \dots, Q_H$ be positive semidefinite matrices on X. Suppose $Q_0(x,x) \le 2$ and $Q_j(x,x) \le 1$ for $j \ge 1$. At layer j, let ρ_j be a controlled preactivation law of finite entropy*

$$D_j = D(\rho_j \| \mathcal{N}(0, Q_{j-1})),$$

and write $\widetilde{Q}_j = \mathbb{E}_{\rho_j}[\tanh(G)\tanh(G)^{\mathsf{T}}]$. *Assume, for some* $0 \le \Delta \le 1$,

$$\|\widetilde{Q}_j - Q_j\|_{\max} \le \Delta \qquad (1 \le j \le H).$$

Let z *lie in the range of* Q_H, *suppose* $|z(x_0)| \ge 1/2$ *at some* $x_0 \in X$, *and put*

$$r = \tfrac{1}{2} z^{\mathsf{T}} Q_H^{\dagger} z, \qquad A = r + \sum_{j=1}^{H} D_j.$$

There is a universal C *such that, for every* $\epsilon \ge 0$ *with* $A \le V_* + \epsilon$,

$$\sum_{j=1}^{H} \sqrt{D_j} \le C \left\{ \log(H+2) + \sqrt{H(\epsilon + CH\sqrt{\Delta})} \right\}.$$

In particular, $\Delta \le c\tau^2/H^2$ *permits the replacement of* $CH\sqrt{\Delta}$ *by* τ, *after a universal adjustment of* c. *The hypothesis uses only one nonzero-margin output; no signed pair or cube-edge condition is needed.*

Proof. Set $q_j = Q_j(x_0, x_0)$ and $\widetilde{q}_j = \widetilde{Q}_j(x_0, x_0)$. The last variance is positive, and RKHS Cauchy–Schwarz gives $r \ge 1/(8q_H) \ge V(q_H)$. For each layer, data processing to coordinate x_0 and the scalar Bellman inequality give

$$V(q_{j-1}) \le D_j + V(\widetilde{q}_j).$$

When $q_{j-1} = 0$, finite entropy forces the corresponding controlled Gaussian coordinate to be zero, so $\widetilde{q}_j = 0$; the displayed inequality remains valid with $V(0) = V_*$.

The uniform scalar-value modulus (Lemma 36.2) gives

$$V(q_{j-1}) \le D_j + V(q_j) + C\sqrt{\Delta}.$$

Telescoping from any hidden index i through the readout yields

$$r + \sum_{j=i+1}^{H} D_j \ge V(q_i) - C(H-i)\sqrt{\Delta}.$$

Consequently, using $V_* - V(q) \le C_0 q$ on $[0,1]$,

$$\sum_{j=1}^{i} D_j \le \epsilon + C_0 q_i + CH\sqrt{\Delta} \qquad (1 \le i \le H). \tag{10}$$

For $j \ge 2$, scalar strict dissipation at the reference variance $q_{j-1} \le 1$ gives

$$D_j \ge \widetilde{q}_j - q_{j-1} + c q_{j-1}^2 \ge q_j - q_{j-1} + c q_{j-1}^2 - \Delta.$$

Thus the inflated costs $E_j = D_j + \Delta$ satisfy exact dissipation. With

$$e = \epsilon + C_1 H\sqrt{\Delta}$$

and C_1 sufficiently large, (10) implies

$$\sum_{j=1}^{i} E_j \le e + C_0 q_i, \qquad \sum_{j=1}^{H} E_j \le V_* + e.$$

If $e \geq 1/4$, Cauchy–Schwarz immediately bounds $\sum_j \sqrt{D_j}$ by $C\sqrt{He}$.

Suppose $e < 1/4$. Apply the last-exit dyadic-band proof of Lemma 12.1 to the states $q_1, \dots, q_H$ and costs $E_2, \dots, E_H$, using the preceding prefix inequalities. Its small-band cutoff is $\max\{e, (H+1)^{-1}\}$. That argument bounds

$$\sum_{j=2}^{H} \sqrt{E_j} \leq C\{\log(H+2) + \sqrt{He}\}.$$

Zero states are allowed here: they occur only in the initial low-band part, where the argument uses Cauchy–Schwarz, while every reciprocal telescoping division is performed on a band with a strictly positive lower endpoint. If the minimum state is positive, the original proof applies without change. The discarded first transition contributes at most $\sqrt{D_1} \leq \sqrt{A} < \sqrt{3}$. Since $D_j \leq E_j$, this proves the lemma. The case $H = 1$ follows from the same bound on the first transition. □

14 Stable transfer of covariance controls

There is a covariance metric in which free Gaussian feature propagation is nonexpansive and an entropy-controlled step has expansion at most $1+\sqrt{2D}$. Together with the preceding entropy-budget bound, this gives a depth factor $C(H+2)^C \exp(C\sqrt{H\varepsilon})$ when the action exceeds the scalar baseline by at most ε. This is polynomial in H when $H\varepsilon = O(\log^2(H+2))$. Stability does not by itself give a finite-rank or product-norm approximation.

Fix a finite input set X and nonnegative weights w_x. They may sum to one, in which case they define a test distribution. Put $\|u\|_w^2 = \sum_x w_x u_x^2$. For positive semidefinite matrices Q, Q' on X, define the weighted Gaussian Bures distance by

$$d_w(Q, Q')^2 = \min_{(G,G')} \mathbb{E}\|G - G'\|_w^2,$$

where the minimum ranges over centered joint Gaussian couplings with marginals $N(0, Q)$ and $N(0, Q')$. The minimum exists. For strictly positive weights this is the usual Gaussian quadratic Wasserstein distance after coordinate rescaling; zero weights give the corresponding pseudometric. For uniform control, we also use the maximum-coordinate Gaussian coupling distance

$$\Delta_\infty(Q, Q') = \min_{(G,G')} \max_{x \in X} (\mathbb{E}|G_x - G'_x|^2)^{1/2},$$

with the same class of centered joint Gaussian couplings. Its minimum also exists, since the feasible cross-covariance matrices form a compact set. Gaussian gluing and Minkowski's inequality give the triangle inequality. Both definitions allow singular covariance blocks. In the statements below, $\mathfrak{d}$ may denote either d_w or Δ_∞, with one choice fixed throughout each application; the transferred law may depend on that choice.

Lemma 14.1 (Second moments give a Gaussian coupling)**.** *For any jointly distributed real vectors (U, U') of finite second moment, let $K = \mathbb{E}UU^{\mathsf{T}}$ and $K' = \mathbb{E}U'U'^{\mathsf{T}}$. Then*

$$d_w(K, K')^2 \leq \mathbb{E}\|U - U'\|_w^2.$$

Likewise,

$$\Delta_\infty(K, K') \leq \max_x (\mathbb{E}|U_x - U'_x|^2)^{1/2}.$$

No zero-mean assumption on U or U' is needed.

Proof. The uncentered second-moment block of (U, U') is positive semidefinite. Use that block as the covariance of a centered joint Gaussian. Its weighted difference cost is exactly the right-hand side. □

Theorem 14.2 (Transfer of one controlled transition)**.** *Let $\rho \ll N(0, Q)$ have finite entropy $D = D(\rho\|N(0, Q))$, and let $K = \mathbb{E}_\rho[\tanh(G)\tanh(G)^\mathsf{T}]$. For every positive semidefinite Q' there is a preactivation law $\rho' \ll N(0, Q')$ whose corresponding feature second moment K' satisfies*

$$D(\rho'\|N(0, Q')) \le D, \qquad \mathfrak{d}(K, K') \le (1 + \sqrt{2D})\mathfrak{d}(Q, Q').$$

The controlled law is transferred; its density formula is not required to remain fixed when the reference covariance changes.

Proof. Choose an optimal centered joint Gaussian coupling (G, G') for the chosen distance between the reference laws. Denote this joint law by μ and multiply it by the density $d\rho/dN(0, Q)$ evaluated at G. The resulting joint law $\widetilde{\rho}$ has first marginal ρ, joint entropy $D(\widetilde{\rho}\|\mu) = D$, and a second marginal ρ' of entropy at most D by data processing.

Write the reference joint Gaussian as $(G, G') = TZ$, where Z is standard Gaussian on the support dimension and T is injective. The tilted law corresponds to a law η for Z of the same entropy D. Let A be the linear map taking this whitened variable to the weighted difference $(\sqrt{w_x}(G_x - G'_x))_x$. First consider $\mathfrak{d} = d_w$. Then

$$\operatorname{tr}(AA^\mathsf{T}) = d_w(Q, Q')^2.$$

Gaussian quadratic transport, as in (42), couples $Z_\eta \sim \eta$ and $Z_0 \sim N(0, I)$ with $\mathbb{E}\|Z_\eta - Z_0\|_2^2 \le 2D$. Hence Minkowski's inequality and $\|A\|_{\mathrm{op}} \le \|A\|_F$ give

$$(\mathbb{E}_\eta\|AZ\|_2^2)^{1/2} \le \|A\|_F + \sqrt{2D}\|A\|_{\mathrm{op}} \le (1 + \sqrt{2D})d_w(Q, Q').$$

Coordinatewise tanh is one-Lipschitz, so applying it to the tilted coupling does not increase this difference norm. Finally apply Lemma 14.1 to the resulting feature pair.

For $\mathfrak{d} = \Delta_\infty$, choose its optimal joint Gaussian coupling at the start and apply the same transport estimate to each scalar row $A_xZ = G_x - G'_x$ separately. Since the operator and Frobenius norms of a scalar row agree, the single tilted joint coupling satisfies

$$(\mathbb{E}_{\widetilde{\rho}}|G_x - G'_x|^2)^{1/2} \le (1 + \sqrt{2D})(\mathbb{E}_\mu|G_x - G'_x|^2)^{1/2} \quad \text{for every } x.$$

Apply tanh and the maximum-coordinate second-moment coupling bound. Thus no factor depending on $|X|$ is needed for uniform control. This also proves the assertion in singular cases. □

Corollary 14.3 (Free blocks are nonexpansive)**.** *Let $\mathcal{F}(Q) = \mathbb{E}_{N(0,Q)}\tanh(G)\tanh(G)^\mathsf{T}$. For every integer $t \ge 0$,*

$$\mathfrak{d}(\mathcal{F}^t(Q), \mathcal{F}^t(Q')) \le \mathfrak{d}(Q, Q').$$

On two input coordinates, for all covariance matrices with diagonal at most two, the map to an entry of $\mathcal{F}^t(Q)$ is uniformly one-half-Hölder in the three input covariance entries, with a constant independent of $t \ge 1$.

Proof. Take $D = 0$ in the theorem and iterate. For the last assertion, in dimension two the elementary bound $d(Q, Q') \le C\|Q - Q'\|_{\max}^{1/2}$ holds for the unweighted distance. One way to see this is to put $S = Q + (Q' - Q)_+ = Q' + (Q - Q')_+$. Couple each Gaussian to $N(0, S)$ by adding independent Gaussian noise, then use the triangle inequality. The squared upper bound is at

most $2\|Q - Q'\|_* \le C\|Q - Q'\|_{\max}$. After at least one feature step, all diagonal entries are at most one. Under an optimal output Gaussian coupling, Cauchy–Schwarz bounds the difference of any second-moment entry by twice the Bures distance. Combining the two estimates proves the claim. □

Corollary 14.4 (Near-baseline path stability)**.** *Suppose a critical path has hidden transition entropies $D_1, \dots, D_H$ and starts at Q_0. For any starting covariance Q_0', there is a transferred path with each hidden entropy no larger than the original one and*

$$\mathfrak{d}(Q_H, Q_H') \le \mathfrak{d}(Q_0, Q_0') \prod_{j=1}^{H} (1 + \sqrt{2D_j}) \le \mathfrak{d}(Q_0, Q_0') \exp\left(\sqrt{2} \sum_{j=1}^{H} \sqrt{D_j} \right).$$

If the original path represents a globally margined Boolean function with total cost at most $V_* + \varepsilon$, where $\varepsilon \ge 0$, then

$$\mathfrak{d}(Q_H, Q_H') \le C(H+2)^C \exp\{C\sqrt{H\varepsilon}\}\, \mathfrak{d}(Q_0, Q_0').$$

The second assertion uses the original path's near-baseline property; the transferred path is not asserted to preserve its global output margins.

Proof. Apply Theorem 14.2 inductively. Lifting a feature law through atanh preserves its entropy, so the theorem applies to every hidden control. Use $\log(1+u) \le u$ for the first exponential bound and Lemma 12.1 for the last bound. □

Lemma 14.5 (Transfer of a readout)**.** *Let $z \in \operatorname{range} Q$ have readout cost $r = \frac{1}{2} z^{\mathsf{T}} Q^{+} z$. For every positive semidefinite Q' there is $z' \in \operatorname{range} Q'$ with*

$$\tfrac{1}{2}(z')^{\mathsf{T}}(Q')^{+} z' \le r, \qquad \|z - z'\|_w \le \sqrt{2r}\, d_w(Q, Q').$$

The same assertion holds with the second estimate replaced by $\|z - z'\|_\infty \le \sqrt{2r}\, \Delta_\infty(Q, Q')$.

Proof. Use an optimal Gaussian coupling (G, G') for the chosen distance and set $\beta = Q^{+} z$, $Y = \beta^{\mathsf{T}} G$, and $z' = \mathbb{E}(G'Y)$. Gaussian linear regression shows that z' has readout cost at most $\frac{1}{2}\mathbb{E}Y^2 = r$. Equivalently this follows from the Schur-complement inequality for the joint covariance, including singular matrices. Since $z = \mathbb{E}(GY)$, Hilbert-space Cauchy–Schwarz gives

$$\|z - z'\|_w \le (\mathbb{E}Y^2)^{1/2} (\mathbb{E}\|G - G'\|_w^2)^{1/2}.$$

This is the required weighted estimate. Applying the same scalar Cauchy–Schwarz bound at every coordinate gives the stated uniform estimate for a coupling attaining Δ_∞. □

For probability weights the readout estimate is in $L^2(\mathcal{D})$. Using Δ_∞ instead gives a uniform output estimate, without a factor depending on the number of inputs. Combining path and readout transfer produces a path with no larger total action whose output differs uniformly by at most

$$\sqrt{2r}\, C(H+2)^C \exp\{C\sqrt{H\varepsilon}\} \Delta_\infty(Q_0, Q_0').$$

In particular, an error smaller than the original margin preserves all output signs. None of these estimates guarantees that a high-rank free Gaussian covariance admits a low-rank approximation in this metric. For example, with uniform weights on m inputs, the identity covariance has squared Bures distance at least $1 - n/m$ from every covariance of rank at most n, and the same lower bound holds for Δ_∞^2. To check this, the unweighted Gaussian Bures distance squared from I_m is $\sum_i (1 - \sqrt{\lambda_i(Q')})^2 \ge m - n$; the maximum coordinate coupling cost is at least the average coupling cost. Thus the stability theorem supplies a controlled propagation factor, while a finite representation or compression theorem remains a separate requirement.

15 Finite realization by preserving an anchor covariance block

A finite representation does not require approximation of the full Gaussian covariance in Bures distance. We preserve a selected block exactly and compare each remaining query jointly with that block. Entropy selects a small block on which the controls can be conditioned; importance sampling then preserves its covariance without adding depth. The resulting network has arbitrary finite real weights. This section bounds its width, not its product norm.

Theorem 15.1 (Finite realization with an entropy-length bound)**.** *Let $X = \{-1, 1\}^d$, and consider a critical variational path with $H = L - 1 \geq 1$ hidden layers, entropy costs $D_1, \dots, D_H$, and output z of readout action r. Set*

$$A_0 = \max\left\{1, \sum_{j=1}^{H} D_j + r\right\}, \qquad S = \sum_{j=1}^{H} \sqrt{D_j}.$$

There is an ordinary depth-L tanh network with finite real weights, width at most

$$n \leq C(d + L + A_0)^{30} \exp(20S),$$

and output $\widehat{z}$ satisfying $\|z - \widehat{z}\|_\infty \leq 3/32$ on X. In particular, if $f(x)z(x) \geq 1/2$ everywhere, the finite network has signed margin at least $13/32 > 1/4$.

If the original path is globally margined and has total action at most $V_* + \varepsilon$, then its width is at most

$$\mathrm{poly}(d, L, A_0) \exp\{C\sqrt{L\varepsilon}\},$$

where the polynomial has universal degree.

15.1 Selecting anchors for all controls and for the readout

Write $Q_0, \dots, Q_H$ for the original path's second moments. Lift its controlled feature laws through coordinatewise atanh, obtaining preactivation laws $\rho_j \ll N(0, Q_{j-1})$ of entropy D_j. For $T \subseteq X$ put

$$I_j(T) = D\big((\rho_j)_T \| N(0, Q_{j-1,TT})\big).$$

The empty-set value is zero. These quantities are monotone under inclusion and bounded above by D_j. The entropy increment upon adjoining x is exactly the average conditional entropy of G_x given G_T, relative to its conditional Gaussian reference.

Represent the terminal kernel by vectors h_x in a finite Hilbert space, with $\langle h_x, h_y\rangle = Q_H(x, y)$, and choose a readout vector u with

$$z(x) = \langle u, h_x\rangle, \qquad \|u\|^2 = 2r.$$

Let P_T be projection onto the span of the vectors h_a, $a \in T$. Fix $0 < \delta \leq 1/8$ and $t > 0$. Starting at $T = \varnothing$, repeatedly add an input whenever either

$$I_j(T \cup \{x\}) - I_j(T) > \delta \quad \text{for some } j, \qquad \text{or} \qquad |\langle u - P_T u, h_x\rangle| > t.$$

In the second case, adjoining the unit residual direction of h_x increases $\|P_T u\|^2$ by more than t^2, since $\|h_x\|^2 = Q_H(x, x) \leq 1$. Therefore each addition increases

$$\sum_j I_j(T) + \frac{\delta}{t^2}\|P_T u\|^2$$

by more than δ. The procedure terminates with

$$|T| \le \frac{\sum_j D_j}{\delta} + \frac{2r}{t^2}. \tag{11}$$

At termination all the conditional entropy increments are at most δ, and the projected readout $z_T(x) = \langle P_T u, h_x \rangle$ obeys

$$\|z - z_T\|_\infty \le t, \qquad z_T(x) = Q_H(x,T)\beta, \qquad \beta^{\mathsf{T}} Q_{H,TT}\beta \le 2r \tag{12}$$

for some finite coefficient vector β. There is no required bound on $\|\beta\|_1$. Both conditions are rechecked after every addition: their individual pointwise residuals need not decrease.

15.2 An anchor-preserving Gaussian comparison

For positive semidefinite kernels K, K' with the same TT block C, define

$$v_K(x) = C^{+1/2}K(T,x), \qquad \sigma_K(x) = \big(K(x,x) - \|v_K(x)\|^2\big)^{1/2},$$

and

$$d_T(K,K') = \max_{x\in X} \Big(\|v_K(x) - v_{K'}(x)\|^2 + |\sigma_K(x) - \sigma_{K'}(x)|^2\Big)^{1/2}.$$

All cross vectors lie in the range of C, so these expressions are well-defined even when C is singular. This is a pseudometric on kernels with a fixed anchor block. For each query x, it is the least Gaussian RMS difference between the query coordinates when the two Gaussian anchor vectors are coupled identically. The minimum uses a common Gaussian for the two scalar conditional residuals. Different queries may use different extensions of the same anchor coupling.

We will also use its second-moment version. If, for each x, a coupling of feature vectors at $T\cup\{x\}$ has identical anchor features, then its query RMS difference bounds d_T at that query. Indeed the uncentered second-moment block of this coupling is a valid centered Gaussian covariance, and the two Gaussian anchor vectors coincide almost surely. This argument does not require zero-mean feature laws.

Replace ρ_j by its Rao–Blackwellized law ρ_j^T: it has the same anchor marginal as ρ_j and the Gaussian conditional law elsewhere, given those anchors. Its density relative to $N(0, Q_{j-1})$ is a function $w_j(G_T)$, and its entropy is $I_j(T) \le D_j$. Its output anchor second moments remain exactly $Q_{j,TT}$. At any query, conditional Gaussian quadratic transport and the anchor selection give a coupling with identical G_T and

$$\mathbb{E}|G_x - G'_x|^2 \le 2\,\mathrm{Var}_{N(0,Q_{j-1})}(G_x \mid G_T)\delta \le 4\delta.$$

The final bound uses $Q_{0,xx} = (d+1)/d \le 2$ at the input and $Q_{j-1,xx} \le 1$ at later layers. Zero conditional variances cause no difficulty, because finite conditional entropy then forces equality. Applying tanh and the second-moment observation bounds the resulting d_T error by $2\sqrt{\delta}$.

Lemma 15.2 (Propagation with an exact anchor block)**.** *Suppose $K_{TT} = K'_{TT}$, and $w(G_T)$ is a density integrating to one under their common Gaussian anchor law. Its entropy is D. Let $\mathsf{F}_w(K)$ and $\mathsf{F}_w(K')$ be the feature second moments obtained by tilting the two Gaussian laws by this same density and then applying tanh. Their anchor blocks agree, and*

$$d_T(\mathsf{F}_w(K), \mathsf{F}_w(K')) \le (1 + \sqrt{2D})\, d_T(K,K').$$

Proof. For a fixed query couple the reference Gaussian anchor vectors identically and optimally couple the scalar conditional residuals. Tilt this joint law by $w(G_T)$. Both marginals are the required controlled laws, and the joint entropy relative to its Gaussian reference equals D. Whiten the reference Gaussian on its support. For any scalar row a, Gaussian quadratic transport and Minkowski's inequality give

$$(\mathbb{E}_{\mathrm{tilted}}|aZ|^2)^{1/2} \le (1+\sqrt{2D})\|a\|_2.$$

Apply this to the difference of the two query coordinates, then use tanh's Lipschitz property and the second-moment observation. The anchor features coincide pointwise, so the output anchor block is exactly common. Although query couplings differ, the two marginal controlled laws are fixed by the same anchor density and do not depend on the queried input. □

15.3 Finite quadrature that preserves the anchor block exactly

Lemma 15.3 (Importance sampling and exact covariance correction)**.** *Let ν be any probability law on $[-1,1]^X$, with second-moment kernel K, and let $|T| = q$. For $0 < \eta \le 1/8$, there are*

$$n \le C(q+1)^4\eta^{-2}\{d + \log(q+2) + 1\}$$

feature vectors $h_1, \dots, h_n$ chosen by importance sampling from ν, positive weights $p_1, \dots, p_n$, and a positive definite matrix B on $\mathbb{R}^n$ such that, on putting

$$F_x = \left(\frac{h_i(x)}{\sqrt{np_i}}\right)_{i=1}^n, \qquad K'(x,y) = F_x^{\mathsf{T}} B F_y,$$

one has $K'_{TT} = K_{TT}$ and $d_T(K,K') \le 6\sqrt{\eta}$. The matrix satisfies $\|B\|_{\mathrm{op}} \le 2$.

Proof. Write $K_{TT} = U\Lambda U^{\mathsf{T}}$ on its range, of dimension $s \le q$, and put $a(h) = \Lambda^{-1/2}U^{\mathsf{T}}h_T$. Then $\mathbb{E}_\nu aa^{\mathsf{T}} = I_s$, and $h_T = U\Lambda^{1/2}a(h)$ almost surely. Sample under the importance law $p(h)\nu(dh)$, where

$$p(h) = \frac{1+\|a(h)\|^2}{s+1}.$$

This density is positive and normalized, and both $\|a(h)/\sqrt{p(h)}\|$ and $|h(x)|/\sqrt{p(h)}$ are at most $\sqrt{s+1}$. For independent samples form the matrix E with rows $a(h_i)^{\mathsf{T}}/\sqrt{np_i}$, and set

$$M = E^{\mathsf{T}}E, \quad \widehat{v}_x = E^{\mathsf{T}}F_x, \quad \widehat{k}_x = \|F_x\|^2, \quad v_x = \mathbb{E}_\nu[a(h)h(x)].$$

Entrywise Hoeffding bounds and a union bound give a realization with

$$\|M - I_s\|_{\mathrm{op}} \le \eta, \quad \sup_x \|\widehat{v}_x - v_x\| \le \eta, \quad \sup_x |\widehat{k}_x - K(x,x)| \le \eta. \tag{13}$$

For completeness, require each Gram entry to have error at most $\eta/(s+1)$ and each cross entry error at most $\eta/\sqrt{s+1}$. The variables have magnitude at most $s+1$, and there are at most $s^2 + s2^d + 2^d$ entries. The asserted width, with a universal constant enlarged, makes their joint success probability positive. Since K is positive semidefinite, $\|v_x\| \le \sqrt{K(x,x)} \le 1$.

Define

$$B = I_n + E(M^{-2} - M^{-1})E^{\mathsf{T}}.$$

If an eigenvalue of M is λ, B acts on its corresponding left singular direction by λ^{-1}; on the orthogonal complement it is the identity. Thus B is positive definite and $\|B\|_{\mathrm{op}} \le 2$. Writing $C = \Lambda^{1/2} U^{\mathsf{T}}$, we have $F_T = EC$ and

$$E^{\mathsf{T}} B E = I_s, \qquad K'_{TT} = C^{\mathsf{T}} C = K_{TT}.$$

The normalized cross vector for K' is, in the U coordinates, $v'_x = M^{-1}\widehat{v}_x$. The errors above give

$$\|v'_x - v_x\| \le 4\eta, \qquad |K'(x,x) - K(x,x)| \le 10\eta.$$

To check the second estimate, use

$$K'(x,x) = \widehat{k}_x + \widehat{v}_x^{\mathsf{T}}(M^{-2} - M^{-1})\widehat{v}_x, \qquad \|M^{-2} - M^{-1}\| \le 4\eta, \quad \|\widehat{v}_x\| \le 1 + \eta.$$

Since $\|v'_x\| \le 3$, the two conditional residual variances differ by at most 26η. The square-root inequality $|\sqrt{a} - \sqrt{b}|^2 \le |a - b|$ now yields

$$d_T(K, K')^2 \le 16\eta^2 + 26\eta \le 28\eta < 36\eta.$$

If $s = 0$, take $p = 1$, let E be empty, and set $B = I_n$; only the diagonal estimate is needed. □

15.4 Recursive network construction and error bound

We now prove Theorem 15.1. Choose the common anchor set by the first subsection. Start with the exact original input covariance Q_0. Suppose the previously stored ordinary tanh neurons give a finite feature embedding F_x and covariance factor B as in the quadrature lemma. A Gaussian incoming row with the corresponding coefficient covariance has preactivation field of covariance $F_x^{\mathsf{T}} B F_y$. Tilt this Gaussian row by $w_j(G_T)$. This is a normalized law: the finite covariance has exactly the original anchor block $Q_{j-1,TT}$. Its activated field is still the output of an ordinary tanh neuron on the stored preceding neurons. Apply Lemma 15.3 to this activated-feature law, select its finite rows, and construct the next factor B_j.

The weights $1/\sqrt{np_i}$ and the factor $B_j^{1/2}$ are absorbed into the incoming coefficients of the next layer. They introduce no additional nonlinear layer. At the first layer the incoming field is realized by the original Gaussian input weights, tilted and importance sampled in the same way. All selected coefficients are finite almost surely. No upper bound on their magnitude is used. The sampling proves existence of a deterministic network; it does not assert that its weight law is the original prior.

Let $\widehat{Q}_j$ be the corrected finite kernel after layer j. Its anchor block is exactly $Q_{j,TT}$. Set $e_j = d_T(Q_j, \widehat{Q}_j)$, with $e_0 = 0$. Compare the original control first to its Rao–Blackwellized version at Q_{j-1}, then propagate the latter control from $\widehat{Q}_{j-1}$, and finally apply finite quadrature. With $\eta = \delta$, the triangle inequality and the preceding lemmas give

$$e_j \le (1 + \sqrt{2D_j})e_{j-1} + 8\sqrt{\delta}.$$

Consequently

$$e_H \le 8H\sqrt{\delta}\prod_j (1 + \sqrt{2D_j}) \le 8H\sqrt{\delta}\, e^{\sqrt{2}S}.$$

At the final layer use the anchor-supported readout from (12). Because the anchor block is exact, the same coefficient vector β gives the finite output $\widehat{z}(x) = \widehat{Q}_H(x, T)\beta$ and

$$|z_T(x) - \widehat{z}(x)| \le \sqrt{2r}\, d_T(Q_H, \widehat{Q}_H).$$

For example, this follows by coupling the two query Gaussians with identical anchors and applying Cauchy–Schwarz to the common scalar $\beta^{\mathsf{T}} G_T$. This readout is an ordinary linear combination of the final stored neuron outputs; its covariance factor is absorbed into those linear coefficients.

Take

$$t = \frac{1}{16}, \qquad \delta = \eta = \left[256H\sqrt{\max\{2r,1\}}\, e^{\sqrt{2}S}\right]^{-2}.$$

Then

$$\|z - \widehat{z}\|_\infty \le t + 8\sqrt{2r}H\sqrt{\delta}\, e^{\sqrt{2}S} \le \frac{1}{16} + \frac{1}{32} = \frac{3}{32}.$$

The anchor count is at most $A_0/\delta + 512r$. The quadrature width is therefore at most

$$CH^{12}A_0^{10}e^{12\sqrt{2}S}\{d + \log(H + A_0 + 2) + S + 1\}.$$

The inequality $S \le \sqrt{HA_0}$ absorbs this last factor into a universal polynomial. Enlarging constants gives the displayed bound with polynomial degree 30 and exponential coefficient 20. Layers may be padded with zero outgoing weights to a common width. Finally the near-baseline assertion follows from Lemma 12.1.

Corollary 15.4 (A statement for the infimum rate)**.** *If the critical function rate satisfies $J_L(f) \le V_* + \varepsilon$, $\varepsilon \ge 0$, then f has a depth-L tanh realization with margin at least $1/4$ and width*

$$\mathrm{poly}(d, L, 1 + J_L(f)) \exp\{C\sqrt{L\varepsilon}\}.$$

No attainment of the variational infimum is assumed.

Proof. Choose an admissible path of action at most $J_L(f) + 1/L$ and apply the theorem with excess $\varepsilon + 1/L$. The extra factor $\exp\{C\sqrt{L\varepsilon + 1}\}$ is at most a universal constant times $\exp\{C\sqrt{L\varepsilon}\}$ after enlarging C. □

16 The upper bound by rank-one purification

Lemma 16.1 (Purifying an exact target component)**.** *There are universal $p_0, c, C > 0$ with the following property. Suppose $Q = vff^T + R$, where $R \succeq 0$ has constant diagonal $0 \le r \le v \le p_0$. One Gaussian covariance-inflation step of entropy at most Cv^2 produces a feature covariance*

$$Q' = vff^T + R', \qquad R' \succeq 0, \qquad R'(x,x) = r' \le (1 - cv)r.$$

Consequently T such steps have cost at most CTv^2, preserve the exact coefficient v, and reduce the residual diagonal by e^{-cvT}.

Proof. Inflate the target component by ϵvff^T. Write $\psi_r(a) = \mathbb{E}\tanh(a + \sqrt{r}Z)$ and

$$v'(\epsilon) = \mathbb{E}\psi_r(\sqrt{(1+\epsilon)v}Z)^2.$$

This is continuous, $v'(0) \le v$, and the first-Hermite lower bound from the preceding construction gives

$$v'(\epsilon) \ge (1+\epsilon)v[1 - 2(v + r + \epsilon v)].$$

For sufficiently small $v + r$, its value at $\epsilon = 16(v + r)$ is at least v. Choose an intermediate ϵ with $v'(\epsilon) = v$. Its entropy cost is at most $\epsilon^2/4 \le 256v^2$.

The residual diagonal is the average conditional variance

$$r' = \mathbb{E}_A \operatorname{Var}_Z[\tanh(A + \sqrt{r}Z)], \qquad A \sim \mathcal{N}(0, (1+\epsilon)v).$$

Gaussian Poincare gives

$$r' \le r\mathbb{E}\operatorname{sech}^4(\sqrt{u}Z), \qquad u = (1+\epsilon)v + r.$$

The expansion $\mathbb{E}\operatorname{sech}^4(\sqrt{u}Z) = 1 - 2u + O(u^2)$ implies that it is at most $1-u$ for all sufficiently small u. Since $u \ge v$ and $u \le 2(v+r)$, reducing p_0 gives $r' \le (1-v)r$. The conditional-mean covariance supplies the exact rank-one component vff^T; the remaining covariance is positive semidefinite and has the displayed constant diagonal. Iterate. □

17 The initial target component and the sufficient depth

The upper-bound construction can retain its dependence on the initial target component, rather than using the worst eigenvalue for every function. Let $Q_1 = Q(\mathsf{M}_{Q_0})$ be the first unperturbed feature covariance and define

$$a_f = (f^T Q_1^{-1} f)^{-1}, \qquad S_f = 1 + \log(1/a_f).$$

The matrix Q_1 is positive definite by Lemma 5.1. The rank-one inequality $Q_1 \succeq a_f ff^T$ holds by Cauchy–Schwarz, and its diagonal implies $0 < a_f \le 1$. Thus $S_f \ge 1$.

Theorem 17.1 (Target-dependent sufficient depth)**.** *There is a universal C such that, for every Boolean function f and integer depth $L \ge CS_f$,*

$$J_L(f) \le V_* + C\frac{S_f^2 + [\log(2L)]^2}{L}.$$

In particular, the uniform upper bound can be sharpened to

$$J_L(f) \le V_* + C\frac{(d+1)^2 + [\log(2L)]^2}{L} \qquad (L \ge C(d+1)).$$

Proof. The proof of Lemma 8.4 uses the dimension only through the logarithm of the initial tracked component. Starting from $Q_1 \succeq a_f ff^T$, the free-cooling lemma reaches diagonal in $[\delta/2, \delta]$ with tracked component at least $c\delta a_f$. Choose the smaller exact component $v_0 = \delta e^{-AS_f}$, with a fixed sufficiently large A. The same holding construction then has cost at most $64H\delta^2$ for any specified H. The prescribed growth sequence in that lemma reaches any $p \ge \delta$ in at most CS_f/δ layers, with cost at most $C(p + S_f\delta)$. Its length depends only on v_0, δ, p, and hence is independent of the holding interval. These assertions follow directly from the displayed growth sequence: below δ its length is $O(\delta^{-1}\log(\delta/v_0)) = O(S_f/\delta)$, and its entropy sum is $O(p + \delta\log(\delta/v_0))$.

Choose large universal constants B_1, B_2 and put

$$\delta = B_1 S_f/L, \qquad p = B_2\{S_f + \log(2L)\}/L.$$

Choose $B_2 \ge B_1$. The weaker depth requirement $L \ge CS_f$ already ensures $0 < \delta \le p \le p_0$ after choosing the universal C large enough: since $S_f \ge 1$ and $L \ge C$,

$$\delta \le B_1/C, \qquad p \le B_2\left\{C^{-1} + \frac{\log(2L)}{L}\right\} \le B_2\frac{1 + \log(2C)}{C}.$$

The last inequality uses monotonicity of $\log(2L)/L$ for large L. Here p_0 is the universal constant in Lemma 16.1. The total holding cost is allowed to be large; it remains explicitly in the final upper bound. Cooling and prescribed growth take less than $L/3$ layers if B_1 is large enough. Purify for $T = \lceil 4\log(2L)/(cp) \rceil$ layers; this leaves the exact component p and residual diagonal at most $\delta(2L)^{-4}$, at cost $O(p\log(2L))$. Append a scalar path of at most C_0/p steps and cost at most $V(p)+p$. Its stable lift adds only $O(\delta(2L)^{-4}/p) = O(L^{-4})$ readout cost. Purification and the scalar path together occupy less than $L/3$ layers if B_2 is sufficiently large. Fill the unused hidden layers by exact holding before prescribed growth. The number of inserted layers is an integer and all subsequent stage lengths are independent of it, so the final depth is exactly L.

The total excess over V_* is at most a universal multiple of

$$L\delta^2 + S_f\delta + p + p\log(2L) + L^{-4} \le C\frac{S_f^2 + [\log(2L)]^2}{L},$$

where $S_f \log(2L) \le \{S_f^2 + [\log(2L)]^2\}/2$. Finally, the first-layer spectrum bound and $\|f\|^2 = 2^d$ give $a_f \ge e^{-13(d+1)}/2^d \ge e^{-14(d+1)}$. Thus $S_f \le 15(d+1)$, yielding the uniform corollary. □

Corollary 17.2 (The depth scale for constants and dictators)**.** *There are universal constants $K_0, C > 0$ such that constants and dictators satisfy $J_L(f) \le K_0$ at every $L \ge C[\log(d+1)]^2$. More precisely,*

$$J_L(f) \le V_* + C\frac{[\log(d+1)]^2 + [\log(2L)]^2}{L} \qquad (L \ge C[\log(d+1)]^2).$$

Together with Theorem 39.1, this makes $\Theta((\log d)^2)$ the necessary and sufficient depth scale for attaining some fixed universal rate budget on these simplest targets. This is an absolute-rate statement, not a sample-complexity statement.

Proof. Write $q = (d+1)/d \in [1,2]$ and let $h_1(q)$ be the first normalized Hermite coefficient of $z \mapsto \tanh(\sqrt{q}z)$. Gaussian integration by parts gives

$$h_1(q) = \sqrt{q}\,\mathbb{E}\,\mathrm{sech}^2(\sqrt{q}Z) \ge c > 0 \qquad (1 \le q \le 2).$$

In the nonnegative Hermite expansion of Q_1, the first term is

$$\frac{h_1(q)^2}{d+1}\left(\mathbf{1}\mathbf{1}^T + \sum_{i=1}^{d} x_i x_i^T\right).$$

It follows that $Q_1 \succeq c(d+1)^{-1} f f^T$ for every constant or dictator, so $S_f \le C\log(d+1)$. Apply the theorem. The displayed excess is bounded by a universal constant on the specified depth range, after enlarging C, which proves the rate-budget assertion. The necessary depth follows from (41) with $k = 1$. □

Corollary 17.3 (The one-hidden-layer cost)**.** *There are universal $0 < c < C < \infty$ such that every Boolean target satisfies $J_2(f) \ge c\sqrt{d}$, whereas constants and dictators satisfy $J_2(f) \le C\sqrt{d}$.*

Proof. At $L = 2$, Theorem 39.1 gives $d \le 8J(1+\sqrt{2J})^2 \le 16J + 32J^2$. This implies $J \ge \sqrt{d}/64$ for every integer $d \ge 1$. For the constant target, shift only the first-layer Gaussian bias by $a = d^{-1/4}$. Its entropy cost is $da^2/2 = \sqrt{d}/2$. At every input the feature mean is the same positive number

$$b = \mathbb{E}\tanh(a + \sqrt{q}Z), \qquad q = (d+1)/d.$$

Since $0 < a \le 1$ and $1 \le q \le 2$, differentiation in a on this compact parameter range gives $b \ge c_0 a$ for a universal $c_0 > 0$. The resulting second moment dominates $b^2\mathbf{1}\mathbf{1}^T$, so the readout $z = \mathbf{1}/2$ costs at most $1/(8b^2)$. The total is at most $C\sqrt{d}$. Output sign symmetry and the augmented-coordinate permutation symmetry transfer the same bound to all constants and dictators. □

Corollary 17.4 (Functions of a small set of inputs)**.** *Suppose f depends on at most k of the d original input coordinates, where $0 \le k \le d$, and put*

$$R_k = (k+1)\log\frac{e(d+1)}{k+1}.$$

Then $S_f \le 14R_k$. In particular,

$$J_L(f) \le V_* + C\frac{R_k^2 + [\log(2L)]^2}{L} \qquad (L \ge CR_k^2).$$

Thus all such functions attain a fixed universal absolute rate budget at depth $O(R_k^2)$.

Proof. Set $D = d+1$ and $m = 2^d$. Let λ_T denote the eigenvalue of Q_1/m on the Walsh character χ_T, rather than the eigenvalue of the raw matrix Q_1. The Hermite proof of Lemma 5.1 gives, with $n = |T|$ if $|T|$ is odd and $n = |T|+1$ otherwise,

$$\lambda_T \ge e^{-12n}\frac{n!}{D^n}.$$

For $|T| \le k$, one has $1 \le n \le k+1 \le D$. The function $u \mapsto u\log(eD/u)$ is increasing on $(0, D]$, so

$$\lambda_T \ge \exp\{-12(k+1) - (k+1)\log(eD/(k+1))\}.$$

Write $f = \sum_T \widehat{f}(T)\chi_T$. Its nonzero coefficients have $|T| \le k$ and Parseval gives $\sum_T \widehat{f}(T)^2 = 1$. Since $Q_1\chi_T = m\lambda_T\chi_T$ and $\|\chi_T\|^2 = m$, the exact normalization is

$$f^T Q_1^{-1} f = \sum_T \frac{\widehat{f}(T)^2}{\lambda_T}.$$

The displayed coefficient bound implies $S_f \le 1 + 12(k+1) + R_k \le 14R_k$. Apply Theorem 17.1. □

18 Controlled Fourier seeds and the depth needed by parity

A weak component already present in the unmodified first-layer kernel can be amplified at small entropy cost. There is another option: change the first feature law slightly to insert a stronger component. For parity, this changes the sufficient depth from a degree-dependent bound to the same order as for constants and dictators. More generally, the Fourier ℓ^1 norm controls a useful family of initial components.

Use the normalized Fourier expansion on the Boolean cube,

$$f = \sum_{S\subseteq[d]} \widehat{f}(S)\chi_S, \qquad \chi_S(x) = \prod_{i\in S} x_i, \qquad A(f) = \sum_S |\widehat{f}(S)|.$$

For Boolean f, Parseval implies $A(f) \ge 1$.

Theorem 18.1 (A controlled Fourier-seed bound)**.** *There are universal constants C_0, C_1 such that, if*

$$T_f = \log(d + L + 2) + \log A(f), \qquad L \ge C_0 T_f,$$

then

$$J_L(f) \le V_* + C_1\frac{T_f^2}{L}.$$

In particular, a depth at least $C[\log(d+2) + \log A(f)]^2$ puts f below a fixed universal absolute critical-rate budget.

Corollary 18.2 (Parity and juntas)**.** *Every parity, irrespective of its degree, attains a fixed universal critical-rate budget at depth $O(\log^2(d+2))$. The universal lower bound in Theorem 39.1 makes $\Theta(\log^2(d+2))$ the necessary and sufficient depth scale for attaining some fixed universal budget, as d grows. Every function depending on at most k coordinates attains such a budget at depth $O((k+\log(d+2))^2)$. These are rate statements. Corollary 22.4 uses them to obtain polynomial sample bounds, already at the stated logarithmic minimum learner depth.*

18.1 A common polynomial-cost family of parity laws

Lemma 18.3 (Parity mean and constant diagonal)**.** *For every $S \subseteq [d]$, there is a first-feature law ν_S with*

$$D(\nu_S \| \mathsf{M}_{Q_0}) \le D_* := C(d+1)^3 \log^2(d+2), \qquad \mathbb{E}_{\nu_S} H = c_S \chi_S, \qquad c_S \ge \frac{1}{8(d+1)}.$$

Its second-moment diagonal is constant and belongs to $[F(1),1]$, where $F(q) = \mathbb{E}\tanh^2(\sqrt{q}Z)$ as above. Increase the universal C if necessary so that $D_* \ge 1$.

Proof. First construct a law with a signed mean margin, then symmetrize it. Let $k = |S| \ge 1$ and $u(x) = \sum_{i \in S} x_i$. At its possible values $u_j = -k + 2j$, the parity equals $f_j = (-1)^{k-j}$. Thus

$$\chi_S(x) = c_0 + \sum_{j=1}^{k} c_j \operatorname{sign}(u(x) - (-k+2j-1)), \qquad |c_0| \le 1, \quad c_j = (f_j - f_{j-1})/2 \in \{-1, 1\}.$$

Put $M = 4 + \log(64(k+1))$. Shift a first-row Gaussian weight vector to have mean coefficient M on each coordinate in S, zero on the other inputs, and bias $-M(-k+2j-1)$. Its preactivation mean has the required sign and absolute value at least M at every cube input. Its variance remains $q_0 = (d+1)/d \le 2$. For a centered Gaussian G of variance at most two,

$$1 - \mathbb{E}\tanh(t+G) \le 2e^{-2t+4} \qquad (t \ge 0).$$

The expected feature v_j therefore approximates its ideal step to maximum error $1/[16(k+1)]$. A constant bias shift supplies a constant mean $c_{\text{const}} \ge 1/2$.

Take $\lambda = 1/[4(k+1)]$. Mix the appropriately signed detector laws with probabilities $\lambda|c_j|$, and the appropriately signed constant law with probability $\lambda|c_0|/c_{\text{const}}$. Their sum is at most $\lambda(k+2) \le 1$; use the unmodified law for the remaining probability. Its feature mean is zero. The resulting mean m_S obeys

$$\chi_S(x) m_S(x) \ge \lambda \left(1 - \frac{k}{16(k+1)}\right) \ge \frac{1}{8(k+1)} \qquad \text{for every } x.$$

Every detector shift has Gaussian relative entropy at most

$$\frac{d}{2} M^2 \{k + (-k+2j-1)^2\} \le C(d+1)(k+1)^2 \log^2(k+2).$$

Convexity and data processing give the common bound D_* for this feature law. The case $k = 0$ uses only a positive constant bias shift, with constant positive mean and entropy $O(d)$.

For $b \in \{-1,1\}^d$, transform features by

$$(T_b H)(x) = \chi_S(b) H(b_1 x_1, \dots, b_d x_d).$$

Input weight sign changes and a simultaneous sign change of all weights implement this transformation. It preserves the raw first Gaussian law. Average the transformed controlled laws over b. Their entropies are unchanged individually, so convexity bounds the mixture's entropy by D_*. Its mean is exactly

$$\mathbb{E}_b \chi_S(b) m_S(bx) = c_S \chi_S(x), \qquad c_S = \mathbb{E}_x \chi_S(x) m_S(x) \geq \frac{1}{8(k+1)}.$$

Transitivity of cube translations makes its diagonal second moment constant. Each original component was a Gaussian mean shift with unchanged covariance. For centered Gaussian G, $|G+t|$ stochastically dominates $|G|$, so $\mathbb{E}\tanh^2(G+t) \geq \mathbb{E}\tanh^2(G)$. Every diagonal before and after symmetrization is consequently at least $F(q_0) \geq F(1)$ and at most one. Only the first feature law has been symmetrized; no convexity of entire covariance trajectories is asserted. □

18.2 From a Fourier mixture to a small initial entropy cost

Proof of Theorem 18.1. Negating a whole feature preserves the raw first law and negates its mean. Choose the corresponding sign of each law in Lemma 18.3, and set

$$Z_f = \sum_S \frac{|\widehat{f}(S)|}{c_S} \leq 8(d+1)A(f).$$

Mix these laws with probabilities $|\widehat{f}(S)|/(c_S Z_f)$. The resulting law ν_f has entropy at most D_*, mean exactly f/Z_f, and second moment

$$Q(\nu_f) \succeq Z_f^{-2} f f^T.$$

Its diagonal is constant in $[F(1), 1]$. The component diagonals may have different values, but each is constant over inputs. Convexity introduces no factor for the number of Fourier components.

Now replace the original first law by

$$\widetilde{\nu}_f = (1-\theta)\mathsf{M}_{Q_0} + \theta \nu_f, \qquad \theta = (D_* L)^{-1}.$$

Its entropy is at most $1/L$, its diagonal remains constant in $[F(1), 1]$, and

$$Q(\widetilde{\nu}_f) \succeq v f f^T, \qquad v = \frac{1}{D_* L Z_f^2}.$$

This uses the second moment of the mixture; it does not square the mixing probability. Hence

$$S_v := 1 + \log(1/v) \leq C[\log(d+L+2) + \log A(f)] = C T_f.$$

We spell out the initial-condition adjustment in the existing steering argument. Lemma 7.1 is stated for an initial diagonal in $[F(1), F(2)]$, but its proof works on $[F(1), 1]$. Indeed $F(q) < q$, $(q - F(q))/q^2$ has a positive lower bound on $(0, 1]$, and

$$R(q) = \frac{q[1-F(q)]^2}{F(q)} > 0, \qquad \log R(q) = O(q^2) \quad (q \downarrow 0).$$

Compactness extends the required estimates to the enlarged interval. The same telescoping proof reaches diagonal δ in $O(1/\delta)$ free layers while retaining a component at least $c\delta v$.

The proof of Theorem 17.1 then applies with S_v in place of S_f and with the first-law entropy $1/L$ added. For clarity, set $\delta = B_1 S_v / L$ and $p = B_2(S_v + \log(2L))/L$ with the sufficiently large universal

constants used there. Prescribed rank-one growth costs $O(p + S_v\delta)$ and uses $O(S_v/\delta)$ layers. Exact holding in the unused layers costs $O(L\delta^2)$. Purification costs $O(p\log(2L))$, leaves residual $O(\delta L^{-4})$, and the scalar path costs at most $V(p)+p \le V_* + p$, with $O(L^{-4})$ additional readout cost. All stage lengths fit when $L \ge CS_v$, and holding gives the exact specified depth. Adding the initial entropy yields

$$J_L(f) \le V_* + \frac{1}{L} + C\frac{S_v^2 + \log^2(2L)}{L} \le V_* + C_1\frac{T_f^2}{L}.$$

The strict-margin rate equals its closed-margin variational infimum by Corollary A.2, so this construction has precisely the prior-rate interpretation in the theorem.

Finally let $B = \log(d+2) + \log A(f) \ge 1$. One has $T_f \le B + \log(L+1) + C$. For every $L \ge CB^2$, increasing C if necessary makes T_f/L satisfy the depth requirement and bounds T_f^2/L by a universal constant. This proves the fixed-budget assertion. □

Proof of Corollary 18.2. A parity has $A(f) = 1$. For a k-junta the Fourier support has at most 2^k elements, so Parseval and Cauchy–Schwarz give $A(f) \le 2^{k/2}$. Substitute these bounds into the theorem. The matching necessary depth for any fixed rate budget follows from Theorem 39.1. □

The controlled mixture explains why a small eigenvalue in the original first-layer kernel cannot by itself give a large critical-rate lower bound. The mixture inserts a stronger component at entropy $1/L$. The theorem gives no lower bound ordering the competing extensions of a training sample. Section 33 explains the additional comparisons needed for a learning conclusion.

19 Exact Boolean feature means with a common diagonal

The next construction propagates exact Boolean gate vectors through a controlled feature law. Both the mean and the diagonal second moment are exact. In particular, approximation errors do not accumulate across the circuit layers. The number of mixture components can be exponential in the state width; only its logarithm enters the entropy bound.

Lemma 19.1 (Exact finite-state feature repair)**.** *Let X be a finite set, and let $H : X \to \{-1,1\}^s$ contain a constant coordinate $H_0 = 1$. Regard H also as the matrix whose rows are indexed by X. Suppose*

$$Q \succeq rHH^{\mathsf{T}}, \qquad r > 0, \qquad Q(x,x) = q \le 2 \quad (x \in X).$$

Let $g(x) = \operatorname{sign}(w \cdot H(x))$, where $|w \cdot H(x)| \ge 1$ everywhere. There is a feature law ν_g such that

$$\mathbb{E}_{\nu_g} U(x) = ag(x), \qquad \mathbb{E}_{\nu_g} U(x)^2 = q_* \quad (x \in X), \qquad a \ge \tfrac{1}{2}, \quad q_* \ge \tfrac{1}{2},$$

and

$$D(\nu_g \| \mathsf{M}_Q) \le \frac{C(s+1)^2\{\|w\|_2^2 + (s+1)^2\}}{r}.$$

Here $\mathsf{M}_Q = \tanh_* N(0,Q)$, and C is universal. For fixed H and Q, the same a and q_* work for every gate g. The law ν_g is a finite mixture of Gaussian mean shifts followed by coordinatewise tanh*. The covariance Q may be singular.*

Proof. For every coefficient vector c, the covariance inequality gives $Hc \in \operatorname{range} Q$ and

$$(Hc)^{\mathsf{T}} Q^{+} (Hc) \le \|c\|_2^2 / r.$$

Indeed, $rQ^{+/2}HH^{\mathsf{T}}Q^{+/2} \preceq I$ on the covariance range, which bounds the operator norm of $\sqrt{r}\,Q^{+/2}H$ by one. A Gaussian mean shift by Hc therefore has relative entropy at most $\|c\|_2^2/(2r)$. The same upper bound holds after applying tanh.

Write $T = H(X)$ for the distinct realized states and $M = |T|$. The constant coordinate implies $M \le 2^{s-1}$. Set

$$F(q) = \mathbb{E}\tanh^2(\sqrt{q}Z), \qquad c_* = 1 - F(2) > 0, \qquad \varepsilon = \frac{c_*^2}{10^6 M}.$$

Choose $A \ge 1$ with $4e^{-2A+4} \le \varepsilon$ and $A \le C(s+1)$. If a Gaussian variable has variance at most two and mean of absolute value at least A, its expected tanh differs from the mean's sign by at most $\varepsilon/2$, and its second moment belongs to $[1-\varepsilon, 1]$. For positive means this follows from

$$1 - \tanh u \le 2e^{-2u}, \qquad \operatorname{sech}^2 u \le 4e^{-2u};$$

negative means follow by symmetry.

For each $p \in T$ define

$$d_p(H) = \frac{s - p \cdot H}{2}.$$

This is zero at state p, is a positive integer at every other state, and is at most s. Using the constant coordinate, both $d_p(H)$ and $1 - 2d_p(H)$ are linear combinations of the coordinates of H with coefficient norm at most Cs.

We use the following feature laws, all obtained from a Gaussian of covariance Q by a mean shift and then tanh:

- P_g has preactivation mean $Aw \cdot H$. Write its feature mean as m_g; then $\|m_g - g\|_\infty \le \varepsilon/2$.
- P_p has preactivation mean $A(1 - 2d_p(H))$. Its feature mean approximates $2\mathbf{1}_{\{H=p\}} - 1$ to error at most $\varepsilon/2$.
- P_c has the constant preactivation mean As. Write its constant feature mean as m_c and second moment as s_c.
- R_c is the equal mixture of P_c and its output-negated law; its mean is zero and its second moment is s_c.
- R_p is the equal mixture of the shifts with preactivation means $Ad_p(H)$ and $-Ad_p(H)$. Its mean is zero. Its second moment is $F(q)$ at state p and belongs to $[1-\varepsilon, 1]$ elsewhere.

The diagonals of P_g, P_p, P_c, R_c all belong to $[1-\varepsilon, 1]$. Each law above has entropy at most

$$E = \frac{CA^2(\|w\|_2^2 + s^2)}{r}.$$

For the mixtures this follows from entropy convexity. Output negation preserves the raw law and is realized by negating the mean shift, so it preserves these entropy bounds. Because Q has constant diagonal, every mean and second moment just described depends only on the state $H(x)$.

Repairing the mean. Let $B_p = (P_p + P_c)/2$. Its mean on T approximates the point indicator of p to error at most $\varepsilon/2$. The square matrix V with these means as columns consequently satisfies

$$\|V - I\|_{\infty\to\infty} \le M\varepsilon/2 < 1/2, \qquad \|V^{-1}\|_{\infty\to\infty} \le 2.$$

Put $t_1 = 4M\varepsilon$ and solve

$$Vc = (1 - t_1)(g - m_g).$$

Then $\|c\|_\infty \le \varepsilon$ and $\sum_p |c_p| \le M\varepsilon < t_1$. Define

$$\nu_1 = (1 - t_1)P_g + \sum_p |c_p| \operatorname{sign}(c_p) B_p + \Big(t_1 - \sum_p |c_p|\Big) R_c,$$

where multiplication of a law by -1 denotes its pushforward by output negation; terms with $c_p = 0$ are omitted. This is a probability mixture. Its mean is exactly $(1 - t_1)g$, its diagonal $s_1(x)$ belongs to $[1 - \varepsilon, 1]$, and its entropy is at most E.

Repairing the diagonal with positive weights. Let $u_p(x)$ be the second moment of R_p, and on the state set T form the matrix

$$D_{x,p} = s_c - u_p(x).$$

For a centered Gaussian, the absolute value after a mean shift is stochastically increasing in the absolute mean. Since $|Ad_p(H)| \le As$, the entries of D are nonnegative. Its diagonal is the common number

$$c = s_c - F(q) \ge c_* - \varepsilon \ge c_*/2,$$

and every off-diagonal entry is at most ε. Thus $D = cI + E_0$ with $\|E_0\|_{\infty\to\infty} \le M\varepsilon$.

Set

$$t_2 = 32M\varepsilon/c_*, \qquad b_x = 4\varepsilon - (1 - t_2)(s_c - s_1(x)).$$

Both t_1, t_2 lie in $(0, 1/10)$, and $3\varepsilon \le b_x \le 5\varepsilon$. Let $\lambda = D^{-1}b$. With $\delta = \|E_0\|_{\infty\to\infty}/c \le 2c_*/10^6$, the Neumann series gives

$$\|\lambda - b/c\|_\infty \le \frac{\delta}{1 - \delta}\frac{5\varepsilon}{c}.$$

The right side is less than $3\varepsilon/(10c)$, so every λ_p is strictly positive. Moreover,

$$\|\lambda\|_\infty \le \frac{5\varepsilon}{c(1 - \delta)} \le \frac{12\varepsilon}{c_*}, \qquad \sum_p \lambda_p < t_2.$$

The probability mixture

$$\nu_g = (1 - t_2)\nu_1 + \sum_p \lambda_p R_p + \Big(t_2 - \sum_p \lambda_p\Big) R_c$$

has exact mean ag, where $a = (1 - t_1)(1 - t_2) \ge 1/2$, and exact diagonal

$$(1 - t_2)s_1 + t_2 s_c - D\lambda = s_c - 4\varepsilon =: q_* \ge 1/2.$$

Neither a nor q_* depends on the gate. Entropy convexity gives $D(\nu_g \| \mathsf{M}_Q) \le E$. Finally, $A \le C(s+1)$ gives the asserted polynomial numerator. □

20 Controlled feature laws for arbitrary Boolean circuits

The Fourier construction can be replaced by a layerwise simulation of a Boolean circuit. The controlled laws used in this simulation are rare mixture components. Their probabilities become small with circuit depth, but the logarithm of the resulting target component grows only linearly with that depth. An exact repair of feature means and diagonal second moments makes this compatible with the subsequent scalar steering. This section bounds the critical prior rate. Section 22 combines it with finite representation to obtain sample bounds.

Definition 20.1 (Layered Boolean threshold implementation)**.** *Let $H^{(0)}(x) = (1, x_1, \dots, x_d)$. A depth-$h$ implementation consists of state vectors $H^{(j)} : \{-1,1\}^d \to \{-1,1\}^{C_j}$, $1 \le j \le h$, each containing a constant coordinate, with*

$$H_i^{(j)}(x) = \operatorname{sign}(w_{j,i} \cdot H^{(j-1)}(x)), \qquad |w_{j,i} \cdot H^{(j-1)}(x)| \ge 1 \quad \text{for every } x.$$

One coordinate of $H^{(h)}$ is the target f. Write $C = \max_j C_j$, $W = \max\{d+1, C\}$, and assume $\|w_{j,i}\|_2 \le B$ with $B \ge 1$. The constant coordinate allows affine gates without adding internal biases to the neural-network model. A bounded-fan-in AND/OR/NOT circuit has such an implementation with universal B, after a fixed scaling of each gate's coefficients.

Theorem 20.2 (A circuit-dependent critical-rate bound)**.** *There are universal constants C_0, C_1, C_2 with the following property. Suppose f has the implementation above, with $h \ge 1$, and put*

$$T = \log(d+2) + h \log(W + B + h + L + 2).$$

If the learner depth satisfies $L - h \ge C_0 T$, then

$$J_L(f) \le V_* + C_1 \frac{T^2}{L-h}.$$

In particular, every

$$L \ge C_2 h^2 \log^2(W + B + h + 2)$$

puts f below a fixed universal absolute critical-rate budget. The prior remains the original independent Gaussian prior; no robustness restriction is imposed on its support.

Corollary 20.3 (Bounded-fan-in Boolean circuits)**.** *For a layered bounded-fan-in AND/OR/NOT circuit of width C and depth $h \ge 1$, including constant rails, define*

$$T_{\mathrm{B}} = \log(d+2) + h \log(C + h + L + 2).$$

If $L - h \ge C_0 T_{\mathrm{B}}$, then

$$J_L(f) \le V_* + C_1 \frac{T_{\mathrm{B}}^2}{L-h}.$$

Consequently every

$$L \ge C_2 [\log(d+2) + h \log(C + h + 2)]^2$$

puts f below a fixed universal absolute critical-rate budget. The constants may depend on the fixed fan-in bound and gate library.

The local ingredient is Lemma 19.1, proved in the preceding section. It constructs an exact Boolean feature mean and an exactly constant diagonal at polynomial entropy cost. The exponentially many states used in its repair do not cause exponential cost: each component has polynomial entropy cost, and convexity bounds the cost of their mixture by the same bound. Individual repair weights are exponentially small; their total mass is a small constant.

20.1 A low-cost circuit prefix

Lemma 20.4 (Circuit prefix)**.** *For every $L \ge h+1$, the first h hidden feature laws can be chosen at total relative entropy at most $1/L$, so that their final second moment has constant diagonal in $(0,1]$ and satisfies*

$$Q_h \succeq r_h H^{(h)}(H^{(h)})^T, \qquad 1+\log(1/r_h) \le C[\log(d+2) + h\log(W+B+h+L+2)].$$

Proof. The input covariance has the exact representation

$$Q_0 = d^{-1}H^{(0)}(H^{(0)})^T, \qquad Q_0(x,x) = (d+1)/d \le 2.$$

Set $r_0 = 1/d$, and choose the universal polynomial

$$P = C_*(W+1)^2\{B^2 + (W+1)^2\} \ge 1$$

large enough to bound the numerator in Lemma 19.1 for every gate. Suppose Q_{j-1} has constant diagonal and dominates $r_{j-1}H^{(j-1)}(H^{(j-1)})^T$. For each gate in the next state, that lemma gives a feature law with mean $a_j H_i^{(j)}$, where $a_j \ge 1/2$, and entropy at most P/r_{j-1} relative to $\mathsf{M}_{Q_{j-1}}$. For this fixed state and reference covariance, all the laws have the same a_j and the same constant diagonal.

Average these controlled laws uniformly over the C_j gates. Their mixture ν_j has constant diagonal and satisfies

$$Q(\nu_j) \succeq \frac{1}{4C_j}H^{(j)}(H^{(j)})^T, \qquad D(\nu_j \| \mathsf{M}_{Q_{j-1}}) \le \frac{P}{r_{j-1}}.$$

Use the diluted law

$$\widetilde{\nu}_j = (1-\theta_j)\mathsf{M}_{Q_{j-1}} + \theta_j \nu_j, \qquad \theta_j = \frac{r_{j-1}}{PhL}, \qquad r_j = \frac{r_{j-1}}{4C_j PhL}.$$

Here $0 < \theta_j \le 1$ because $r_{j-1} \le 1$. Entropy convexity bounds the layer cost by $1/(hL)$, and its second moment dominates $r_j H^{(j)}(H^{(j)})^T$. The uncontrolled law has constant diagonal whenever its reference does, so dilution preserves this property. Every hidden diagonal is at most one, and its constant rail makes it strictly positive. Induction therefore applies through all h layers. Finally,

$$\log(1/r_h) = \log d + \sum_{j=1}^{h} \log(4C_j PhL) \le \log d + h\log(4WPhL),$$

which gives the stated estimate. □

20.2 Steering from an arbitrary positive constant diagonal

Lemma 20.5 (Steering after a prescribed prefix)**.** *Suppose a prefix of h hidden layers has entropy cost E and ends at a second moment Q with constant diagonal $q \in (0,1]$ and $Q \succeq vff^T$, where $0 < v \le 1$. Write $S = 1+\log(1/v)$ and $R = L-h$. For $R \ge CS$, it can be completed to the exact learner depth L with*

$$J_L(f) \le E + V_* + C\frac{S^2 + \log^2(2R)}{R}.$$

Proof. Only the initial cooling step of Theorem 17.1 needs adjustment. For $q \in (0,1]$, the estimates in Lemma 7.1 hold uniformly: $q - F(q) \geq cq^2$, and the product of the factors

$$\frac{q[1-F(q)]^2}{F(q)}$$

along a free trajectory is bounded below by a universal positive constant. The latter follows from a logarithm of order $O(q^2)$ at zero, positivity on $(0,1]$, and the uniformly bounded sum of q_j^2. If the current diagonal exceeds a desired level δ, cool to the first diagonal below δ. This takes $O(1/\delta)$ layers, ends with diagonal at least $c\delta$, and retains a tracked component at least $c\delta v$; the initial diagonal is at most one. If $q \leq \delta$ already, skip cooling. Its existing component v is also at least $c\delta v$.

In either case reduce the tracked component to the exact value $v_0 = \delta e^{-AS}$ for a sufficiently large universal A. The residual diagonal is at most δ. The holding and prescribed-growth arguments of Lemma 8.4 use precisely this upper bound; they do not require an initial diagonal bounded below by δ. Consequently holding costs $O(H\delta^2)$ for a prescribed H, and growth to $p \geq \delta$ takes $O(S/\delta)$ layers with entropy $O(p + S\delta)$.

Choose $\delta = B_1 S/R$ and $p = B_2\{S + \log(2R)\}/R$, as in the target-component theorem. Its purification, stable scalar lifting, and exact-depth padding then fit among the remaining $R-1$ hidden layers for $R \geq CS$, after increasing the universal constants. The cost after the prefix is at most

$$V_* + C\{R\delta^2 + S\delta + p + p\log(2R) + R^{-4}\} \leq V_* + C\frac{S^2 + \log^2(2R)}{R}.$$

The prefix cost adds E, as claimed. □

Proof of Theorem 20.2. Apply Lemma 20.4 and retain the coordinate corresponding to f, giving $Q_h \succeq r_h f f^T$ with $S = 1 + \log(1/r_h) \leq CT$. Apply Lemma 20.5 with $E \leq 1/L$ and $R = L - h$. Since $h \geq 1$, the definition of T also absorbs $\log(2R)$. This proves the displayed rate bound. The equality between strict margin rates and their variational infima gives its prior-probability interpretation.

For the fixed-budget claim, put $K = W + B + h + 2$ and note $K \geq h + 2$. At $L \geq Ch^2 \log^2 K$, increasing C ensures $L - h \geq L/2$, the required depth condition, and a universal bound for $h^2[\log K + \log L]^2/L$. Indeed $(\log K + \log L)^2/L$ is decreasing on this range, and its value at the lower endpoint is $O(h^{-2})$. The rate bound is therefore uniformly bounded there. □

Proof of Corollary 20.3. For each gate apply Lemma 19.1 only to its input wires and the constant coordinate. This restricted state has bounded width, its threshold coefficient norm is universally bounded, and $Q \succeq rHH^T$ implies domination of the restricted state's second moment with the same r. The controlled gate law therefore costs at most P_*/r with universal P_*. Its mean amplitude is at least $1/2$ and its diagonal is constant. These constants need not agree between different gates: a mixture of laws whose diagonals are individually constant still has constant diagonal, and the uniform amplitude lower bound suffices for covariance domination.

Repeat the prefix proof with $P = P_*$ and $C_j \leq C$. This gives

$$1 + \log(1/r_h) \leq 1 + \log d + h\log(4CP_*hL) \leq C_* T_{\mathrm{B}}.$$

The same steering proves the rate bound. For the final assertion put $A = \log(d+2) + h\log(C+h+2)$. Then $A \geq h\log(h+2)$ and $T_{\mathrm{B}} \leq A + h\log(L+1) + C_* h$. At $L = C_2 A^2$, one has $h\log(L+1) = O(A)$, apart from the harmless constant $h\log C_2$, since $h\log A = O(A)$ for $A \geq h\log(h+2)$. Increasing C_2 gives the depth requirement and a universal bound on $T_{\mathrm{B}}^2/(L-h)$ at that point. The bound persists for all larger L: the function $(A + h\log(L+1) + C_* h)^2/L$ is decreasing once $A + h\log(L+1) + C_* h > 2h$, and $L - h \geq L/2$ on the specified range. □

Combining this construction with the Fourier and unmodified-kernel constructions permits the best of their upper bounds for any particular target. The new construction avoids exponential dependence on the teacher's depth in the critical prior-rate upper bound. The learning theorem in Section 22 adds the necessary capacity estimate for competing functions. Its general teacher bound still has an exponent depending on teacher depth.

21 Boolean functions implemented by real-valued tanh teachers

The circuit-rate bound also applies to Boolean targets specified by a bounded-weight tanh teacher. Finite precision is enough because the teacher's final output has a fixed margin. The necessary precision grows linearly with teacher depth, and each numerical layer has a constant-depth threshold implementation of polynomial size. No stability condition on the teacher's hidden states is needed.

Lemma 21.1 (Finite-precision threshold compilation)**.** *There is a universal integer a with the following property. Suppose a tanh network on $\{-1,1\}^d$ has depth $D \ge 1$, hidden width at most C, parameter magnitudes at most $B \ge 1$, and a linear readout z_T satisfying*

$$f(x)z_T(x) \ge \tfrac{1}{2} \qquad \text{for every } x, \qquad f : \{-1,1\}^d \to \{-1,1\}.$$

Depth counts affine maps, including the linear readout. Biases may be present; the original model with a constant input and no internal biases is a special case. Then f has a layered Boolean threshold implementation of depth $h \le a(D+1)$, width at most K^a, and gate coefficient Euclidean norm at most K^a, where

$$K = d + C + B + D + 2.$$

Every Boolean gate has margin at least one, and constant rails and layering overhead are included in these bounds.

Proof. We use the standard arithmetic threshold-circuit theorem: addition, comparison, division, and iterated multiplication of polynomially many polynomial-bit integers have constant-depth, polynomial-size circuits of unweighted majority and Boolean gates; see Hesse–Allender–Barrington [2] for the nontrivial division and iterated-multiplication assertions. The depth and polynomial degree are universal. Thus this theorem provides majority gates, rather than arbitrary threshold gates with uncontrolled coefficient magnitudes.

First consider approximating a single scalar tanh. For a dyadic input u with $|u| \le R$ and desired error 2^{-p}, put $t = 2|u|$ and

$$E_m(t) = \sum_{j=0}^{m-1} \frac{t^j}{j!}, \qquad m = \lceil 40(R+p+2) \rceil.$$

The Taylor remainder satisfies

$$0 \le e^t - E_m(t) \le e^{2R} \frac{(2R)^m}{m!} \le e^{2R} \left(\frac{2eR}{m}\right)^m < 2^{-p-4}.$$

Write the sum over a common denominator. All its powers, factorial constants, integer products, and sums involve polynomially many bits in R, p, and the input bit length. They can therefore be computed in constant threshold depth by the arithmetic theorem. Since $E_m(t) \ge 1$, the map $v \mapsto (v-1)/(v+1)$ has derivative at most one for $v \ge 1$. Division followed by rounding to $p+2$ fractional bits gives

$$\operatorname{sign}(u) \frac{E_m(2|u|) - 1}{E_m(2|u|) + 1}$$

to total error less than 2^{-p} from $\tanh u$. Choose the rounding convention to keep the result in $[-1,1]$. This is a constant-depth majority circuit whose size is polynomial in the stated numerical parameters, with universal degree.

For the teacher, set

$$K_0 = 4(d+C+1)(B+1), \qquad p = \lceil (D+3)\log_2 K_0 \rceil + 5.$$

Round every parameter to p fractional bits and hardcode these bits in the circuit. This is a nonuniform existence statement and requires no computability assumption on the original real parameters. Starting from the exact Boolean input, evaluate each affine map on the stored dyadic values exactly, and evaluate each tanh to error at most 2^{-p} by the preceding construction. All stored hidden values have magnitude at most one, so every preactivation has magnitude at most $R = (d+C+1)(B+1)$. Parameter rounding contributes at most $(d+C+1)2^{-p}$ at any layer. Since tanh is one-Lipschitz, a conservative induction bounds the error after j affine maps by $K_0^{j+1}2^{-p}$, including the linear readout. The final error is therefore less than $1/16$. Its sign is exactly f by the margin assumption.

The arithmetic at every layer has constant threshold depth and polynomial size in d, C, B, D: the precision p, the range R, all intermediate bit lengths, and the number of Taylor terms are polynomial in these parameters. Stack these blocks over D layers and append a sign comparison. The resulting depth is at most $a(D+1)$ and its total size is at most K^a after increasing a universal a. Make it layered by identity gates and propagate a constant wire. This introduces only polynomial overhead and a universal constant factor in depth.

Finally, an unweighted majority or Boolean gate with fan-in at most the total circuit size can be written as an integer threshold gate with coefficients of polynomial magnitude and margin at least one. For example, place the threshold halfway between the two neighboring allowed integer sums, and multiply the score by two if necessary. The constant rail carries the threshold. Its coefficient norm is polynomial in the circuit size. Increasing a once more proves all three bounds simultaneously. □

Corollary 21.2 (Polynomial teacher-depth dependence of the critical rate)**.** *For a Boolean target with the tanh-teacher implementation in Lemma 21.1, there is a universal A such that, whenever*

$$L \geq A(D+1)\log(d+C+B+D+L+2),$$

the original unrestricted Gaussian learner satisfies

$$J_L(f) \leq V_* + A\frac{(D+1)^2\log^2(d+C+B+D+L+2)}{L}.$$

In particular, every

$$L \geq A(D+1)^2\log^2(d+C+B+D+2)$$

puts f below a fixed universal absolute critical-rate budget.

Proof. Apply the compiler lemma. Its threshold depth is $h \leq a(D+1)$, while its state width and normalized gate coefficient norms are polynomial in $K = d+C+B+D+2$. The parameter T in Theorem 20.2 is consequently at most $A_0(D+1)\log(K+L)$. Increase the universal A so that the displayed depth condition ensures $L-h \geq L/2$ and $L-h \geq C_0T$. The theorem gives the rate estimate. Its fixed-budget conclusion, or the same squared-logarithm comparison used in its proof, gives the final assertion. □

The result concerns the rate of the target's full Boolean sign pattern under the original Gaussian prior. It removes an exponential teacher-depth cost from this rate upper bound. Corollary 22.5 combines it with the capacity theorem to give a sample bound polynomial in learner depth for each fixed teacher depth. Joint polynomial dependence on teacher width and depth remains open.

22 A sample bound from critical-rate sublevel sets

A finite representation theorem gives a learning bound even when its weights have no useful numerical bound. The relevant fact about tanh is that the class of networks with a fixed finite architecture has polynomial VC dimension in its number of parameters. This allows us to count the Boolean functions below a critical-rate threshold. The resulting bound applies to the original Gaussian prior; it does not condition that prior on an additional stability event.

Lemma 22.1 (Counting finite tanh networks)**.** *Let $\mathcal{T}_{d,L,M}$ be the Boolean sign functions on $\{-1,1\}^d$ represented by depth-L tanh networks with at most M neurons per hidden layer and a linear readout. Their real parameters are unrestricted. There is a universal constant C such that*

$$\log|\mathcal{T}_{d,L,M}| \le C(d+1)(d+L+1)^4(M+1)^8.$$

Here $L \ge 2$ and logarithms are natural.

Proof. Pad every hidden layer to width M with zero neurons. For an upper bound we allow arbitrary biases at every layer, whether or not they were present in the original representation. The resulting fixed architecture has at most

$$W \le C(d+L+1)(M+1)^2$$

weights and biases. The classical sigmoid-network bound of Karpinski and Macintyre [1] gives VC dimension at most CW^4, with a universal constant independent of depth and parameter values. It applies to tanh because $\tanh(t) = 2\sigma(2t)-1$, where $\sigma(t) = (1+e^{-t})^{-1}$: rescale the incoming weights and absorb the additive constants into the biases in the next layer and the final linear readout. This preserves the fixed architecture up to a universal factor in its parameter count. The threshold at the readout is absorbed into its bias.

If $v \ge 1$ bounds the VC dimension, Sauer's lemma bounds the number of dichotomies on $m = 2^d$ points by $(m+1)^v$; the same bound holds when $v \ge m$, since there are at most 2^m dichotomies. Hence $\log|\mathcal{T}_{d,L,M}| \le v(d+1)\log 2$. Substitute the parameter bound. In particular, finite parameter precision and a bound on weight magnitudes are unnecessary. □

Theorem 22.2 (Critical-rate sublevel capacity)**.** *There are universal constants $c, C > 0$ such that, for every $d \ge 1$, $L \ge 2$, and $a \ge 0$, the class*

$$\mathcal{F}_{L,a} = \{g : \{-1,1\}^d \to \{-1,1\} : J_L(g) \le a\}$$

satisfies

$$\log\max\{1, |\mathcal{F}_{L,a}|\} \le C\big[(d+1)(L+1)(a+2)\big]^c \exp\left(C\sqrt{L(a-V_*)_+}\right). \tag{14}$$

Consequently, for every $\eta \ge 0$,

$$\log\max\{1, |\mathcal{F}_{L,V_*+\eta}|\} \le C\big[(d+1)(L+1)(\eta+3)\big]^c e^{C\sqrt{L\eta}}. \tag{15}$$

The polynomial degree and exponential constant are universal.

Proof. For every member g of the class choose an admissible path of cost $A \le a + 1/L$; this needs no attainment of the variational infimum. It satisfies

$$A \le V_* + \eta_a, \qquad \eta_a = (a - V_*)_+ + 1/L.$$

Lemma 12.1 therefore gives

$$S := \sum_{j=1}^{L-1} \sqrt{D_j} \le C \log(L+1) + C\sqrt{L(a-V_*)_+} + C.$$

The lemma is valid for all nonnegative excesses. For excess at least $1/4$, its proof simply uses $\sum_j D_j \le V_* + \eta_a \le 9\eta_a$ and Cauchy–Schwarz; there is no small-excess restriction here.

Theorem 15.1 supplies a depth-L tanh realization of the same Boolean function with width

$$M \le C\big[(d+1)(L+1)(a+2)\big]^{c_0} e^{C\sqrt{L(a-V_*)_+}}$$

for universal constants. Its positive margin ensures exact equality of sign functions on the whole cube. This width bound is common to all members of $\mathcal{F}_{L,a}$, although the networks and their real weights may differ. Round the displayed upper bound upward and apply Lemma 22.1. Absorb fixed powers into c, C. Finally substitute $a = V_* + \eta$ and use $V_* < 2$. □

Theorem 22.3 (Learning from the target's critical excess rate)**.** *There are universal $c, C > 0$ with the following property. Fix a Boolean target f, dimension $d \ge 1$, learner depth $L \ge 2$, any input distribution $\mathcal{D}$ on the cube, and accuracy and confidence parameters $\varepsilon, \delta \in (0,1)$. Suppose $J_L(f) \le a < \infty$. A sufficient number of independent training examples, labeled by f, is*

$$n \ge \frac{C}{\varepsilon}\left\{\big[(d+1)(L+1)(a+2)\big]^{c} e^{C\sqrt{L(a-V_*)_+}} + \log(1/\delta)\right\}. \tag{16}$$

Then there is a finite N_0, independent of the realized sample, such that for every learner width $N \ge N_0$, with sample probability at least $1-\delta$, the original globally margin-conditioned critical posterior with exact training fit satisfies

$$\mathcal{D}\{x : p_N(f(x) \mid x) \le 2/3\} \le \varepsilon.$$

The result allows arbitrary posterior mixtures of tied rate minimizers. It does not give a quantitative bound on N_0 or an efficient sampling algorithm for the posterior.

Proof. Use the target-relative class $\mathcal{A}_f = \{g : J_L(g) \le J_L(f)\} \subseteq \mathcal{F}_{L,a}$. Each member whose error exceeds $\varepsilon/6$ agrees with all n independent labels with probability at most $e^{-n\varepsilon/6}$. A union bound and Theorem 22.2 show that, under (16), with probability at least $1-\delta$, every sample-consistent member of $\mathcal{A}_f$ has error at most $\varepsilon/6$.

Write π_N for the original globally margin-conditioned critical prior on Boolean functions. There are finitely many such functions at fixed d. The exact rate formula therefore gives

$$\frac{\pi_N(\mathcal{A}_f^c)}{\pi_N(f)} \longrightarrow 0 \qquad (N \to \infty).$$

Indeed every function outside $\mathcal{A}_f$ has rate strictly greater than $J_L(f)$, and the minimum of these finitely many positive gaps is positive whenever the complement is nonempty. For every sample labeled by f, its posterior normalizer is at least $\pi_N(f)$. Thus, at every sufficiently large width, posterior mass outside $\mathcal{A}_f$ is at most $\varepsilon/6$, simultaneously for all realized samples. On the good-sample event, posterior Gibbs error is at most $\varepsilon/3$. At an input where correct-label probability is at most $2/3$, the wrong-label probability is at least $1/3$. Integrating proves the displayed confidence-error bound. □

Corollary 22.4 (Fourier variation and every parity)**.** *Put*

$$A_F(f) = \sum_{S \subseteq [d]} |\widehat{f}(S)|, \qquad T_f = \log(d + L + 2) + \log A_F(f).$$

There are universal constants $C_0, c, C > 0$ such that, whenever $L \geq C_0 T_f$, a sufficient sample size in Theorem 22.3 is

$$n \geq \frac{C}{\varepsilon} \left\{ [(d + L + 2) A_F(f)]^c + \log(1/\delta) \right\}.$$

In particular, every signed parity $f = \pm\chi_S$, including constants, is learned with a sample bound polynomial in $d, L, 1/\varepsilon$ and $\log(1/\delta)$ whenever $L \geq C_0 \log(d + L + 2)$. This statement is uniform in the parity degree. The prior has no additional robustness restriction.

Proof. The controlled Fourier-seed theorem gives $J_L(f) \leq a := V_* + C_1 T_f^2 / L$. Consequently $\sqrt{L(a - V_*)} = \sqrt{C_1} T_f$. The exponential factor in (16) is therefore at most $[(d + L + 2) A_F(f)]^C$. Since $A_F(f) \geq 1$, the polynomial prefactor involving $d, L, a + 2$ is bounded by a fixed power of the same quantity. Apply the theorem. All signed parities have $A_F(f) = 1$. Only the original seed condition $L \geq C_0 T_f$ is needed: if $C_1 T_f^2 / L \geq 1/4$, the large-excess branch of the square-root entropy lemma still gives the required bound. No squared-logarithmic minimum depth is imposed by the learning argument. □

Corollary 22.5 (Bounded-weight teachers and growing learner depth)**.** *Suppose f has a depth-D, width-C_T tanh teacher with parameter magnitudes at most $B \geq 1$ and output margin at least $1/2$ on the cube. Put $R = d + C_T + B + D + L + 2$. There are universal constants c, C_0, C such that, whenever*

$$L \geq C_0 (D + 1) \log R,$$

a sufficient sample size for the original critical learner is

$$n \geq \frac{C}{\varepsilon} \left\{ R^{c(D+1)} + \log(1/\delta) \right\}.$$

This is polynomial in the learner depth L for every fixed teacher depth D. The degree depends linearly on D, so the result does not settle the desired bound polynomial jointly in teacher width and depth.

Proof. Corollary 21.2 supplies

$$J_L(f) \leq V_* + C \frac{(D + 1)^2 \log^2 R}{L}$$

under the stated depth hypothesis. In Theorem 22.3, the exponential factor is at most $e^{C(D+1) \log R} = R^{C(D+1)}$; its polynomial prefactor is absorbed in the same expression. □

The theorem concerns the width-first critical scaling $\gamma = 1$. Its finite-network realizations are certificates used to count competing functions; the learner still uses the original unrestricted Gaussian prior and may be much wider than those certificates. The argument makes no assertion that the posterior is computationally tractable, that the certificate has a small product of weight norms, or that the result automatically transfers to a different order of limits.

23 Deep sequential computations with polynomial sample bounds

Some computations have long sequential implementations while admitting sample bounds without a teacher-depth exponent. Decision trees and binary-state branching programs have small Fourier variation. A different argument handles permutation programs with solvable transition groups, even when their Fourier variation is exponential. All results in this section concern the original unrestricted critical prior and inherit the sufficiently-large-width convention of Theorem 22.3. Their every-realizable-width versions are collected in Theorem 2.2 and Corollary 2.6; these use the separate finite-width target-mass constructions.

Proposition 23.1 (Decision trees and binary-state programs)**.** *A Boolean function computed by a decision tree with s leaves satisfies $A_F(f) \le s$. A function computed by a deterministic layered branching program of width at most two and length T satisfies $A_F(f) \le T+1$. The branching program may query different input coordinates in its two states, and may query a coordinate repeatedly.*

Consequently, there are universal $c, C > 0$ such that a sufficient sample bound for either class is

$$n \ge \frac{C}{\varepsilon}\left\{(d+L+K+2)^c + \log(1/\delta)\right\}, \qquad L \ge C\log(d+L+K+2),$$

where $K = s$ for the tree and $K = T+1$ for the branching program. The guarantee is distribution-free on the cube, uses exact training fit, and requires correct-label posterior probability greater than $2/3$ except on mass at most ε.

Proof. The indicator of a nonempty decision-tree leaf is a conjunction of input literals, after removing repeated queries and contradictory branches. Its Fourier ℓ^1 norm is one. Expressing f as the sum of its signed leaf indicators proves the first assertion.

For the second, encode the two states by $\{-1, 1\}$ and let $s_i(x)$ be the state after step i. Let $t_i^+(x)$ and $t_i^-(x)$ be the next states selected from the two current states. Each is a unary Boolean function of the coordinate queried at that state, hence a constant or a signed input bit. Put

$$a_i = \frac{t_i^+ - t_i^-}{2}, \qquad b_i = \frac{t_i^+ + t_i^-}{2}.$$

Then $s_i = a_i s_{i-1} + b_i$, and the triangle inequality gives $A_F(a_i), A_F(b_i) \le 1$. Fourier ℓ^1 norm is submultiplicative: multiplication is convolution on the group of subsets under symmetric difference. Therefore

$$A_F(s_i) \le A_F(s_{i-1}) + 1.$$

The initial state is constant, so induction gives $A_F(s_T) \le T+1$. The final output map is either constant or a signed copy of the state. Neither case increases this bound. Apply Corollary 22.4. □

Here width two counts discrete control states; it does not mean two continuously valued neural activations. For a layered decision tree of width C and depth h, its number of leaves is at most $1 + Ch$. Thus the tree bound is jointly polynomial in its width and depth, regardless of tree imbalance. Sharing in a general circuit can make a decision-tree representation much larger.

23.1 Solvable-group programs

A length-T program over a finite group G chooses a group element $g_i(x)$ at step i according to one queried input bit, and outputs

$$f(x) = \phi(g_1(x)\cdots g_T(x)), \qquad \phi : G \longrightarrow \{-1, 1\}.$$

The queried coordinates may repeat. The group and the output table are part of the specified implementation. The derived length of a solvable group is the number of abelian quotients in its derived series.

Lemma 23.2 (A uniform threshold compiler for solvable groups)**.** *There is a universal C such that the following holds. Suppose G is a finite solvable group of order $q \geq 2$ and derived length $\ell \geq 1$. Every length-T group program has a layered Boolean threshold implementation with*

$$h \leq C\ell, \qquad C_T \leq q^C (T+1)^C, \qquad B_T \leq q^C (T+1)^C.$$

Here h is depth, C_T is width including constant rails and layering overhead, and B_T bounds the Euclidean norms of gate coefficient vectors. Gate scores have absolute value at least one on every realized Boolean input. The constants are independent of the group.

Proof. We prove the stronger assertion that one circuit computes one-hot encodings of every prefix product from one-hot encodings of the letters. A lookup involving at most three elements of a group of order at most q is a constant-depth Boolean circuit of size $O(q^4)$: form the relevant conjunctions, then their disjunctions for each output value. Its coefficients and fan-ins are polynomial in q, with universal degrees.

For an abelian group, use a decomposition into cyclic groups of orders $q_j \leq q$, with at most $\log_2 q$ factors. For one cyclic factor, let $v_i \in \{0, \dots, q_j - 1\}$ be the input values. For every prefix length $i \leq T$, form the integer score

$$U_i = \sum_{r=1}^{i} \sum_{a=0}^{q_j - 1} a \, \mathbf{1}_{\{v_r = a\}}.$$

Compute $z_{i,s} = \mathbf{1}_{\{U_i \geq s\}}$ for $1 \leq s \leq (q_j - 1)i$. For each residue u, put $\psi_u(t) = 2\mathbf{1}_{\{t \equiv u \pmod{q_j}\}} - 1$. On this integer interval the exact identity

$$\psi_u(U_i) = \psi_u(0) + \sum_{s=1}^{(q_j-1)i} [\psi_u(s) - \psi_u(s-1)] z_{i,s}$$

shows that one further threshold layer computes the residue indicator. For each prefix, combining the cyclic coordinates uses one conjunction per group element: there are at most q valid tuples, with at most $\log_2 q$ coordinates each. Computing all prefixes uses $q^{O(1)}(T+1)^2$ gates. The size, fan-ins, and coefficients are polynomial in q, T, with universal degrees, and the depth is universally bounded. Half-integer threshold offsets, followed by multiplication by two, give integer margins at least one. Switching between $\{0,1\}$ and $\{-1,1\}$ conventions preserves these polynomial bounds.

For the inductive step, take a nontrivial abelian normal subgroup $A \triangleleft G$ such that $H = G/A$ has smaller derived length; the last nontrivial term of the derived series suffices. Fix a section $s : H \to G$ with $s(e) = e$, and write $g_i = a_i s(h_i)$ uniquely. Set $H_i = h_1 \cdots h_i$, with $H_0 = e$. Define the finite tables

$$\alpha_h(a) = s(h) a s(h)^{-1}, \qquad c(h, h') = s(h)s(h')s(hh')^{-1} \in A.$$

The group need not be a split extension. Compute all H_i in parallel using one copy of the stronger induction hypothesis, and then put

$$t_i = \alpha_{H_{i-1}}(a_i) c(H_{i-1}, h_i) \in A.$$

These are finite lookups. Multiplying the decompositions gives

$$g_1 \cdots g_i = (t_1 \cdots t_i)s(H_i), \qquad 1 \le i \le T.$$

Indeed a prefix $A_{i-1}s(H_{i-1})$ times $a_i s(h_i)$ equals $A_{i-1}\alpha_{H_{i-1}}(a_i)c(H_{i-1}, h_i)s(H_i)$. All prefix products of the t_i lie in the abelian group A and are computed simultaneously by the preceding construction. Final lookups combine them with their corresponding H_i.

Each extension step increases depth by a universal constant and adds $q^{O(1)}(T+1)^2$ gates to one quotient-prefix circuit. More explicitly, for universal c, C, the raw size satisfies

$$F(G, T) \le F(G/A, T) + Cq^c(T+1)^2.$$

The quotient order decreases by at least a factor two at every step, so summing this geometric bound gives $F(G, T) \le Cq^c(T+1)^2$. Every new gate has fan-in and coefficients polynomial in q, T with universal degrees. No step enlarges previously constructed coefficients. The same uniform bound therefore holds for gate coefficient norms. Layering with identity gates and constant rails costs only a polynomial in raw size and depth. Since $\ell \le \log_2 q$, it is absorbed in the stated width bound and does not increase coefficient norms except by a universal polynomial factor.

Finally, choosing between two group letters requires only literals and constants in their one-hot encoding. The output map ϕ is a finite lookup. These additions preserve all bounds. □

Theorem 23.3 (Learning solvable-group programs)**.** *There are universal $c, C > 0$ such that the following holds. Let f be a length-T program over a solvable group of order $q \ge 2$ and derived length $\ell \ge 1$, and put $R = d + q + T + L + 2$. If*

$$L \ge C\ell \log R,$$

a sufficient sample size for the original critical posterior is

$$n \ge \frac{C}{\varepsilon}\left\{R^{c\ell} + \log(1/\delta)\right\}.$$

The prediction and width guarantees are those of Theorem 22.3. In particular, for bounded derived length the bound is jointly polynomial in group order, program length, and learner depth.

Proof. Apply Lemma 23.2. The parameter in Theorem 20.2 satisfies

$$T_c = \log(d+2) + h\log(\max\{d+1, C_T\} + B_T + h + L + 2) \le C\ell \log R.$$

The depth hypothesis ensures $L - h \ge L/2$ and $L - h \ge C_0 T_c$ after increasing the universal constant. Thus

$$J_L(f) \le V_* + C\frac{\ell^2 \log^2 R}{L}.$$

The exponential factor in Theorem 22.3 is at most $R^{C\ell}$. Its remaining polynomial prefactor is absorbed in the same power, since $\ell \le \log_2 q$ and the displayed rate bound is polynomial in R. □

Corollary 23.4 (Permutation programs of width at most four)**.** *There are universal $c, C > 0$ such that every length-T permutation branching program of width at most four has sufficient sample size*

$$n \ge \frac{C}{\varepsilon}\left\{(d+T+L+2)^c + \log(1/\delta)\right\}, \qquad L \ge C\log(d+T+L+2).$$

Its original sequential depth T is unrestricted.

Proof. A width-w permutation program is a group program over S_w, with its output determined by the action of the product on a fixed initial state. For $w \le 4$, the order and derived length are universally bounded. The normal series $\{e\} \triangleleft V_4 \triangleleft A_4 \triangleleft S_4$ has abelian successive quotients, so S_4 and its subgroups are solvable. Apply the theorem. Width one gives a constant target, already covered by Corollary 22.4. □

The permutation condition is material: the group proof does not cover arbitrary noninvertible transformation programs of width three or four. Nor does it prove a lower bound for nonsolvable groups. These are restricted positive results for deep computations, and leave general small-circuit learning unresolved.

Large Fourier variation can coexist with polynomial learning. For $b_i \in \{0,1\}$ and $x_i = (-1)^{b_i}$, consider the three-state program

$$f(x) = 2\mathbf{1}_{\{\sum_i b_i \equiv 0 \pmod 3\}} - 1.$$

With $\alpha = (1+\sqrt{3})/2 > 1$, it satisfies

$$A_F(f) \ge \tfrac{1}{3}\alpha^d - \tfrac{2}{3}.$$

Indeed the roots-of-unity formula gives, for a nonempty Fourier set of size k,

$$\widehat{f}(S) = \frac{4}{3\,2^d} 3^{k/2} \cos(d\pi/3 - k\pi/2),$$

with an additional $-1/3$ for the empty set. Retain the even k terms, use $|\cos(d\pi/3)| \ge 1/2$, and apply the binomial theorem. Nevertheless this is a program over the cyclic group of order three, so Theorem 23.3 gives polynomial sample complexity at logarithmic learner depth. Large Fourier variation alone is therefore insufficient for an adverse critical-prior learning bound.

24 Low-dimensional target representations with arbitrarily deep suffixes

A low-rank first layer gives another target class for which the critical learner has a polynomial sample bound without a restriction on the prior. The teacher's remaining depth and width can be arbitrary. The useful parameters are its projection rank and its product norm. The theorem below follows the variational route and has a sufficiently large width qualification. Corollary 2.6 gives its every-realizable-width version using the same target compiler.

Theorem 24.1 (Low-rank target representations)**.** *Suppose a Boolean target f has a finite tanh representation with $f(x)z(x) \ge 1$ on the full cube. Let $P \ge 1$ bound the product of its layer Frobenius norms, and suppose its first weight matrix, including the constant input coordinate, has rank at most $r \ge 1$. Put*

$$S = 1 + \log(d + L + 2) + r \log(P + r + 2).$$

There are universal constants C_0, C_1, C_2 such that

$$L \ge C_0 S \quad \Longrightarrow \quad J_L(f) \le V_* + C_1 S^2 / L.$$

Under that depth condition, a sufficient sample size for the original critical posterior is

$$n \ge \frac{C_2}{\varepsilon}\left\{(d + L + 2)^{C_2}(P + r + 2)^{C_2 r} + \log(1/\delta)\right\}.$$

The posterior conclusion and width-first quantifiers are those of Theorem 22.3. In particular, the width threshold is finite and independent of the realized sample, but is not quantified here. For fixed r, the bound is polynomial and independent of the teacher's depth and width.

Proof. Constant functions have constant-size threshold implementations, so consider a nonconstant target. There are at most P^2 relevant coordinates. Indeed, if coordinate i is relevant, a bichromatic cube edge and the network's Lipschitz bound give

$$\|W^1 e_i\|_2 \prod_{j\geq 2} \|W^j\|_F \geq 1.$$

Square and sum this inequality over relevant coordinates.

Freeze every irrelevant coordinate to an independent uniform sign. Its first-layer column is absorbed into the bias column. The expected squared Frobenius norm of the resulting first matrix is the original squared norm, so some freezing does not increase it. The new matrix's range is contained in the original range, and therefore its rank also does not increase. The function is unchanged. Write $k \leq P^2$ for the reduced input dimension, $m = k + 1$ and $R = \sqrt{m}$. We keep the original ambient d in the learner's prior normalization.

Factor the reduced first matrix as UV^{T}, where the columns of V are orthonormal and their number is at most r; decrease r for the construction if necessary. The network output has the form

$$z(x) = g(V^{\mathsf{T}}\bar{x}), \qquad \bar{x} = (x, 1),$$

where g is P-Lipschitz. This follows because $\|U\|_{\mathrm{op}} = \|W^1\|_{\mathrm{op}}$ and every tanh map is one-Lipschitz. Oppositely labeled inputs consequently obey

$$\|V^{\mathsf{T}}\bar{x} - V^{\mathsf{T}}\bar{y}\|_2 \geq 2/P. \tag{17}$$

Choose powers of two a, η with

$$\frac{1}{128P\sqrt{r}} < a \leq \frac{1}{64P\sqrt{r}}, \qquad \frac{a}{200m} < \eta \leq \frac{a}{100m}.$$

Round each entry of V^{T} to a multiple of η, obtaining B. On every cube input this changes each projection coordinate by at most $m\eta/2 \leq a/200$. Each coordinate of $B\bar{x}$ is an integer multiple of η, and a/η is an integer.

For each positive input choose a nearest center in the grid $a\mathbb{Z}^r$, with arbitrary deterministic tie breaking. Retain all centers obtained in this way. Their number satisfies

$$M \leq (CPR\sqrt{r})^r \leq (CP^2\sqrt{r})^r.$$

For each selected center c, define a box predicate by $|(B\bar{x})_j - c_j| \leq a$ for every j. Every positive input lies in its selected box. No negative input lies in a selected box: if y is a positive input selecting c and x lies in that box, their rounded projections differ in every coordinate by at most $3a/2$. Their true projections then differ in Euclidean norm by less than $2a\sqrt{r} \leq 1/(32P)$, contradicting (17). Thus the target is the disjunction of these box predicates.

The construction has bounded integer coefficients and internal margins. On the projection lattice, the lower and upper box inequalities have the same signs as

$$(B\bar{x})_j - (c_j - a - \eta/2), \qquad (c_j + a + \eta/2) - (B\bar{x})_j,$$

respectively. Every score has absolute value at least $\eta/2$. Multiplying by $2/\eta$ gives integer coefficients and margin at least one. Their norms are bounded by a universal polynomial in P, r, since $m \le P^2+1$ and $|c_j| \le R+a$. An AND of the $2r$ signed inequalities is implemented by the score $\sum_i t_i - (2r-1)$, and an OR of the M signed box outputs by $\sum_c u_c + (M-1)$. These also have integer margin at least one. Constant rails supply the offsets.

We have obtained a threshold circuit with three nonconstant stages, width at most $C(rM+P^2+1)$ and coefficient norms bounded by a fixed polynomial in P, r, M. Apply the controlled circuit seed theorem and the steering-after-prefix lemma. The logarithmic resource parameter is at most

$$C\{1+\log(d+L+2)+\log(M+2)+\log(P+r+2)\} \le CS.$$

The original input coordinates can be restored as zero columns; only their Gaussian normalization contributes the displayed d. This proves the rate and learner-depth bounds. Finally, Theorem 22.3 bounds the sample factor by a polynomial prefactor times e^{CS}, which has the form stated in the theorem after increasing C_2. □

Corollary 24.2 (An approximate first-layer bottleneck)**.** *The theorem also applies when the first matrix W^1 has a rank-r truncated singular-value decomposition A such that*

$$\sqrt{d+1}\,\|W^1 - A\|_{\mathrm{op}} \prod_{j\ge 2} \|W^j\|_{\mathrm{op}} \le 1/8,$$

with P replaced by $8P/7$.

Proof. The replacement changes every output by at most $1/8$ and does not increase the first Frobenius norm. Multiply the readout by $8/7$ to restore unit margin, and apply the theorem. □

This is a restriction on a target representation, not on prior support. It does not show that product norm alone permits a critical rate excess of order $\log^2(d+L+P)/L$. A product bound controls the sum of logarithms of layer stable ranks, but a margin-preserving rank-one truncation generally needs a much smaller discarded singular value than that budget alone supplies.

25 Optimizing the circuit prefix and blocking Boolean layers

The circuit construction can spend more than $1/L$ entropy in its prefix. Optimizing that expenditure for the sample bound improves the teacher-width and depth dependence. Blocking several Boolean layers into a truth-table computation gives a further improvement. These bounds still do not give polynomial sample complexity for general small deep circuits. The optimized budgets here are rate-based; Theorem 2.7 states the separate finite-width bound and its slightly different parameter U_b.

Proposition 25.1 (A prefix budget adapted to learning)**.** *Consider a layered Boolean threshold implementation of depth $q \ge 1$. Let C_j be its state width at stage j. Suppose the exact feature repair for every gate at that stage has entropy cost at most P_j/r when the incoming covariance dominates rHH^{T}, where $P_j \ge 1$. For $L \ge q$, put*

$$T = 1 + \log d + \log(2L) + \sum_{j=1}^{q} \log\left(\frac{4C_jP_jL}{q}\right).$$

There are universal constants C_0, C_1 such that

$$L \ge C_0 T \quad \Longrightarrow \quad J_L(f) \le V_* + C_1 T^2/L.$$

Consequently a sufficient sample size for the original critical learner is

$$n \geq \frac{C}{\varepsilon}\{e^{CT} + \log(1/\delta)\}.$$

The statistical conclusion has the same width-first quantifiers as Theorem 22.3.

Proof. In the proof of Lemma 20.4, replace the entropy budget $1/(qL)$ at each stage by $\eta = q/L$. Starting with $r_0 = 1/d$, use the rare-mixture weight $\theta_j = \eta r_{j-1}/P_j$ and tracked component

$$r_j = \frac{\eta r_{j-1}}{4C_j P_j}.$$

Inductively $r_j \leq 1$, so $0 < \theta_j \leq 1$. Entropy convexity gives cost at most η per stage. The prefix therefore costs at most q^2/L, has constant diagonal, and retains the target component with

$$\log(1/r_q) = \log d + \sum_{j=1}^{q} \log\left(\frac{4C_j P_j L}{q}\right).$$

In particular $T \geq q$. Increasing C_0 ensures $L - q \geq L/2$ and the hypothesis of Lemma 20.5. That lemma gives

$$J_L(f) \leq V_* + \frac{q^2}{L} + C\frac{(1 + \log(1/r_q))^2 + \log^2(2L)}{L - q} \leq V_* + C_1 T^2/L.$$

Apply Theorem 22.3. Its exponential term is e^{CT}; its polynomial prefactor is absorbed in that expression because $T \geq \log(d + L + 2) - C$. The entropy and compression bounds allow arbitrary nonnegative excess, so the prefix cost q^2/L need not be bounded by a universal constant. □

For a bounded-fan-in Boolean gate library, the repair constants P_j are universal. A depth-h, width-C implementation therefore has the rate and learning bounds above with

$$T \leq C_2\{\log(d + L + 2) + h\log((C + 2)(2 + L/h))\}.$$

This improves the earlier fixed-budget prefix factor $h\log(hL)$.

Theorem 25.2 (Boolean layer blocking)**.** *Suppose f has a layered AND/OR/NOT circuit of fan-in at most two, width C, and depth $h \geq 1$, including constant rails. For an integer $1 \leq b \leq h$, put $k = \lceil h/b \rceil$ and*

$$T_b = \log(d + L + 2) + k\{\log(C + 2) + 2^b + \log(2 + L/k)\}.$$

There are universal constants C_0, C_1, C_2 such that, if $L \geq C_0 T_b$, then

$$J_L(f) \leq V_* + C_1 T_b^2/L, \qquad n \geq \frac{C_2}{\varepsilon}\{e^{C_2 T_b} + \log(1/\delta)\}$$

is sufficient for posterior confidence-error at most ε with sample probability at least $1 - \delta$, at every sufficiently large learner width. The width threshold is independent of the realized sample.

Proof. Divide the h layers into k blocks. Each output of a block depends on at most 2^b signals at its entrance, so a full DNF uses at most 2^{2^b} terms. Use one threshold layer for those conjunctions and one for their disjunction. Constants are treated as constant gates. This gives exactly $q = 2k$ threshold stages after harmless identity padding. Their widths are at most $C_3(C + 1)2^{C_3 2^b}$.

Normalize the integer threshold scores to have margin one. Each conjunction has at most 2^b inputs and each disjunction at most 2^{2^b} inputs. Their coefficient norms are polynomial in those quantities. Lemma 19.1, applied to the gate's actual inputs and its constant rail, therefore gives $P_j \leq 2^{C_4 2^b}$. Apply Proposition 25.1. Its parameter is at most $C_5 T_b$. The stated depth hypothesis ensures $L \geq q$ and all other required inequalities. □

Corollary 25.3 (An adaptive choice of block length)**.** *In the theorem put*

$$a = \log(C+2) + \log(2 + L/h), \qquad T_* = \log(d + L + 2) + a\left(1 + \frac{h}{\log(a+2)}\right).$$

The same rate and sample bounds hold with T_b replaced by T_*, after increasing the universal constants, under $L \geq C_0 T_*$.

Proof. Choose $b = \min\{h, \max\{1, \lfloor \log_2(a+2) \rfloor\}\}$. Then $2^b \leq a+2$, $k \leq 1 + h/b$, and

$$\log(2 + L/k) \leq \log(2 + bL/h) \leq \log b + \log(2 + L/h).$$

Moreover b is at least a universal constant times $\min\{h, \log(a+2)\}$, and $\log b \leq Ca$. Substitution gives $T_b \leq CT_*$. □

Corollary 25.4 (General circuits and formulas)**.** *Let a bounded-fan-in Boolean circuit have depth h and size at most s, and put $S = \max\{16, d + s + 2\}$. If*

$$L \geq C \log(S + L)\left(1 + \frac{h}{\log\log(S + L)}\right),$$

then a sufficient sample size is

$$\frac{C}{\varepsilon}\left\{(S+L)^{C(1+h/\log\log(S+L))} + \log(1/\delta)\right\}.$$

For a formula of size s, its original depth can be replaced by $C \log S$ in these displays, after increasing constants. At learner depths polynomial in S, this yields $\exp\{O(\log^2 S/\log\log S)\}$ samples, apart from the statistical factors.

Proof. Layer a circuit by inserting identity gates along skipped wires; the width is $O(S)$. In the preceding theorem choose $b = \min\{h, \max\{1, \lfloor \log_2 \log(S+L) \rfloor\}\}$. The claimed parameter bound follows directly from the formula for T_b.

A binary formula of size m has a rooted subformula G with size between $m/3 - O(1)$ and $2m/3 + O(1)$. Replace that occurrence by zero and one to obtain outer formulas F_0, F_1. The identity

$$F = (G \text{ AND } F_1) \text{ OR } ((\text{NOT } G) \text{ AND } F_0)$$

permits parallel recursive construction of three formulas of size at most $2m/3 + O(1)$, followed by constant-depth logic. Simplify unary gates and constants first. The resulting circuit has polynomial size and depth $O(\log m)$ by the corresponding size and depth recurrences. Apply the circuit result to this balanced implementation. □

A fixed-width deterministic branching program of length t likewise has a polynomial-size bounded-fan-in circuit of depth $O(\log t)$: multiply its constant-dimensional Boolean transition matrices in a balanced tree. Each transition entry is an input literal or a constant, and repeated queries remain explicit at the leaves. This gives the same quasipolynomial sample bound. It does not justify treating the program's work register as the full hidden state of a sequential network; the queried input information cannot be discarded.

The bounds improve the teacher-depth dependence but remain quasipolynomial for general polynomial-size logarithmic-depth circuits. They do not establish a jointly polynomial theorem for arbitrary small deep circuits, formulas, or constant-width branching programs.

26 Finite-width entropy certificates on input witnesses

The finite-width argument uses forward conditional entropy and a finite set of input probes. It does not replace a continuous empirical measure by an absolutely continuous law of the same moments. Instead, a short revealed prefix of an exchangeable layer supplies a conditional one-row law whose moments approximate the empirical moments. This permits an additive action error, with a polynomial width requirement.

Lemma 26.1 (Exchangeable entropy decrement)**.** *Let $P_N \ll \mu^{\otimes N}$ be exchangeable, with $B = D(P_N\|\mu^{\otimes N}) < \infty$, and let $1 \le K \le N-2$. Independently choose k uniformly from $\{0,\dots,K-1\}$ and put $\nu = P_N(X_{k+1} \in \cdot \mid X_1,\dots,X_k)$. Then*

$$\mathbb{E}D(\nu\|\mu) \le \frac{B}{N-K}.$$

For every measurable $\phi : E \to [-1,1]$,

$$\mathbb{E}\left|\frac{1}{N}\sum_{i=1}^{N}\phi(X_i) - \nu(\phi)\right|^2 \le 8(K/N)^2 + \frac{2}{N-K} + 2\sqrt{\frac{2B}{K(N-K)}}. \tag{18}$$

The same assertions hold conditionally on a past variable, and may then be averaged over that variable.

Proof. Write $a_j = \mathbb{E}D(P_N(X_{j+1} \in \cdot \mid X_1,\dots,X_j)\|\mu)$. The entropy chain rule gives $\sum_{j=0}^{N-1} a_j = B$. Exchangeability and the conditional entropy chain rule give

$$a_{j+1} - a_j = I(X_{j+2}; X_{j+1} \mid X_1,\dots,X_j) \ge 0.$$

Consequently $a_j \le B/(N-j)$ and

$$\frac{1}{K}\sum_{j=0}^{K-1} I(X_{j+2}; X_{j+1} \mid X_1,\dots,X_j) \le \frac{B}{K(N-K)}.$$

Given the realized prefix, denote the relative entropy of the conditional pair law to the product of its marginals by I_k. The unrevealed empirical average has conditional mean $\nu(\phi)$ and conditional variance at most $1/(N-k) + \sqrt{2I_k}$: the diagonal terms are at most one, and Pinsker's inequality bounds each off-diagonal covariance by $\sqrt{2I_k}$. Replacing this average by the full empirical average changes it by at most $2K/N$. Squaring, averaging and using concavity of the square root proves (18). □

Let μ_N denote the original, unconditioned critical Gaussian network prior. A *function event* means an event determined by the network's real-valued input–output function. In the following lemma it is enough that the event be invariant under hidden-neuron permutations. Covariances are uncentered second moments throughout.

Lemma 26.2 (An approximate path from a finite-width event)**.** *Fix a nonempty set Z of m cube inputs. Let E be a function event on which every output at Z has absolute value at least $1/2$, and suppose $\mu_N(E) \ge e^{-NA} > 0$. Let $H = L - 1 \ge 1$, $0 < \Delta \le 1$ and $0 < \tau \le 1$. There is a universal polynomial p such that, whenever*

$$N \ge p(m, H, A+1, \Delta^{-1}, \tau^{-1}),$$

there is a realization in E, with empirical hidden kernels $Q_0, \dots, Q_H$ on Z and output z, and controlled laws $\rho_j \ll \mathcal{N}(0, Q_{j-1})$, for which

$$\left\| \mathbb{E}_{\rho_j}[\tanh(G)\tanh(G)^{\mathsf{T}}] - Q_j \right\|_{\max} \le \Delta,$$

$$\sum_{j=1}^{H} D(\rho_j \| \mathcal{N}(0, Q_{j-1})) + \frac{1}{2} z^{\mathsf{T}} Q_H^{\dagger} z \le A + \tau.$$

Here Q_0 is the original input kernel, $Q_0(x,x) \le 2$, $Q_j(x,x) \le 1$ for $j \ge 1$, and $z \in \operatorname{range}(Q_H)$.

Proof. Let $P = \mu_N(\cdot \mid E)$, so $D(P\|\mu_N) \le NA$. Disintegrate P in forward order by hidden weight blocks and then the readout. Let B_j and B_{out} be the resulting conditional relative entropies. Thus

$$\sum_j \mathbb{E}B_j + \mathbb{E}B_{\text{out}} \le NA.$$

Given all preceding weight blocks, the current block is exchangeable in its rows under P. Indeed, permuting these rows and the corresponding columns of the next block preserves the prior and E; all future blocks have been integrated out. Its reference law is the original product Gaussian row law. On Z a reference row has preactivation law exactly $\mathcal{N}(0, Q_{j-1})$.

At each hidden layer apply Lemma 26.1, using an independently randomized prefix length. Push the resulting conditional row law to its preactivations on Z, calling it ρ_j, and put $D_j = D(\rho_j \| \mathcal{N}(0, Q_{j-1}))$ and $\widetilde{Q}_j = \mathbb{E}_{\rho_j} \tanh(G)\tanh(G)^{\mathsf{T}}$. Data processing and the lemma give

$$\mathbb{E}\sum_j D_j \le \frac{\sum_j \mathbb{E}B_j}{N-K}$$

and, after summing the m^2 bounded covariance observables,

$$\mathbb{E}\sum_j \|\widetilde{Q}_j - Q_j\|_F^2 \le m^2 \left\{ 8H(K/N)^2 + \frac{2H}{N-K} + 2\sqrt{\frac{2HNA}{K(N-K)}} \right\}. \tag{19}$$

The last term uses Cauchy–Schwarz across layers.

We retain the actual random readout. Given the hidden blocks, its output under the reference prior is Gaussian with covariance Q_H/N. For $r = \frac{1}{2} z^{\mathsf{T}} Q_H^{\dagger} z$ and $0 < s < 1$, Gaussian integration, including singular Q_H, gives

$$\log \mathbb{E}_{\mu_N}\left[e^{(1-s)Nr} \mid \text{hidden blocks} \right] = \frac{\operatorname{rank}(Q_H)}{2} \log(1/s) \le \frac{m}{2} \log(1/s).$$

Entropy duality therefore implies

$$\mathbb{E}r \le \frac{\mathbb{E}B_{\text{out}}/N + (m/(2N))\log(1/s)}{1-s}.$$

In particular, if $K/N \le \eta$ and $s = \eta \le 1/16$,

$$\mathbb{E}\left(\sum_j D_j + r \right) \le \frac{A + (m/(2N))\log(1/\eta)}{1-\eta}. \tag{20}$$

The projected readout has rank at most m, so its additive term is $m\log(1/\eta)/(2N)$ before the displayed factor $1/(1-\eta)$. It can therefore be made smaller than any prescribed action tolerance by increasing width; no fixed positive readout penalty is introduced.

For explicit polynomial choices take

$$\eta = \min\{1/16, \tau/[32(A+1)]\}, \qquad \kappa = \min\{\eta, \Delta\sqrt{\eta}/(32m\sqrt{H})\}, \qquad K = \lfloor \kappa N \rfloor.$$

It suffices to require, for a large universal C,

$$N \geq C\left\{\kappa^{-1} + \frac{Hm^2}{\eta\Delta^2} + \frac{HAm^4}{\kappa\eta^2\Delta^4} + \frac{m\log(1/\eta)}{\tau}\right\}.$$

Then $1 \leq K \leq N-2$ and (19) is at most $\eta\Delta^2$. With probability at least $1-\eta$, all moment errors are at most Δ. Dividing the expectation in (20) by $1-\eta$ shows that some realization in this good event has action at most $A+\tau$. All preceding conditions are polynomial in the stated arguments. The kernels and output of this realization have the remaining properties by their empirical Gram representation. No conditioning on future weight blocks was used in defining the layer costs. □

Theorem 26.3 (Finite-width representation on an input witness)**.** *There is a universal C with the following property. Put*

$$\epsilon = (A - V_*)_+, \qquad R = (d+L+A+2)^C \exp(C\sqrt{L\epsilon}).$$

The universal constant can be chosen so that, for every set Z of m cube inputs and every function event E satisfying $\mu_N(E) \geq e^{-NA} > 0$ and $|z(x)| \geq 1/2$ on Z,

$$N \geq R(m+1)^C$$

implies that the sign restriction to Z of some realization in E is represented by an ordinary depth-L tanh network of width at most R. Its weights are arbitrary finite real numbers.

Proof. The empty witness is trivial. Apply Lemma 26.2 with $\tau = 1/(10L)$ and a moment tolerance Δ to be chosen. Write D_j for the extracted layer costs and r for its terminal readout cost. Lemma 13.1 gives

$$S := \sum_j \sqrt{D_j} \leq C\{\log(L+1) + \sqrt{L\epsilon+1}\},$$

provided $\Delta \leq c/L^4$.

For positive semidefinite m by m matrices whose entries differ by at most Δ, the square-root Gaussian coupling gives

$$\Delta_\infty(Q, \widetilde{Q}) \leq \|Q^{1/2} - \widetilde{Q}^{1/2}\|_F \leq \|Q - \widetilde{Q}\|_*^{1/2} \leq m\sqrt{\Delta}.$$

Here the middle inequality is the standard square-root trace inequality. Starting from the original input kernel, repeatedly transfer the extracted control law to the constructed preceding kernel using the controlled Gaussian-coupling transfer theorem. This does not increase its relative entropy. The constructed output kernels Q'_j obey

$$e_j := \Delta_\infty(Q_j, Q'_j) \leq (1 + \sqrt{2D_j})e_{j-1} + m\sqrt{\Delta}, \qquad e_0 = 0.$$

Thus $e_H \le Hm\sqrt{\Delta}\, e^{\sqrt{2}S}$. The readout transfer has cost at most r and changes each output by at most $\sqrt{2r}\, e_H$. Choose Δ small enough that this is at most $1/16$. Together with $\Delta \le c/L^4$, the choice can be made with

$$\Delta^{-1} \le (m+1)^C (d+L+A+2)^C e^{C\sqrt{L\epsilon}}.$$

Consequently the polynomial width requirement in the extraction lemma is bounded by $R(m+1)^C$, on enlarging C.

The resulting consistent controlled path has the same input kernel, layer costs at most D_j, and outputs with the original signs and margin at least $7/16$ on Z. Apply the anchored compression theorem (Theorem 15.1) to this path. Its proof applies on any finite set, with $\log|Z|$ in place of the cube union-bound factor d; since $|Z| \le 2^d$, the same stated bound using d applies. The uniform compression error is at most $3/32$, so all signs are preserved. Its width is at most $\mathrm{poly}(d, L, A+2)e^{CS}$, which is bounded by R. □

Corollary 26.4 (A finite-width exceptional-mass bound)**.** *There are universal $C, c > 0$ such that, for $A \ge 0$ and*

$$R = (d+L+A+2)^C \exp(C\sqrt{L(A-V_*)_+}),$$

one can choose a class $\mathcal{G}_A$ of Boolean functions with $\mathrm{VCdim}(\mathcal{G}_A) \le R$ for which, at every $N \ge R^C$,

$$\mu_N\{\textit{global margin, sign}(z) \notin \mathcal{G}_A\} \le \exp(dR\log 2 - NA).$$

The class is independent of the training sample and may be chosen independent of N.

Proof. Let M be the witness-compression width from Theorem 26.3, and let v be a universal upper bound for the VC dimension of depth-L, width-M tanh networks on the cube. The standard polynomial VC bound for tanh networks gives $v \le \mathrm{poly}(d, L, M)$. Take $m = \lfloor v \rfloor + 1$ and enlarge R so that $m \le R$. If $m > 2^d$, use the class of all Boolean functions and the conclusion is immediate.

Otherwise define $\mathcal{G}_A$ to contain those functions whose restriction to every m-element input set is represented by a width-M, depth-L tanh network. If this class shattered an m-element set, the representing networks would also shatter it, contrary to their VC bound. Hence its VC dimension is less than m. For each fixed m-element set Z, let E_Z be the event of a globally margined network whose sign restriction at Z has no such representation. The preceding theorem implies $\mu_N(E_Z) < e^{-NA}$ whenever $N \ge R^C$. A union bound over at most $(2^d)^m$ choices of Z gives the claimed exceptional-mass estimate. All constructions depend on A, d, L, but not on the training sample or on N. □

27 An intrinsic sample bound at every finite width

The exceptional-mass theorem yields a finite-width learning statement in terms of the target's actual prior mass. This formulation separates the capacity question from the further question of estimating that mass from a teacher implementation.

For the original unconditioned critical Gaussian prior μ_N, write

$$A_{N,f} = \{w : f(x)z_w(x) > 1/2 \text{ for every } x \in \{-1,1\}^d\}.$$

When $\mu_N(A_{N,f}) > 0$, define the intrinsic target cost

$$a_N(f) = -\frac{1}{N}\log \mu_N(A_{N,f}).$$

This is the raw prior mass, before global-margin conditioning. The learner conditions the original prior on an absolute output margin at every input and on the exact training signs, as throughout the paper.

Theorem 27.1 (Every-width learning from intrinsic target cost)**.** *There are universal $c, C > 0$ such that the following holds. Let $d \geq 1$, $L \geq 2$, $N \geq 1$, and suppose $\mu_N(A_{N,f}) > 0$. Put $a = a_N(f)$. For any input distribution D on the cube and $0 < \varepsilon, \delta < 1$, a sufficient sample size is*

$$n \geq \frac{C}{\varepsilon}\left\{[(d+L+a+2)(1+\log(1/\varepsilon))]^c \exp(C\sqrt{L(a-V_*)_+}) + \log(1/\delta)\right\}. \tag{21}$$

With probability at least $1-\delta$ over the iid training inputs, the posterior predictive probabilities satisfy

$$D\{x : p_N(f(x) \mid x) \leq 2/3\} \leq \varepsilon.$$

The theorem holds at the stated width N, with no sufficiently-large-width qualification. Its dependence on N is through the target's actual cost $a_N(f)$.

Proof. Set $A = a + 1/L$. By Corollary 26.4, there is a sample-independent class $\mathcal{G}_A$ with VC dimension at most R and an exceptional-mass estimate valid for $N \geq R^{C_0}$, where, after enlarging universal constants, we may use

$$R = (d+L+a+2)^{C_1} \exp(C_1\sqrt{L(a-V_*)_+}) \geq 2.$$

Indeed $\sqrt{L(A-V_*)_+} \leq \sqrt{L(a-V_*)_+} + 1$. Sauer's lemma on the 2^d cube inputs, or the trivial count when the VC dimension exceeds the domain size, gives

$$\log|\mathcal{G}_A| \leq C(d+1)R. \tag{22}$$

First suppose

$$N \geq R^{C_0}, \qquad N \geq L\{dR\log 2 + \log(6/\varepsilon)\}. \tag{23}$$

For every training sample, the raw mass of the posterior-conditioning event is at least $\mu_N(A_{N,f}) = e^{-Na}$. The posterior mass outside $\mathcal{G}_A$ is consequently at most

$$\frac{\exp(dR\log 2 - NA)}{e^{-Na}} = \exp(dR\log 2 - N/L) \leq \varepsilon/6.$$

For a fixed Boolean function g with test error greater than $\varepsilon/6$, its chance of fitting all n iid training labels is at most $e^{-n\varepsilon/6}$. A union bound using (22) shows that, with probability at least $1-\delta$, every consistent member of $\mathcal{G}_A$ has error at most $\varepsilon/6$, provided

$$n \geq \frac{6}{\varepsilon}\{C(d+1)R + \log(1/\delta)\}.$$

The posterior expected test error is then at most $\varepsilon/3$, including its exceptional mass. On an input where the correct-label posterior probability is at most $2/3$, the posterior error probability is at least $1/3$. Averaging therefore proves the asserted confidence criterion.

It remains to consider widths that do not satisfy (23). For a universal C_2, every such width satisfies

$$N \leq R^{C_2}\{1+\log(1/\varepsilon)\}.$$

The VC dimension of all depth-L, width-N tanh sign networks is at most a universal polynomial in d, L, N. Applying the finite-cube count once more, the logarithm of the number of Boolean sign functions in the entire posterior support is at most

$$\mathrm{poly}(d, L, N) \le R^{C_3}\{1 + \log(1/\varepsilon)\}^{C_3}.$$

A union bound over this whole support shows that every consistent function has test error at most $\varepsilon/6$ once (21) holds with sufficiently large universal constants. This implies the desired posterior guarantee as above. Substituting the displayed formula for R covers both width ranges and proves the theorem. □

Scope of the finite-width conclusion. A bound on the limiting rate $J_L(f)$ alone does not give the same bound on $a_N(f)$ at a specified finite width. The theorem is quantitative in the actual raw target cost. The following sections supply separate finite-width estimates of that cost for Fourier and circuit constructions. For example, if a separate argument gives $a_N(f) \le V_* + K \log^2 Q/L$ and $d + L + a_N(f) \le Q^K$ for a universal K, then the displayed sample bound is polynomial in Q, $1/\varepsilon$, and $\log(1/\delta)$ at that width. The target-based conclusions in Section 2 combine those estimates with a VC bound for smaller widths.

28 Finite-width realization of a planted scalar channel

The construction below gives a finite-width lower bound on the raw probability of a target. It follows a scalar component and diagonal variances, rather than approximating an entire hidden covariance matrix. In particular its width depends on the logarithm of the input-set size.

Let $X = \{-1, 1\}^d$, $m = |X|$, $H = L - 1 \ge 1$, and let $f : X \to \{-1, 1\}$. We first specify the controlled data used by the construction. All auxiliary variables below are absent from the final network; they are used only in its probabilistic construction.

Definition 28.1 (A scalar-channel program)**.** *The initial controlled row law, including auxiliary variables, has entropy at most D_1 relative to a reference law whose weight marginal is the original first-row Gaussian. Write $h(x) = \tanh(w \cdot \bar{x})$. It has a score b_1 and numbers $v_1 > 0, r_1 \ge 0$ satisfying*

$$\mathbb{E}b_1^2 = 1, \quad |b_1| \le B, \quad \mathbb{E}[b_1 h(x)] = \sqrt{v_1} f(x), \quad \mathbb{E}h(x)^2 = v_1 + r_1 \le 1.$$

For $2 \le j \le H$, a controlled law of (Z, U) has entropy at most D_j relative to a reference law whose Z marginal is $\mathcal{N}(0, 1)$. With an independent standard Gaussian G, put

$$T_{j,x} = \tanh\big(\sqrt{v_{j-1}} f(x) Z + \sqrt{r_{j-1}} G\big)\,.$$

There is a score $b_j = b_j(Z, U)$ such that

$$\mathbb{E}b_j^2 = 1, \quad |b_j| \le B, \quad \mathbb{E}[b_j T_{j,x}] = \sqrt{v_j} f(x), \quad \mathbb{E}T_{j,x}^2 = v_j + r_j \le 1,$$

where $v_j > 0$ and $r_j \ge 0$. Define the cost and accumulated entropy distortion by

$$A = \sum_{j=1}^{H} D_j + \frac{1}{8 v_H}, \qquad S = \sum_{j=2}^{H} \sqrt{D_j}, \qquad B \ge 1.$$

The auxiliary part of each reference law may depend on its Gaussian coordinate. Its marginal on the actual incoming weights is always the original Gaussian law.

Theorem 28.2 (A quantitative raw target-probability bound)**.** *There is a universal C such that a scalar-channel program has the following consequence. For every $\tau > 0$ and every integer width*

$$N \geq CB^4H^4e^{4S}(A+1)^6(1+\tau^{-1})^4\left[\log(2m)+\log(H+A+\tau^{-1}+3)\right], \tag{24}$$

the original critical Gaussian network satisfies

$$\mu_N\{f(x)z_w(x) > 1/2 \text{ for all } x \in X\} \geq e^{-N(A+\tau)}.$$

No conditioning or support restriction is added to this prior.

Proof. We give the metric, the finite realization, and the probability comparison separately. All expectations in the construction refer to the indicated controlled law.

Projection onto a bounded score. For bounded features $h(x)$ and a scalar score b with $\mathbb{E}b^2 = 1$, set

$$c(x) = \mathbb{E}[bh(x)], \qquad q(x) = \mathbb{E}h(x)^2, \qquad s(x) = \sqrt{q(x) - c(x)^2}.$$

For N independent rows let

$$M = \frac{1}{N}\sum_i b_i^2, \quad \widehat{c}(x) = \frac{N^{-1}\sum_i b_i h_i(x)}{\sqrt{M}}, \quad \widehat{q}(x) = \frac{1}{N}\sum_i h_i(x)^2, \quad \widehat{s}(x) = \sqrt{\widehat{q}(x) - \widehat{c}(x)^2}.$$

Suppose $|M-1|$, all errors in $N^{-1}\sum_i b_i h_i(x)$, and all errors in $\widehat{q}(x)$ are at most $\alpha \leq 1/4$. Then Cauchy–Schwarz and elementary square-root estimates give

$$\max_x \sqrt{(\widehat{c}(x) - c(x))^2 + (\widehat{s}(x) - s(x))^2} \leq 8\sqrt{\alpha}. \tag{25}$$

Indeed $|\widehat{c} - c| \leq 4\alpha$, $|c| \leq 1$, and $|(\widehat{q} - \widehat{c}^2) - (q - c^2)| \leq 13\alpha$. The inequality $(\sqrt{u} - \sqrt{v})^2 \leq |u - v|$ finishes the bound.

The coefficients $a_i = b_i/(N\sqrt{M})$ satisfy $\|a\|_2^2 = 1/N$. Consequently the actual empirical feature kernel has the exact decomposition

$$K(x,y) = \frac{1}{N}\sum_i h_i(x)h_i(y) = \widehat{c}(x)\widehat{c}(y) + R(x,y), \qquad R \succeq 0. \tag{26}$$

No approximate covariance inequality is needed.

Propagation of the component metric. Suppose the current finite features have coefficient $c(x) = a^T h(x)$, $\|a\|^2 = 1/N$, and residual diagonal $s(x)^2$ in (26). Compare them with the ideal coefficient $\sqrt{v}f(x)$ and residual diagonal r using

$$e = \max_x \sqrt{(c(x) - \sqrt{v}f(x))^2 + (s(x) - \sqrt{r})^2}.$$

Set $u = \sqrt{N}a$, a unit vector. Sample an incoming Gaussian coordinate Z with the prescribed controlled auxiliary law, and an independent standard Gaussian vector $\xi \in \mathbb{R}^N$. The actual incoming row is

$$w = \{Zu + (I - uu^T)\xi\}/\sqrt{N}.$$

Under its reference law this row is exactly $\mathcal{N}(0, I/N)$. Its preactivation at x is distributed as $c(x)Z + s(x)G$, with G standard Gaussian independent of (Z, U). For comparison with the ideal preactivation use the same Z, U, G at this one input.

Data processing and Gaussian entropy duality imply

$$\max\{1, \mathbb{E}Z^2\} \le e^{2\sqrt{D_j}}.$$

For completeness, writing $t = \mathbb{E}Z^2$, one has $D_j \ge (t - 1 - \log t)/2$; for $t \ge 1$, the latter is at least $(\log t)^2/4$. Independence and the zero mean of G therefore bound the preactivation RMS difference by $e^{\sqrt{D_j}}e$. The same bound holds after tanh.

Projection onto the common unit vector b_j in the coupled L^2 space is contractive. The difference of the norms of its two orthogonal residuals is bounded by their L^2 distance. Hence the population next coefficient and residual RMS, compared with $\sqrt{v_j}f$ and $\sqrt{r_j}$, have metric error at most $e^{\sqrt{D_j}}e$. This argument uses only one-input marginal couplings; the actual sampled network has its usual joint law on all inputs. No simultaneous choice of the comparison residual correlations is required.

Applying (25) at the new layer now gives the recursion

$$e_1 \le 8\sqrt{\alpha}, \qquad e_j \le e^{\sqrt{D_j}}e_{j-1} + 8\sqrt{\alpha}, \qquad e_H \le 8He^S\sqrt{\alpha}. \tag{27}$$

Uniform concentration at finite width. At every stage, conditional on the complete previous construction, the new rows and their scores are independent. The observables b^2, $bh(x)$, and $h(x)^2$ are bounded by $B^2, B, 1$ in absolute value. Hoeffding's inequality and a union bound over the $2m + 1$ observables, then over the H stages, show that all the required moment estimates hold with probability at least $1 - \kappa/2$ if

$$N \ge CB^4\alpha^{-2}\log\frac{8H(2m+1)}{\kappa}. \tag{28}$$

The same assertion follows by successive conditioning, since the bound is uniform in the realized previous layers. Stop using the controlled construction if a score Gram M is outside $[1/2, 3/2]$; thereafter use the original prior. This defines the construction on every outcome, and the good moment event never triggers stopping.

Choose

$$\kappa = \min\left\{\frac{1}{4}, \frac{\tau}{32(A+1)}\right\}, \qquad \alpha = \left(\frac{\kappa\sqrt{v_H}}{64He^S}\right)^2.$$

On the good event, (27) gives $e_H \le \kappa\sqrt{v_H}/8$. In particular the final coefficient satisfies $f(x)c(x) \ge \sqrt{v_H}(1 - \kappa/8)$ everywhere.

Readout and change of measure. When this final inequality holds, shift the Gaussian readout mean to ta, where $\|a\|^2 = 1/N$ is its final score coefficient vector and

$$t = \frac{1+\kappa}{2\sqrt{v_H}}.$$

The mean output then has signed margin at least $1/2 + \kappa/4$. At all other histories use zero readout mean. The conditional readout noise has variance $K(x,x)/N \le 1/N$ at every input. A Gaussian tail bound and union bound make its maximum absolute value smaller than $\kappa/8$ with probability at least $1 - \kappa/2$, provided $N \ge C\kappa^{-2}\log(4m/\kappa)$. The complete controlled construction, denoted Q_N, therefore produces the strict target event E_f with probability at least $1 - \kappa$.

Augment the original network prior with the same reference auxiliary kernels used above. Conditional on the complete past, each reference incoming weight row remains its original Gaussian, including under the history-dependent orthogonal decomposition. Chain rule and data processing

bound each controlled hidden block's relative entropy by ND_j; stopping can only reduce this bound. The shifted readout has relative entropy $Nt^2/2$. Consequently, after discarding all auxiliary variables,

$$D(Q_N\|\mu_N) \le N\left[\sum_j D_j + \frac{(1+\kappa)^2}{8v_H}\right] \le N(A+3A\kappa).$$

Binary relative entropy for the event E_f gives

$$-\frac{1}{N}\log \mu_N(E_f) \le \frac{A+3A\kappa+N^{-1}\log 2}{1-\kappa} \le A+\tau$$

when $N \ge 8/\tau$. Finally $v_H^{-1} \le 8A$ and the definition of κ show that (24) implies all three numerical width conditions above. This proves the theorem. □

Corollary 28.3 (Beginning with an approximate finite prefix)**.** *Suppose a controlled finite prefix has relative entropy at most NE against the corresponding original Gaussian blocks. With probability at least $1-\kappa/4$, suppose its feature kernel has a measurable coefficient $c(x) = a^T h(x)$, $\|a\|^2 = 1/N$, and residual diagonal $s(x)^2$, such that*

$$\max_x \sqrt{(c(x)-\sqrt{v_0}f(x))^2 + (s(x)-\sqrt{r_0})^2} \le \frac{\kappa\sqrt{v_M}}{64e^S}.$$

Here a scalar-channel continuation has $M \ge 1$ remaining hidden stages, starts at v_0, r_0, ends at v_M, has score bound B, and has costs $D_1, \dots, D_M$. Put

$$A = E + \sum_{j=1}^M D_j + \frac{1}{8v_M}, \qquad S = \sum_{j=1}^M \sqrt{D_j}, \qquad \kappa = \min\{1/4, \tau/[32(A+1)]\}.$$

For widths satisfying (24) with H replaced by M, after a universal change of its constant, the full network has raw target probability at least $e^{-N(A+\tau)}$. The prefix and its ideal comparison parameters may depend on N, provided these displayed bounds hold uniformly.

Proof. On a bad prefix use the original subsequent prior. On a good prefix run the controlled scalar continuation. Its metric recursion gives

$$e_M \le e^S\{e_0 + 8M\sqrt{\alpha}\}.$$

Use $\sqrt{\alpha} = \kappa\sqrt{v_M}/(128Me^S)$ and assign failure probability $\kappa/4$ to each of the prefix, subsequent moment estimates, and readout noise. Then $e_M < \kappa\sqrt{v_M}/8$, and total target success probability is at least $1-\kappa$. The joint entropy is at most $N(A+3A\kappa)$ by chain rule. The proof of Theorem 28.2 now applies without any other change. □

Remark 28.4 (Reducing a tracked component)**.** *Write $\psi_r(t) = \mathbb{E}\tanh(t+\sqrt{r}G)$ and let $v_{\max} = \mathbb{E}\psi_r(\sqrt{v}Z)^2 > 0$. The full conditional mean component uses the score*

$$b = \psi_r(\sqrt{v}Z)/\sqrt{v_{\max}}.$$

Any smaller prescribed component $0 < v' \le v_{\max}$ can instead be obtained with the unit score

$$b = \sqrt{\frac{v'}{v_{\max}}}\frac{\psi_r(\sqrt{v}Z)}{\sqrt{v_{\max}}} + \sqrt{1-\frac{v'}{v_{\max}}}\,U,$$

where U is an independent random sign. Its coefficient is $\sqrt{v'}f$, and $|b| \le 1+(v')^{-1/2}$. The auxiliary sign adds no entropy. This implements every deliberate reduction of a tracked component in the scalar steering construction.

Corollary 28.5 (Finite-width target mass for Fourier seeds)**.** *There are universal $c, C_0, C_1 > 0$ with the following property. Put*

$$A_F(f) = \sum_T |\widehat{f}(T)|, \qquad R = (d + L + 2)A_F(f).$$

If $L \geq C_0 \log R$, then for every $N \geq R^c$ the original critical prior satisfies

$$\mu_N(E_f) \geq \exp\left[-N\left\{V_* + C_1 \frac{\log^2 R}{L}\right\}\right].$$

In particular the sufficient width is polynomial for every parity, independently of its degree. This corollary is a target-probability lower bound, not by itself a posterior learning theorem.

Proof. Use the controlled Fourier mixture with exact mean f/Z_f, $Z_f \leq 8(d+1)A_F(f)$, and common diagonal. Its rare component has probability $\theta = (D_*L)^{-1}$, where $D_* = C(d+1)^3 \log^2(d+2)$. Retain its Bernoulli mixture tag and use $b_1 = \mathbf{1}_{\{\text{rare}\}}/\sqrt{\theta}$. Then $\mathbb{E}b_1^2 = 1$, the coefficient is exactly $\sqrt{\theta} f/Z_f$, and $v_1 = \theta/Z_f^2$. Using the same component tags in the reference law bounds the joint row entropy by $\theta D_* \leq 1/L$.

The established cooling, holding, prescribed growth, purification, and scalar continuation form a scalar-channel program. Use the conditional-mean scores at each stage and Remark 28.4 for any deliberate reduction of the tracked component. Its total cost satisfies

$$A \leq V_* + C \log^2 R/L.$$

All tracked positive components are at least R^{-C}: the first one is inverse polynomial; cooling retains at least a constant times the cooling scale times that component; the holding value is $v_0 = \delta e^{-CS_0}$ with $S_0 = O(\log R)$ and $\delta \geq c/L$; growth and purification retain at least v_0. The final scalar path is increasing from $p \geq c/L$ and has $O(1/p)$ steps. Its stable lift retains a fixed positive fraction of every scalar state, because purification leaves residual $O(\delta L^{-4})$. Thus its tracked components are also inverse polynomial. All scores are consequently bounded by $B \leq R^C$.

The square-root entropy budget also applies when the numbers D_j are upper bounds on the actual feature entropies. Indeed its proof uses only scalar dissipation, Bellman suffix lower bounds, and the total cost; all three remain valid with these larger nonnegative costs. Applying that proof to this program gives

$$S \leq C \log(L+1) + C\sqrt{L(A - V_*)_+} \leq C \log R.$$

Take $\tau = 1/L$ in Theorem 28.2. Its sufficient width is at most R^c after enlarging a universal exponent, and its extra $1/L$ cost is absorbed into the displayed constant C_1. □

The finite-width bound is close to the cost of the explicit scalar-channel construction. It does not assert a bound within $1/L$ of the unknown minimum rate $J_L(f)$ for an arbitrary target. A collective bound on competing functions is a separate ingredient in any learning consequence.

29 Finite-width circuit seeds from Gaussian mean shifts

A Boolean circuit can be planted into a finite Gaussian network using small, prescribed groups of neurons. Every weight retains its original Gaussian variance. Only the means change, and the total relative entropy of these changes is small. Saturation corrects the approximation error at each gate; no exact finite-state repair is needed.

Lemma 29.1 (A finite-width Boolean circuit prefix)**.** *Let $H^{(0)}(x) = (1, x_1, \dots, x_d)$ and let $H^{(j)}(x) \in \{-1,1\}^{C_j}$, $1 \le j \le h$, be a layered threshold circuit with a constant wire at every level. Suppose*

$$H_i^{(j)}(x) = \operatorname{sign}(w_{j,i} \cdot H^{(j-1)}(x)), \qquad |w_{j,i} \cdot H^{(j-1)}(x)| \ge 1, \qquad \|w_{j,i}\|_2 \le B,$$

where $B \ge 1$. Put $C = \max_j C_j$ and $W = \max(d+1, C)$. For $L \ge h+1$, $0 < \delta < 1$, and $0 < \eta \le (4B\sqrt{W})^{-1}$, define

$$a = 4 + \log(8/\eta), \qquad p_0 = 1/d, \qquad p_j = \frac{p_{j-1}}{8C_j h L a^2 B^2}, \qquad k_j = \lceil N p_j \rceil.$$

If

$$Np_h \ge 16\eta^{-2}\left[d + \log\frac{8h(C+1)}{\delta}\right], \tag{29}$$

there is a product Gaussian law P on the first h hidden weight matrices such that

$$D(P\|\mu_N^{\text{prefix}}) \le \frac{N}{8L}.$$

Here the covariance of every weight row is unchanged from the critical prior. Under P, with probability at least $1-\delta$, the following hold simultaneously on every input in $\{-1,1\}^d$. Each circuit wire i at layer j has a prescribed group of k_j neurons whose empirical mean $m_{j,i}$ satisfies

$$\|m_{j,i} - H_i^{(j)}\|_\infty \le \eta.$$

If $q_j(x) = N^{-1}\|h_j(x)\|_2^2$, set

$$\theta_j = C_j k_j / N, \qquad \bar{q}_0 = (d+1)/d,$$
$$\bar{q}_j = (1-\theta_j)F(\bar{q}_{j-1}) + \theta_j, \qquad F(q) = \mathbb{E}\tanh^2(\sqrt{q}Z).$$

Then $\|q_j - \bar{q}_j\|_\infty \le 2j\eta$. If a final wire is f, there is a deterministic unit vector $u \in \mathbb{R}^N$ such that, writing $p = k_h/N$ and $c(x) = N^{-1/2}\langle u, h_h(x)\rangle$,

$$|c(x) - \sqrt{p}f(x)| \le \sqrt{p}\eta, \tag{30}$$
$$|q_h(x) - c(x)^2 - (\bar{q}_h - p)| \le (2h+3)\eta, \qquad \bar{q}_h - p \ge 0. \tag{31}$$

Consequently its scalar-channel metric error relative to the ideal seed $(v, r) = (p, \bar{q}_h - p)$ is at most $\sqrt{(2h+4)\eta}$.

Proof. Choose disjoint deterministic groups of size k_j, one for each wire at layer j. They fit within the width: the hypothesis gives $Np_j \ge 1$, and hence

$$\theta_j \le 2C_j p_j \le \frac{1}{4hLa^2B^2} \le \frac{1}{4}.$$

Rows outside these groups retain their original Gaussian law. A row in the first-layer group for gate i has mean $aw_{1,i}$ and covariance I/d. At layer $j \ge 2$, the row mean is $aw_{j,i,g}/k_{j-1}$ on each coordinate of the preceding group for wire g, and zero on the remaining coordinates; its covariance is I/N. These weight means are deterministic, so all the weight rows remain independent under P. The resulting preactivation mean is exactly

$$a\sum_g w_{j,i,g} m_{j-1,g}(x).$$

The relative entropy of each controlled row is at most $a^2B^2/(2p_{j-1})$, including the first layer with $p_0 = 1/d$. For later layers this follows directly from

$$\frac{N}{2}\sum_g k_{j-1}\left(\frac{aw_{j,i,g}}{k_{j-1}}\right)^2 \le \frac{a^2B^2}{2p_{j-1}}.$$

Gaussian additivity therefore gives

$$\frac{1}{N}D(P\|\mu_N^{\text{prefix}}) \le \sum_{j=1}^{h}\frac{\theta_j a^2B^2}{2p_{j-1}} \le \sum_{j=1}^{h}\frac{1}{8hL} = \frac{1}{8L}.$$

This bound does not depend on any concentration event.

Suppose the preceding empirical gate means have error at most η. Every controlled preactivation then has the correct sign and mean magnitude at least

$$a(1 - B\sqrt{W}\eta) \ge 3a/4 \ge a/2.$$

Its variance is at most two in the first layer and at most one thereafter. For a Gaussian variable V of variance at most two whose mean has absolute value at least $a/2$, Gaussian exponential moments and $1 - \tanh t \le 2e^{-2t}$ give

$$|\mathbb{E}\tanh V - \operatorname{sign}(\mathbb{E}V)| \le 2e^{-a+4} \le \eta/4, \qquad 1 - \mathbb{E}\tanh^2 V \le 4e^{-a+4} \le \eta/2.$$

Thus each new gate mean is close to its Boolean value independently of the size of the preceding error, provided that error stays below the stated gate-margin threshold.

Conditionally on the preceding layers, the rows in each group are independent and identically distributed. Hoeffding's inequality, followed by a union over all 2^d inputs, bounds the chance that any empirical gate mean differs from its conditional mean by more than $\eta/2$ by

$$2C_j 2^d \exp(-k_j\eta^2/8).$$

The resulting error relative to the Boolean wire is less than η. The squared features are independent, bounded in $[0,1]$, and need not have identical laws. Their empirical average differs from its conditional mean by at most η everywhere except with probability at most $2\cdot 2^d\exp(-2N\eta^2)$.

The function F is one-Lipschitz on $[0,\infty)$: its heat derivative is

$$F'(q) = \mathbb{E}[1 - 4\tanh^2(\sqrt{q}Z) + 3\tanh^4(\sqrt{q}Z)],$$

whose absolute value is at most one. The conditional diagonal mean is within $\theta_j\eta/2$ of $(1-\theta_j)F(q_{j-1}(x)) + \theta_j$. Therefore, on the concentration events,

$$|q_j(x) - \bar{q}_j| \le |q_{j-1}(x) - \bar{q}_{j-1}| + 3\eta/2 \le 2j\eta.$$

Conditional union bounds over the layers and (29) prove that all these events hold with probability at least $1-\delta$.

Take u to be the target-group indicator divided by $\sqrt{k_h}$. Then $\|u\|_2 = 1$ and $c = \sqrt{p}\,m_{h,\text{target}}$, giving (30). Also $|c^2 - p| \le 3p\eta$, which proves (31); its deterministic residual is nonnegative because $\bar{q}_h \ge \theta_h = C_h p \ge p$. Finally $(\sqrt{u} - \sqrt{v})^2 \le |u - v|$ gives the stated metric bound. Orthogonally projecting the next Gaussian weight row onto u yields an exact independent Gaussian score/residual decomposition, not an approximate positive-semidefinite covariance inequality. □

Remark 29.2 (Seed-relative accuracy and the circuit depth parameter)**.** *Put*

$$R = d + C + B + h + L + 2, \qquad T = \log(d+2) + h\log(C + B + h + L + 2).$$

If subsequent scalar steering needs $\eta \le cR^{-b}p_h^b$ for some fixed universal b, choose $a = KT$ and $\eta = 8e^{4-a}$ with a sufficiently large universal K. The gate-margin condition then holds. Moreover, since $T \ge h\log(h+2)$,

$$h\log T = h\log h + h\log(T/h) \le 2T.$$

Consequently

$$\log(1/p_h) = \log d + \sum_{j=1}^{h} \log(8C_j hLa^2B^2) \le C_0(1 + \log K)T.$$

The required accuracy inequality follows by taking K sufficiently large: the coefficient in $\log(1/\eta)$ is linear in K, whereas the coefficient on the right is logarithmic in K. The same reasoning allows any fixed requirement $\eta \le e^{-c_1 T}$. For $\delta^{-1} \le e^{c_2 T}$ with a fixed universal c_2, the width condition (29) is therefore satisfied by $N \ge e^{C_1 T}$, for a universal constant C_1 depending on the required fixed accuracy exponents. In particular this is $R^{O(h)}$. Arbitrary smaller failure probabilities contribute the additional factor $1 + \log(1/\delta)$ in the explicit width condition.

30 Finite-width target probability for circuit teachers

The finite circuit prefix and scalar-channel realization together give a quantitative width threshold for the original critical prior. The dependence on teacher depth is explicit; no stability restriction is imposed on the learner's support.

Corollary 30.1 (Finite-width target mass for a threshold circuit)**.** *Suppose $f : \{-1,1\}^d \to \{-1,1\}$ is the final wire of the depth-h layered threshold circuit in Lemma 29.1. Its maximum hidden width is C, its gate weight norms are at most $B \ge 1$, and its gate margins on Boolean inputs are at least one. Put*

$$T = \log(d+2) + h\log(C + B + h + L + 2).$$

There are universal constants $c_0, c_1, c_2 > 0$ such that, if $L - h \ge c_0 T$, then for every integer $N \ge \exp(c_1 T)$ the original critical Gaussian prior satisfies

$$\mu_N\{f(x)z_w(x) > 1/2 \text{ for all } x \in \{-1,1\}^d\} \ge \exp\left[-N\left\{V_* + c_2\frac{T^2}{L}\right\}\right].$$

The sufficient width is polynomial in d, C, B, L at every fixed teacher depth h. All statements concern the raw unrestricted prior.

Proof. We keep track of the accuracy needed by the continuation, since it is not enough merely to plant approximately correct gate signs. Choose the prefix parameters

$$a = KT, \qquad \eta = 8e^{4-KT},$$

where the universal constant K will be fixed below. The prefix construction has entropy at most $N/(8L)$ and, except on an event of a prescribed small probability, ends at a kernel with an exact score component whose metric error relative to

$$(v_0, r_0) = (p, \bar{q}_h - p), \qquad p = k_h/N \in [p_h, 2p_h],$$

is at most $\sqrt{(2h+4)\eta}$. The ideal diagonal is at most one. These assertions hold uniformly at all widths satisfying the prefix bound, even though p and $\bar{q}_h$ may depend on N.

The estimate in Remark 29.2 gives

$$S_0 := 1 + \log(1/p) + \log(L+2) \le C(1+\log K)T. \tag{32}$$

Apply the scalar cooling, holding, growth, purification, and final scalar-continuation construction to this ideal seed. There are $M = L - h - 1$ remaining hidden layers. Increasing c_0 by a universal amount makes $M \ge L/2$ and $M \ge CS_0$, so that construction fits into exactly the available number of layers. With the prefix cost included, it has

$$A := \frac{1}{8L} + \sum_{j=1}^{M} D_j + \frac{1}{8v_M} \le V_* + C\frac{S_0^2}{L}. \tag{33}$$

All tracked components in this program are at least e^{-CS_0}. Indeed cooling loses only a polynomial factor in its scale; holding uses a component δe^{-CS_0} with $\delta \ge c/L$; growth and purification retain that lower bound; and the final increasing scalar path starts at $p_{\mathrm{sc}} \ge c/L$. The purified residual is $O(\delta L^{-4})$, so its stable lift retains a fixed fraction of every scalar state. The bounded-score construction of Remark 28.4 therefore gives a common score bound $B_{\mathrm{sc}} \le e^{CS_0}$.

The square-root entropy estimate applies to this continuation alone, with its stated upper entropy costs D_j. To check the initial condition, its ideal starting diagonal is at most one; the Bellman suffix lower bound and total cost (33) give the same prefix budget used in the occupation proof. Thus no Gaussian assumption on the initial seed is needed. Consequently

$$S := \sum_{j=1}^{M} \sqrt{D_j} \le C\log(L+2) + C\sqrt{L(A-V_*)_+} \le CS_0. \tag{34}$$

Set $\tau = 1/L$ and $\kappa = \min\{1/4, \tau/[32(A+1)]\}$. By (32)– (34), all factors in

$$\frac{\kappa\sqrt{v_M}}{64e^S}$$

are bounded below by $\exp[-C(1+\log K)T]$. Fix K sufficiently large that

$$\sqrt{(2h+4)\eta} \le \frac{\kappa\sqrt{v_M}}{64e^S}$$

holds uniformly. This is possible because the logarithmic decay of η is linear in K, whereas the required exponent grows only as $1+\log K$. The gate-margin condition on η is absorbed by the same choice. Choose the prefix failure probability at most $\kappa/4$, using a uniform lower bound on κ if its ideal parameters depend on N.

After fixing these universal constants, the prefix width condition and the sufficient width in Corollary 28.3 are both at most $\exp(c_1 T)$. In the latter bound this follows from $B_{\mathrm{sc}} \le e^{CS_0}$, $S \le CS_0$, $M \le L$, $A \le V_* + CS_0^2/L$, and $\log|X| = d \le e^T$. That corollary yields raw target mass at least $e^{-N(A+1/L)}$. Finally (32) absorbs both the fixed K and the extra $1/L$ into c_2T^2/L. □

The gate-margin and gate-norm assumptions are automatically satisfied with bounded constants for fan-in-two AND and OR gates and unary NOT gates, after including constant wires. Any other fixed Boolean gate basis can first be compiled into this basis with constant overhead. More general threshold gates are covered with the displayed dependence on B. Compiling or grouping a circuit before applying the corollary is allowed, but its actual resulting depth, width, and gate norms must be substituted in T.

31 Removing threshold-weight magnitudes from the teacher parameters

A Boolean threshold gate may be specified using arbitrarily large or arbitrarily precise real coefficients. These numerical parameters are unnecessary for the learning bound. The gate's Boolean input set admits an equivalent separator with polynomially many bits, and constant-depth majority arithmetic evaluates that separator using polynomially bounded gate coefficients. This is an existence result; it does not require an algorithm that receives or computes with the original real coefficients.

Lemma 31.1 (Small-bit integer separators)**.** *Let $S \subseteq \{-1,1\}^k$, where $k \geq 0$, and suppose $g : S \to \{-1,1\}$ agrees with an affine real threshold, with either fixed choice of sign at zero. There are integers $b_0, \dots, b_k$ such that*

$$g(u)\left(b_0 + \sum_{j=1}^{k} b_j u_j\right) \geq 1 \quad (u \in S), \qquad |b_j| \leq (k+1)^{(k+1)/2}.$$

In particular, each coefficient has $O((k+1)\log(k+2))$ bits, with a universal constant. The assertion applies to the full Boolean cube or to only the states realized by earlier gates.

Proof. The empty set is immediate, so assume S is nonempty. If any original scores vanish, change the bias by a sufficiently small quantity whose sign is the specified sign at zero. Finiteness of S ensures that this does not change the sign of any nonzero score. All desired inequalities then become strict. Scale the coefficients so that every signed score is at least one.

Put $m = k+1$ and form the matrix A whose row for $u \in S$ is $g(u)(1, u_1, \dots, u_k)$. Thus A has entries in $\{-1,1\}$ and $Aw \geq \mathbf{1}$ is feasible. Let $r = \operatorname{rank} A$, so $1 \leq r \leq m$, and retain r linearly independent columns to obtain A_J. These columns span the column space of A, so $A_J v \geq \mathbf{1}$ remains feasible.

We verify directly that this latter system has a feasible point with r linearly independent tight inequalities. Start at any feasible point. If its tight rows have rank less than r, choose a nonzero direction orthogonal to all of them. Since A_J has column rank r, some row has a nonzero inner product with the direction. Reverse the direction if necessary so that at least one row has negative inner product. Every such row is currently slack. Move until the first of these finitely many rows becomes tight: the permitted distance is the positive minimum of its slack divided by the negative directional derivative. All old tight rows stay tight, feasibility is preserved, and a new tight row is independent of them because it is not orthogonal to the chosen direction. After at most r iterations the tight rows have rank r.

Choose r independent tight rows, forming a nonsingular integer matrix $M \in \{-1,1\}^{r \times r}$. At the resulting feasible point, $Mv = \mathbf{1}$. Cramer's rule gives

$$v_j = \frac{\det M_j}{\det M},$$

where M_j replaces column j by $\mathbf{1}$. Multiply the feasible coefficient vector by $|\det M|$, and put zero in the omitted coordinates. The retained coefficients are the integers $\operatorname{sign}(\det M)\det M_j$. Their signed scores are at least $|\det M| \geq 1$. All entries of M_j still have absolute value one. Hadamard's determinant bound therefore gives

$$|\det M_j| \leq r^{r/2} \leq m^{m/2},$$

as required. This also covers $k = 0$, where $r = m = 1$. □

Lemma 31.2 (A weight-independent threshold compiler)**.** *There is a universal integer a with the following property. Suppose a Boolean target on $\{-1,1\}^d$ has a layered real threshold circuit of depth $h \geq 1$ and width at most $C \geq 1$. Threshold gates may have arbitrary real coefficients and biases, and may use either fixed tie convention. Put $K = d + C + h + 2$. The target has a layered Boolean threshold implementation with depth at most ah, width at most K^a, and Euclidean norm of every gate coefficient vector at most K^a. Every gate has signed score at least one on its realized Boolean states. Constant wires and layering overhead are included. The bounds are independent of all original coefficient magnitudes and precisions.*

Proof. Apply Lemma 31.1 separately to every gate, either on its full Boolean input cube or on its realized input states. The maximum fan-in is at most $\max\{d, C\}$, after combining duplicate wires; biases are included in the constant coefficient. Thus every gate has an equivalent integer description with polynomially many bits in K. Choosing these descriptions is nonuniform and requires no representation of the original real numbers.

Use the standard constant-depth majority-arithmetic theorem: iterated addition and comparison of polynomially many polynomial-bit integers have polynomial-size, constant-depth circuits of unweighted majority and Boolean gates, with universal depth and polynomial degree. This is part of the usual arithmetic threshold-circuit toolkit; the stronger division and iterated-multiplication theorem is proved in Hesse–Allender–Barrington [2]. To apply it here, hardcode the integer coefficients in binary. Each input bit selects either zero or the corresponding fixed summand. Sum positive and negative parts separately and compare them, including the constant term. All intermediate integers have polynomially many bits. This construction uses the binary coefficients as input constants to the arithmetic circuit; it does not put exponentially large bit-position weights on the new threshold gates.

Replace each original gate by this constant-depth majority circuit. All gates from one original level are evaluated in parallel. The total depth is $O(h)$, and the total number of gates and wires is a universal polynomial in K. Insert identity gates where necessary to obtain a layered circuit and propagate a constant wire. This costs only a polynomial in the size and depth.

Finally, every unweighted majority, AND, OR, or NOT gate has an integer threshold realization with coefficients polynomially bounded in its fan-in. Put its threshold halfway between consecutive allowed integer sums and multiply the score by two if needed. On all Boolean inputs its score has absolute value at least one and the prescribed sign. Its bias is supplied by the constant wire, so its coefficient Euclidean norm is polynomial in the total circuit size. Increasing the universal exponent a proves the stated simultaneous bounds. □

This replacement acts on Boolean states. It does not remove a weight bound from a real-valued tanh teacher: preserving intermediate real activations is a different approximation problem.

32 A sample-dependent posterior ceiling near the common rate

The Fourier construction supplies a low-rate, globally separated extension of *every* consistently labeled sample. Thus it gives an actual-posterior comparison, rather than only a representation bound.

Theorem 32.1 (Near-baseline critical posterior ceiling)**.** *There are universal constants C_0, C_1 such that the following holds. Let S be any consistently labeled sample of at most $n \geq 1$ Boolean inputs, and suppose*

$$L \geq C_0 \log(2 + d + n + L).$$

Put

$$U_{d,n,L} = V_* + C_1 \frac{\log^2(2+d+n+L)}{L}.$$

For the original globally margin-conditioned critical posterior,

$$\Pi_{N,1}\{g : J_L(g) > U_{d,n,L} \mid S\} \longrightarrow 0 \qquad (N \to \infty).$$

The dimension, depth, and realized sample are fixed in this limit. The constants and the rate ceiling are uniform over the sample.

Proof. Let $p_1, \dots, p_m$ be the distinct negatively labeled sample points, where $m \le n$. The Boolean function

$$g_S(x) = 1 - 2\sum_{j=1}^{m} \mathbf{1}_{\{x=p_j\}}$$

agrees with the sample, including any repeated inputs. A point indicator has normalized Fourier ℓ^1 norm exactly one, since

$$\mathbf{1}_{\{x=p\}} = 2^{-d} \prod_{i=1}^{d} (1 + p_i x_i).$$

Therefore $A(g_S) \le 1 + 2m \le 1 + 2n$. Theorem 18.1 gives $J_L(g_S) \le U_{d,n,L}$ after adjusting universal constants.

At fixed d, there are finitely many Boolean functions. Their exact prior rates and the posterior-rate formula are given by Corollary A.2. The denominator's rate is at most $J_L(g_S)$. If any consistent functions have rate greater than $U_{d,n,L}$, the finite set of their rates has a strictly positive gap above this ceiling. Their total posterior mass consequently tends to zero. If there are none, the assertion is immediate. This argument places no restriction on mixtures of tied minimizers. □

Combining this ceiling with the universal lower bound gives the limiting rate interval

$$V_* - C/L \le J_L(g) \le V_* + C \frac{\log^2(2+d+n+L)}{L}$$

for posterior-typical functions, in the stated fixed-domain width limit. Small remaining differences inside this interval can still determine learning.

Corollary 32.2 (A target-relative obstruction to learning)**.** *For a Boolean target f, input distribution $\mathcal{D}$, and $0 < \epsilon < 1$, define*

$$J_{L,\epsilon}(f; \mathcal{D}) = \inf\{J_L(g) : \Pr_{\mathcal{D}}(g \ne f) < \epsilon\}.$$

Under the depth hypothesis of Theorem 32.1, if

$$J_{L,\epsilon}(f; \mathcal{D}) > U_{d,n,L},$$

then every sample of at most n inputs labeled by f yields width-first posterior Gibbs error with lower limit at least ϵ. Its correct-label probabilities $p_N(x)$ satisfy

$$\liminf_{N\to\infty} \Pr_{\mathcal{D}}\{p_N(x) < 2/3\} \ge \max\{0, (3\epsilon - 1)/2\}.$$

Proof. The rate ceiling excludes, asymptotically, every function of error less than ϵ. Every remaining function has error at least ϵ, proving the Gibbs assertion. If e_N is the displayed confidence-error probability, Gibbs error is at most $(1 - e_N)/3 + e_N$. Rearranging and taking lower limits proves the second claim. □

In particular, a polynomial sample guarantee would require suitable weak approximations whose normalized rate excess $L[J_L(g) - V_*]$ is only $O(\log^2(d + C + L))$, when the sample size is polynomial in dimension d, teacher complexity C, and depth L, and the statistical parameters are fixed. This is a necessary condition, not a proved approximation theorem for arbitrary small circuits. One should use a fixed error slack when passing from a Gibbs guarantee to a particular approximant.

The requirement is not vacuous merely because all rates converge to V_*. The uniform all-function upper bound gives normalized excess $O(d^2 + \log^2 L)$, whereas the ceiling for polynomially many examples has only squared-logarithmic dependence. A negative learning result would follow from a suitable approximation-excess lower bound for an explicit small target. A positive learning result must instead control inaccurate extensions within this much narrower interval. Neither conclusion follows from the common leading rate alone.

33 What a rate bound does and does not imply for learning

The controlled-circuit theorem is a target representation bound. A learning theorem additionally needs to compare the inaccurate extensions of a training sample with its true target. The next proposition shows that normalized excess rate alone cannot supply a general complexity measure for learning.

Proposition 33.1 (Normalized action alone does not control learning)**.** *There are depths $L_d = \Theta((d+1)^2)$ and a universal constant C such that every Boolean function satisfies*

$$\max\{0, L_d[J_{L_d}(f) - V_*]\} \le C(d+1)^2.$$

Nevertheless, let $m = 2^d$ and $n \le m/4$. For every learning rule that assigns probabilities to the two labels after seeing n uniform cube inputs and their correct labels, there is a Boolean target f for which

$$\Pr_S\left\{\Pr_X\{p_S(f(X) \mid X) < 2/3\} \ge 1/4\right\} \ge 1 - e^{-m/24}.$$

This holds for iid inputs and for sampling without replacement. Randomness of the learning rule may be included in the probability. Consequently no universal sample bound can be polynomial solely in d, L, and positive normalized excess $L[J_L(f) - V_*]$. The hard target in this statement need not have a small circuit.

Proof. The uniform all-function rate upper bound in Theorem 3.1 gives

$$J_L(f) \le V_* + C\frac{(d+1)^2 + \log^2(2L)}{L} \qquad \text{when } L \ge C_0(d+1).$$

Take L_d to be a sufficiently large universal multiple of $(d+1)^2$, rounded up.

For the learning claim, first choose the entire target truth table using independent fair signs. Condition on the sample inputs, their observed labels, and the learner's randomness. At least $u \ge m - n \ge 3m/4$ inputs are unobserved. Their labels are still independent fair signs, while the prediction probabilities are fixed. At any such input at most one label can have probability at least $2/3$, so the failure indicator has probability at least $1/2$. These indicators are independent across the unobserved inputs. Hoeffding's inequality gives

$$\Pr\{\text{fewer than } u/3 \text{ unseen inputs fail}\} \le e^{-u/18} \le e^{-m/24}.$$

Otherwise at least $u/3 \ge m/4$ inputs fail. Averaging over the random truth table supplies a deterministic target with the asserted sample probability. Repeated observations only increase the

number of unobserved inputs. Finally, any fixed polynomial in the displayed complexity parameters is eventually smaller than $2^d/4$. The obstruction also applies to guarantees valid at all sufficiently large widths with a target-dependent width threshold: at each fixed d there are finitely many targets, so one can take the maximum of their thresholds before applying the learning lower bound. □

Why direct prior-mass comparisons retain a width cost. Write π_N for the original globally margin-conditioned critical prior on Boolean functions, and $J_{\min} = \min_g J_L(g)$. At fixed domain and depth the exact finite-class rates imply

$$-\frac{1}{N}\log \pi_N(f) \longrightarrow J_L(f) - J_{\min}.$$

Thus a point-mass PAC-Bayes comparator retains a linear width cost whenever the target rate exceeds the minimum. More generally, if a comparator law Q_N is supported on a nonempty function class $\mathcal{B}$, then

$$D(Q_N\|\pi_N) \geq -\log \pi_N(\mathcal{B}) = N\left[\min_{g\in\mathcal{B}} J_L(g) - J_{\min}\right] + o(N).$$

The inequality follows by conditioning π_N on $\mathcal{B}$. This does not refute width-independent learning; it explains why the fixed-depth proofs instead control competing functions whose rates are no larger than the target's.

Proposition 33.2 (A target-relative error-witness criterion)**.** *Fix a Boolean target f, a distribution $\mathcal{D}$, and $\epsilon, \delta \in (0,1)$. Define*

$$\mathcal{A}_f = \{g : J_L(g) \leq J_L(f)\}.$$

Suppose there are sets $B_1, \ldots, B_M$ such that every $g \in \mathcal{A}_f$ of error greater than $\epsilon/6$ has some index j with

$$B_j \subseteq \{x : g(x) \neq f(x)\}, \qquad \mathcal{D}(B_j) \geq \epsilon/12.$$

The sets may depend on f and $\mathcal{D}$, but not on the sample. If

$$n \geq \frac{12}{\epsilon}\left[\log M + \log(1/\delta)\right],$$

then, at all sufficiently large learner widths, the original critical posterior has confidence error $\mathcal{D}\{x : p_N(f(x) \mid x) \leq 2/3\} \leq \epsilon$ with sample probability at least $1-\delta$. The width threshold can be chosen uniformly over the realized sample.

Proof. An inaccurate consistent function in $\mathcal{A}_f$ forces the sample to miss one of the B_j. A union bound bounds this event by $M\exp(-n\epsilon/12) \leq \delta$. On the complementary event, all consistent members of $\mathcal{A}_f$ have error at most $\epsilon/6$.

Finiteness of the Boolean function class and the exact prior rates give

$$\frac{\pi_N(\mathcal{A}_f^c)}{\pi_N(f)} \longrightarrow 0.$$

For every sample labeled by f, its posterior normalizer is at least $\pi_N(f)$. Thus, for all sufficiently large widths, posterior mass outside $\mathcal{A}_f$ is at most $\epsilon/6$, uniformly over the sample. On the good-sample event its Gibbs error is at most $\epsilon/3$. At every point of confidence error the posterior wrong-label probability is at least $1/3$, giving the claimed bound. Mixtures of tied rate minimizers are included in this argument. □

The witness condition is a one-sided containment statement about error sets. An ordinary $L^1(\mathcal{D})$ cover does not suffice: a predictor can change its labels on a small-measure set chosen to contain the observed points. Taking one witness for each inaccurate function recovers the usual cardinality proof. A substantially smaller common family could give a learning theorem even when that proof is wasteful; the controlled-circuit rate upper bound alone does not provide such a family.

34 Boolean feature geometry and the limits of a stability argument

A depth lower bound based on exponentially close continuous inputs with separated target values does not automatically transfer to Boolean inputs. The free critical kernel retains polynomial separation between distinct Boolean inputs, even after arbitrarily many layers. Entropy-controlled paths behave differently, however, and can contract a selected pair at stretched-exponential speed. These are statements about feature geometry, not posterior sample lower bounds.

Proposition 34.1 (The free kernel retains linear input geometry)**.** *Let $\mathcal{X} \subseteq \{-1,1\}^d$, put $H(x) = (1,x)$, and define*

$$Q_0 = d^{-1}HH^{\mathsf{T}}, \qquad Q_{s+1} = \mathbb{E}[\tanh(G_s)\tanh(G_s)^{\mathsf{T}}], \qquad G_s \sim N(0,Q_s).$$

There are universal constants $c, C > 0$ such that, for every $s \geq 0$, Q_s has constant diagonal q_s and

$$\frac{c}{s+1} \leq q_s \leq \frac{C}{s+1}, \qquad \frac{Q_s}{q_s} \succeq \frac{c}{d+1}HH^{\mathsf{T}}.$$

Consequently, for every $x, y \in \mathcal{X}$,

$$Q_s(x,x) + Q_s(y,y) - 2Q_s(x,y) \geq \frac{c\, d_{\mathrm{H}}(x,y)}{d(s+1)}, \qquad 1 - \frac{Q_s(x,y)}{q_s} \geq \frac{c\, d_{\mathrm{H}}(x,y)}{d+1},$$

where d_{H} is Hamming distance. The constants do not depend on the subset or any sampling distribution on it.

Proof. Write $F(q) = \mathbb{E}\tanh^2(\sqrt{q}Z)$ for a standard Gaussian Z. The diagonal follows $q_{s+1} = F(q_s)$, starting at $q_0 = (d+1)/d \in [1,2]$. The expansion $F(q) = q - 2q^2 + O(q^3)$ and compactness give $q - F(q) \geq aq^2$ on $(0,2]$, for a universal $a > 0$. Thus $1/q_{s+1} \geq 1/q_s + a$. Also $\sinh^2 t \geq t^2$ implies $\tanh^2 t \geq t^2/(1+t^2)$, and Cauchy–Schwarz yields

$$F(q) \geq q\,\mathbb{E}\frac{Z^2}{1+qZ^2} \geq \frac{q}{1+3q}.$$

Hence $1/q_{s+1} \leq 1/q_s + 3$, proving the diagonal bounds.

For any centered Gaussian G of constant diagonal q, let $h(q) = \mathbb{E}\,\mathrm{sech}^2(\sqrt{q}Z) = 1 - F(q)$. Gaussian integration by parts gives

$$\mathbb{E}[\tanh(G)G^{\mathsf{T}}] = h(q)Q.$$

The residual $R = \tanh(G) - h(q)G$ is therefore orthogonal in L^2 to every coordinate of G, and

$$\mathbb{E}[\tanh(G)\tanh(G)^{\mathsf{T}}] = h(q)^2Q + \mathbb{E}[RR^{\mathsf{T}}] \succeq h(q)^2Q.$$

Representing a singular Gaussian as a linear image of a standard Gaussian proves the same identity without a nondegeneracy assumption.

Track $r_0 = 1/d$ and $r_{s+1} = r_s h(q_s)^2$. Then $Q_s \succeq r_s HH^{\mathsf{T}}$, and r_s/q_s is multiplied by

$$R_0(q_s) = \frac{q_s[1 - F(q_s)]^2}{F(q_s)}.$$

The expansion of F gives $R_0(q) = 1+O(q^2)$ at zero. Positivity and compactness on $(0, 2]$ consequently imply $|\log R_0(q)| \le Aq^2$ for a universal A. Moreover

$$\sum_{s\ge 0} q_s^2 \le a^{-1} \sum_{s\ge 0} (q_s - q_{s+1}) \le 2/a.$$

The product of all ratio factors is bounded below by $e^{-2A/a}$. Since $r_0/q_0 = 1/(d + 1)$, this proves the matrix bound. Apply it to $e_x - e_y$ and use $\|H(x) - H(y)\|_2^2 = 4d_{\mathrm{H}}(x, y)$ to obtain both distance estimates. □

Thus neither physical Boolean distances nor the free normalized feature metric supplies exponentially close pairs when dimension and depth are polynomially related. Increasing d exponentially with depth can make the normalized distances exponentially small, but a resulting lower bound may then be only polynomial in d.

Proposition 34.2 (A bounded-entropy path can contract one Boolean pair)**.** *Fix $T \ge 1$, $J > 0$, and $d \ge 4(1 + 3T)$. For any specified Boolean pair x, y differing in one coordinate, there is a controlled critical feature path of T transitions, defined on the entire cube, whose total hidden relative entropy is at most J and whose final covariance satisfies*

$$Q_T(x, x) = Q_T(y, y) = q_T \ge \frac{3}{4(1 + 3T)}, \qquad 1 - \frac{Q_T(x, y)}{q_T} \le \frac{8(1 + 3T)}{3d} \exp\{-2\sqrt{TJ}\}.$$

Proof. At each stage write the selected pair covariance as $\left(\begin{smallmatrix} q_s & c_s \\ c_s & q_s \end{smallmatrix}\right)$, with common variance $a_s = (q_s + c_s)/2$ and contrast variance $b_s = (q_s - c_s)/2$. Initially $a_0 = 1$, $b_0 = 1/d$. In the reference Gaussian, the pair can be written $(A + B, A - B)$, where A, B are independent with variances a_s, b_s. Put $t = \exp(-\sqrt{J/T})$.

In the full reference Gaussian process, decompose $G = \mathbb{E}[G \mid B] + R$ with R independent of B. Replace it by $t\mathbb{E}[G \mid B] + R$. This changes only the contrast marginal and retains its conditional Gaussian law, so its relative entropy is exactly

$$k(t) = \tfrac{1}{2}(t^2 - 1 - \log t^2) \le J/T.$$

The inequality follows from $e^{-2u} \le 1 - 2u + 2u^2$ for $u \ge 0$. The construction is valid on the Gaussian support even when the full covariance is singular. The contrast remains positive at every finite step, because $t > 0$ and tanh is strictly increasing. Tanh pushforward cannot increase the entropy.

The controlled feature pair is $(\tanh(A + tB), \tanh(A - tB))$. Its diagonal is equal, and

$$q_{s+1} = F(q_s - (1 - t^2)b_s), \qquad b_{s+1} = \tfrac{1}{4}\mathbb{E}\big[(\tanh(A + tB) - \tanh(A - tB))^2\big] \le t^2 b_s.$$

Thus $b_s \le t^{2s}/d$. The function F is increasing and one-Lipschitz: differentiation gives

$$0 \le F'(q) = \mathbb{E}\frac{Z \tanh(\sqrt{q}Z) \operatorname{sech}^2(\sqrt{q}Z)}{\sqrt{q}} \le \mathbb{E}Z^2 = 1.$$

Let $\bar{q}_s$ be the free diagonal sequence from the same initial value. Monotonicity and the Lipschitz bound imply

$$0 \le \bar{q}_s - q_s \le \sum_{j<s} (1 - t^2)b_j \le 1/d.$$

Since $\bar{q}_s \geq 1/(1+3s)$, the assumed dimension gives $q_s \geq 3/[4(1+3s)]$ for $s \leq T$. Finally $1 - c_T/q_T = 2b_T/q_T$ gives the claimed estimate, and summing $k(t)$ over the T transitions gives the entropy bound. □

This proposition does not produce a hard target or a low-cost globally margined readout separating the pair. It shows why the free-kernel separation estimate cannot be imposed on every controlled path. A posterior lower bound would additionally need a function-level approximation obstruction and a comparison with the cost of fitting the training data.

Corollary 34.3 (No depth-uniform stability from vanishing directional entropy)**.** *There are a fixed Boolean target on three inputs and critical paths of increasing depth for which the sum of prefix entropy and subsequent conditional entropy in Theorem 11.2 tends to zero, and the readout becomes exactly an anchored kernel classifier, but the target stays a fixed positive distance from every function invariant under permutations fixing its two anchors. Consequently no modulus tending to zero with that entropy budget alone can bound uniform approximation by the invariant class independently of depth.*

Proof. Take $f(u) = u_1 u_2$ on $\{-1,1\}^3$, with anchors $x = (1,1,1)$ and $y = (-1,1,1)$. They form a bichromatic edge. Their stabilizer contains the transposition of coordinates 2 and 3, under which f is not invariant. For example it exchanges $(1,1,-1)$ and $(1,-1,1)$, whose labels are opposite. Every invariant real function therefore differs from f by at least one at one of these inputs; an invariant Boolean classifier must misclassify at least one of them.

The universal upper bound in Theorem 3.1 supplies, for all sufficiently large L, admissible paths with

$$A_L \leq V_* + \varepsilon_L, \qquad \varepsilon_L = O\left(\frac{(1+\log L)^2}{L}\right) \longrightarrow 0.$$

Allow an additional $o(1/L)$ if the variational infimum is not attained. Apply Theorem 11.2 to the fixed edge. Its minimum pair variance m_L tends to zero, and the prefix plus conditional entropy is at most $\varepsilon_L + C_0 m_L \to 0$. The same bound controls the readout's orthogonal slack, so Corollary 11.3 identifies its classifier exactly with f for large L. The invariant-class obstruction above remains unchanged, proving the claim. □

This failure of depth-uniform stability is consistent with a bound involving $\sqrt{L\varepsilon}$ or another depth-dependent quantity. The example itself has small intrinsic product complexity and is not a counterexample to the proposed critical-rate-to-product bound.

35 A product-complexity bound for invariant terminal kernels

The general passage from critical rate to intrinsic product complexity remains open; Section 15 instead bounds finite width without controlling product norms. There is a depth-independent bound when the terminal feature covariance retains Boolean translation invariance. This is a sufficient condition on a variational path; a general minimizing path need not satisfy it.

For a Boolean function f, define

$$\mathfrak{P}(f) = \inf\left\{\prod_j \|W^j\|_F : f(x) z_w(x) \geq 1 \text{ for every } x \in \{-1,1\}^d\right\}.$$

The infimum allows arbitrary finite depth and width, tanh hidden layers, a linear readout, and biases only through the augmented input.

Lemma 35.1 (Fourier variation and a shallow representation)**.** *Suppose a real function g satisfies $f(x)g(x) \geq 1$ on the cube, and write*

$$g(x) = \sum_{S \subseteq [d]} \widehat{g}(S)\chi_S(x), \qquad \chi_S(x) = \prod_{i \in S} x_i, \qquad A = \sum_S |\widehat{g}(S)|.$$

There is a universal constant C such that

$$\mathfrak{P}(f) \leq C(d+1)^2 A \log\bigl(2(d+1)A\bigr).$$

A depth-two tanh network of finite width attains this bound after increasing C by a universal factor.

Proof. The margin implies $A \geq 1$. Fix a nonempty S and put $k = |S|$. For $j = 1, \ldots, k$, define

$$q_j(x) = \operatorname{sign}\left(-\sum_{i \in S} x_i + k - 2j + 1\right).$$

If exactly t of the selected coordinates equal -1, the expression inside this sign is $2t - 2j + 1$. Consequently

$$\chi_S(x) = b_0 + \sum_{j=1}^{k} (-1)^j q_j(x), \qquad b_0 = 1 + \sum_{j=1}^{k} (-1)^j \in \{0, 1\}.$$

All threshold preactivations are odd integers and have absolute value at least one. For $0 < \delta < 1$, replace their signs by tanh after multiplying the preactivation by

$$\tau = \tfrac{1}{2} \log(2k/\delta).$$

Since $|\tanh t - \operatorname{sign} t| \leq 2e^{-2|t|}$, the resulting parity approximation has uniform error at most δ. The constant term is represented exactly by the augmented-input neuron $\tanh(\operatorname{atanh}(1/2)) = 1/2$, with readout coefficient $2b_0$. The squared Frobenius norm of the input matrix is $O(\tau^2 k^3 + 1)$, and the squared readout norm is $O(k+1)$. Its product is therefore at most

$$C(d+1)^2 \log\bigl(2(d+1)/\delta\bigr).$$

The same bound holds for its shallow variation sum, namely the sum of absolute outgoing coefficients times incoming-row norms, by Cauchy–Schwarz. The empty-set character is supplied exactly by a constant neuron and obeys the same bound.

Take $\delta = 1/(4A)$, and combine these parity approximations with coefficients $\widehat{g}(S)$. The resulting output differs from g uniformly by at most $A\delta = 1/4$. Doubling its readout gives signed margin at least $3/2$, and its variation sum is at most

$$C(d+1)^2 A \log\bigl(2(d+1)A\bigr).$$

It remains to relate variation to a Frobenius product without paying for the number of Fourier characters.

For any finite shallow network with incoming rows w_i and outgoing coefficients v_i, clone row i exactly c_i times and divide its outgoing coefficient by c_i. This leaves its function unchanged and gives product

$$\left\{\left(\sum_i c_i \|w_i\|_2^2\right)\left(\sum_i v_i^2/c_i\right)\right\}^{1/2}.$$

Omit zero rows and zero coefficients. Choosing $c_i = \lceil t|v_i|/\|w_i\|_2 \rceil$ and letting t increase makes this expression converge to $\sum_i |v_i| \|w_i\|_2$. A sufficiently large finite t is within a fixed constant factor. This proves the claim, without any bound on the supplied Fourier support size. □

Proposition 35.2 (An invariant-kernel product bound)**.** *Let K be a positive semidefinite kernel on the Boolean cube, invariant under simultaneous Boolean translations:*

$$K(ax, ay) = K(x, y) \quad \text{for every } a, x, y \in \{-1, 1\}^d,$$

where multiplication is coordinatewise. Write $q = K(x, x)$. If g belongs to its finite-set reproducing-kernel Hilbert space and $f(x)g(x) \geq 1$ everywhere, then, with $B = \|g\|_K \sqrt{q}$,

$$\mathfrak{P}(f) \leq C(d+1)^2 B \log\bigl(2(d+1)B\bigr).$$

In particular, suppose an admissible critical variational path has total action at most J and ends at a translation-invariant covariance Q of diagonal $q \leq 1$. Its globally margined classifier satisfies

$$\mathfrak{P}(f) \leq C(d+1)^2 \sqrt{8Jq}\, \log\bigl(2(d+1)\sqrt{8Jq}\bigr).$$

There is no dependence on the path's depth in these bounds.

Proof. Write $K(x, y) = \kappa(xy)$. Relative to the uniform input measure, the characters diagonalize the kernel operator, with nonnegative eigenvalues

$$\lambda_S = \mathbb{E}_t \kappa(t) \chi_S(t), \qquad \sum_S \lambda_S = q.$$

The Fourier coefficients of an RKHS function vanish whenever $\lambda_S = 0$, and

$$\|g\|_K^2 = \sum_{S : \lambda_S > 0} \frac{|\widehat{g}(S)|^2}{\lambda_S}.$$

Equivalently this norm is $g^T K^\dagger g$ when K denotes the unnormalized kernel matrix. Cauchy–Schwarz gives

$$\sum_S |\widehat{g}(S)| \leq \left(\sum_S \frac{|\widehat{g}(S)|^2}{\lambda_S} \right)^{1/2} \left(\sum_S \lambda_S \right)^{1/2} = \|g\|_K \sqrt{q} = B.$$

The unit margin makes $B \geq 1$. Apply Lemma 35.1; its right side is increasing in the variation bound for arguments at least one.

For the critical path, its readout z has signed margin at least $1/2$ and readout action $\frac{1}{2} z^T Q^\dagger z \leq J$. Thus $g = 2z$ has unit margin and RKHS norm at most $\sqrt{8J}$. The preceding assertion gives the second bound. □

The free Gaussian kernel tower preserves translation invariance, so the proposition applies to its readouts at arbitrary depth. It also applies to controlled paths whose terminal covariance has that invariance. An arbitrary controlled path can create a component proportional to ff^T, which generally breaks the symmetry. Charging the entropy cost of creating such structure remains necessary for a general bridge. The proposition alone is not a posterior sample-complexity theorem.

36 A finite-width upper bound at the limiting critical exponent

The scalar comparison yields a finite-width bound with an explicit subexponential prefactor. In this section the Gaussian scale is critical, $t = 1$, and the required clean output margin is $1/2$. No large-deviation limit is used. Write $H = L - 1$ and let $A_{N,f}$ denote the event $f(x) z_w(x) > 1/2$ on the entire Boolean cube.

Theorem 36.1 (Finite-width universal upper bound)**.** *There is a universal constant $C < \infty$ such that, for every $d, N, H \geq 1$ and every Boolean function f,*

$$\mu_{N,1}(A_{N,f}) \leq \min\left\{1, \exp\left[-NV_* + \frac{CN}{H} + CH\log(2N)\right]\right\}. \tag{35}$$

Here $V_* \in (1,2)$ is the scalar limiting value defined above. The constant does not depend on input dimension, width, or depth. In fact, the same one-sided bound holds for $sz_w(x_0) > 1/2$ at any fixed input x_0 and sign $s \in \{-1,1\}$. The full global-margin event therefore satisfies

$$\mu_{N,1}(A_N) \leq \min\left\{1, 2\exp\left[-NV_* + \frac{CN}{H} + CH\log(2N)\right]\right\}, \qquad A_N = \bigcup_f A_{N,f}.$$

We first record a modulus estimate for the scalar value. Extend V to zero by $V(0) = V_*$.

Lemma 36.2 (A uniform scalar-value modulus)**.** *For a universal C_V, every $0 \leq q \leq q' \leq 1$ satisfies*

$$0 \leq V(q) - V(q') \leq C_V\sqrt{q'-q}.$$

In addition, $0 \leq V_* - V(q) \leq C_V q$ for $0 \leq q \leq 1$.

Proof. The small-state bridge gives $V_* - V(q) \leq 17q$ for $q \leq 1/128$. For larger q, use $0 \leq V(q) \leq V_* < 2$ to obtain the second assertion with, for example, constant 256.

For $0 < q < q'$, we also have

$$V(q) - V(q') \leq 5(q'/q - 1). \tag{36}$$

Indeed choose a path from q' arbitrarily close to optimal and of cost less than two; the preceding uniform scalar upper construction permits this. If it has at least one step, write η for its first preactivation control. Gaussian entropy duality at parameter $1/(4q')$ gives

$$\int z^2\, d\eta(z) \leq q'\left[4D(\eta\|\mathcal{N}(0,q')) + 2\log 2\right] < 10q'.$$

Use the same control and all the same subsequent states when starting at q. The first entropy changes by exactly

$$\frac{1}{2}\log(q/q') + \frac{1}{2}(1/q - 1/q')\int z^2\, d\eta(z) \leq 5(q'/q - 1).$$

If the selected path has zero steps, its cost is $1/(8q') < 2$; the zero-step cost from q is larger by at most $2(q'/q-1)$. Taking the approximation error to zero proves (36).

Put $h = q' - q$. If $q \geq \sqrt{h}$, the relative modulus is at most $5\sqrt{h}$. Otherwise $q' \leq \sqrt{h} + h \leq 2\sqrt{h}$, and the second assertion gives $V(q) - V(q') \leq V_* - V(q') \leq 512\sqrt{h}$. The zero-width interval and $q = 0$ cases are immediate. □

Lemma 36.3 (Occupation with an additive prefix budget)**.** *Suppose $u_0, \ldots, u_H \in (0,1]$, nonnegative costs D_j, and $e \geq 0$ satisfy*

$$D_j \geq u_j - u_{j-1} + cu_{j-1}^2, \qquad \sum_{j=1}^{i} D_j \leq C_0 u_i + e, \qquad \sum_{j=1}^{H} D_j \leq 3.$$

For constants depending only on c, C_0, their minimum $a = \min_j u_j$ satisfies

$$H \leq \frac{C_1}{a} + \frac{C_1 e}{a^2}, \qquad a \leq \frac{C_2}{H} + C_2\sqrt{e/H}.$$

Proof. The reciprocal telescoping inequality gives, on any interval whose pre-states are at least b,

$$c\,\#\{\text{steps}\} \le b^{-2}\sum D_j + 1/u_{\text{end}} - 1/u_{\text{start}}.$$

If a exceeds the fixed threshold $1/256$, the total cost bound gives a universal length bound. Otherwise partition the path at its last exits from $[0, 2^k a]$, stopping when $2^k a \ge 1/256$. The prefix ending at the global minimum has cost at most $C_0 a + e$, so its length is at most $C/a + Ce/a^2$. In each subsequent dyadic block, remove its first transition. All remaining pre-states exceed $b = 2^k a$, the endpoint lies in $(b, 2b]$, and the block cost is at most $2C_0 b + e$. Its remaining length is therefore at most $C/b + Ce/b^2$. The final tail above $1/256$ has universal bounded length by the total cost bound, after again removing one transition. Summing the geometric series, and restoring the removed transitions, proves the first estimate. Solving its quadratic inequality for a gives the second. □

For $q > 0$ and $0 \le y < 1$, define the upper-tail scalar cost

$$I_+(q, y) = \inf\left\{ D(\eta\|\mathcal{N}(0,q)) : \int \tanh^2 z\, d\eta(z) \ge y \right\}.$$

It has the Chernoff dual formula

$$I_+(q, y) = \sup_{\lambda \ge 0}\left\{ \lambda y - \log \mathbb{E}\exp[\lambda \tanh^2(\sqrt{q}Z)] \right\}. \tag{37}$$

For y below the reference mean both sides are zero. Otherwise an exponential tilt with positive parameter realizes mean y; the tilted mean increases continuously from the reference mean to one. This proves the formula directly, without an asymptotic Cramer theorem. Two consequences will be used:

$$I_+(q, y) \ge V(q) - V(y), \tag{38}$$

$$I_+(q, y) \ge y - q + cq^2 \qquad (0 < q \le 1), \tag{39}$$

where $c > 0$ is the strict scalar-potential constant. For the first, any feasible control has next state $q' \ge y$, so the definition and monotonicity of V give $V(q) \le D(\eta\|\mathcal{N}(0,q)) + V(q') \le D(\eta\|\mathcal{N}(0,q)) + V(y)$. For the second, entropy duality and $\log \mathbb{E}\exp[\tanh^2(\sqrt{q}Z)] \le q - cq^2$ give the result. These inequalities also hold at $y = 0$ with the extension $V(0) = V_*$.

Proof of Theorem 36.1. After increasing C, the theorem is trivial if $N = 1$, $H = 1$, or $H > N$, since $V_* < 2$. Thus assume $N \ge 2$ and $2 \le H \le N$, and put $K = H - 1$. Fix one input x_0 and condition on its entire first hidden vector. Define

$$v_j = \frac{1}{N}\sum_{i=1}^{N} h_i^{j+1}(x_0)^2 \quad (0 \le j \le K).$$

All these numbers belong to $[0, 1]$; the conditional initial value v_0 is arbitrary. If $v_0 = 0$, every later hidden vector at x_0 and its readout are zero, so the asserted bound is immediate. Otherwise, conditional on all preceding layers, the next preactivations at x_0 are independent Gaussians of variance v_{j-1}. Thus v_j is exactly an average of N independent $\tanh^2$ Gaussian variables, and, for every $y \ge 0$,

$$\Pr\{v_j \ge y \mid \text{previous layers}\} \le \exp[-NI_+(v_{j-1}, y)].$$

When $v_{j-1} = 0$, the vector remains zero and the same statement has the evident degenerate interpretation. The right-hand-side rate is nonincreasing in its first argument: $\tanh^2(\sqrt{q}Z)$ increases pointwise with q, so its nonnegative exponential moments increase with q.

Take the grid mesh $h = N^{-4}$ and $J = N^4$ bins $[0,h], (h,2h], \dots, ((J-1)h, 1]$. For a path of bin indices, let u_j be its upper endpoint and $l_j = u_j - h$ its lower endpoint, and put $u_0 = v_0$. By conditioning successively on full preceding layers, the probability of this bin path, together with $sz_w(x_0) > 1/2$ for any fixed sign s, is at most

$$\exp[-N\mathcal{A}(u)], \qquad \mathcal{A}(u) = \sum_{j=1}^{K} I_+(u_{j-1}, l_j) + \frac{1}{8u_K}.$$

For bins after the first, the actual event has $v_j > l_j$, which is contained in $v_j \geq l_j$; for the first bin the lower threshold is zero. The conditional readout variance is $v_K/N \leq u_K/N$, giving the last term. Monotonicity in the previous variance justifies replacing its actual value by u_{j-1} at every transition.

We bound the action of every grid path. If $\mathcal{A}(u) \geq V_*$, the claimed lower estimate below is automatic. Otherwise put $D_j = I_+(u_{j-1}, l_j)$ and $\widetilde{D}_j = D_j + h$. By the modulus lemma, $V(l_i) - V(u_i) \leq C_V/N^2$ for every bin. Bellman's inequality on any suffix, together with $V(u_K) \leq 1/(8u_K)$, gives

$$\sum_{j=i+1}^{K} D_j + \frac{1}{8u_K} \geq V(u_i) - \frac{C_V K}{N^2}.$$

Consequently every prefix obeys

$$\sum_{j=1}^{i} \widetilde{D}_j \leq \mathcal{A}(u) - V(u_i) + \frac{C_V K}{N^2} + Kh \leq C_V u_i + e, \qquad e = \frac{C_V K}{N^2} + Kh.$$

Also $\sum_j \widetilde{D}_j \leq V_* + Kh < 3$, and (39) yields the exact inequality

$$\widetilde{D}_j \geq u_j - u_{j-1} + cu_{j-1}^2.$$

Lemma 36.3 therefore supplies some $j \in \{0, \dots, K\}$ with

$$u_j \leq \frac{C_1}{K} + C_1\sqrt{e/K} \leq \frac{C_2}{K} + \frac{C_2}{N}.$$

The suffix bound at this layer gives

$$\mathcal{A}(u) \geq V(u_j) - \frac{C_V K}{N^2} \geq V_* - \frac{C_3}{K} - \frac{C_3}{N} - \frac{C_V K}{N^2}.$$

There are $J^K = N^{4K}$ bin paths. Summing their probability bounds, using $1/K \leq 2/H$, and absorbing $C_3 + C_V K/N + 4K\log N$ into $CH\log(2N)$ proves the claimed one-sided bound conditionally on every first hidden vector with $v_0 > 0$. The proof used no restriction on v_0 beyond $v_0 \leq 1$; the prefix inequality at $i = 0$ is automatic. Integrating this uniform bound over the first layer proves it unconditionally.

For a fixed target, $A_{N,f}$ is contained in $\{f(x_0)z_w(x_0) > 1/2\}$. The full global-margin event is contained in $\{|z_w(x_0)| > 1/2\}$, and the two possible signs cost only a factor of two. This proves both assertions. □

In particular, if $m = 2^d$, $N = m\exp(O(\sqrt{d\log d}))$, and $H = \exp(O(\sqrt{d\log d}))$ is chosen so that $N/H = o(m)$, then the correction $CN/H + CH\log(2N)$ is $o(m)$. This statement alone is an upper bound. In fact, the global-event version rules out matching lower masses with logarithmic error $o(m)$ uniformly over all Boolean functions: summing such lower bounds over their 2^m disjoint events contradicts the displayed upper bound. Section 37 makes this obstruction quantitative. A posterior comparison must retain the function-specific finite-width terms concealed by the common leading rate.

37 Why a common-exponent counting argument is insufficient

Let $m = 2^d$ and

$$E_{N,L} = C\left[\frac{N}{L-1} + (L-1)\log(2N)\right],$$

with C as in the finite-width universal upper bound. The target events $A_{N,f}$ are disjoint and their union is the global-margin event A_N. Therefore

$$\sum_f \mu_{N,1}(A_{N,f}) \le 2\exp(-NV_* + E_{N,L}).$$

This gives an immediate obstruction to transferring the common variational rate into uniform finite-width target masses.

Proposition 37.1 (Unavoidable function-specific logarithmic overhead)**.** *If a number R satisfies*

$$\mu_{N,1}(A_{N,f}) \ge \exp(-NV_* - R) \qquad \text{for every Boolean } f,$$

then necessarily

$$R \ge m\log 2 - \log 2 - E_{N,L}.$$

In particular, in a regime where $E_{N,L} = o(m)$, a uniform lower bound with $R = o(m)$ is impossible.

Proof. Sum the asserted lower bound over the 2^m sign patterns and compare with the displayed upper bound on their union. □

More generally, for every $r \in \mathbb{R}$, the number of functions whose target event has probability at least e^{-NV_*-r} is at most $2e^{E_{N,L}+r}$. Thus the common limiting exponent must conceal substantial function-specific finite-width terms. Their presence is not an artifact of a particular coupling or concentration proof.

This observation does not establish good learning: the missing terms could favor simple targets, or could resemble a nearly uniform prior over truth tables. It also does not establish bad learning. A posterior sample lower bound requires a target-relative comparison that retains these terms; a shared leading exponent alone cannot supply one.

38 An exact symmetry useful for comparing targets

The first-layer bias has the same Gaussian variance as each input coefficient. This gives a slightly larger symmetry than coordinate permutations of the Boolean cube. The result below holds at finite width and every prior scale, not only for the critical rate.

Proposition 38.1 (Projective coordinate symmetry)**.** *Write $y = (1, x_1, \dots, x_d)$ and let π be any permutation of $\{0, \dots, d\}$. Define*

$$a_\pi(x) = y_{\pi(0)}, \qquad T_\pi(x)_j = y_{\pi(j)}/y_{\pi(0)} \quad (1 \le j \le d).$$

The map T_π is a bijection of the Boolean cube. If z is a network drawn from the Gaussian prior, then

$$x \longmapsto a_\pi(x) z(T_\pi(x))$$

has exactly the same law as z. Consequently, writing $f^\pi(x) = a_\pi(x) f(T_\pi(x))$, the target events $A_{N,f}$ and A_{N,f^π} have equal prior probabilities. At critical scaling, $J_L(f) = J_L(f^\pi)$.

In particular, for every m for which the indicated degrees exist, all parity functions of degree $2m$ or $2m+1$ have the same prior probability and the same critical rate. Constants and dictators are the first such pair of degree classes.

Proof. Permute the $d+1$ columns of the first-layer matrix by π. Its Gaussian law is unchanged. The permuted input vector satisfies

$$(y_{\pi(0)}, \dots, y_{\pi(d)}) = a_\pi(x)(1, T_\pi(x)).$$

Every subsequent hidden activation is odd, and there are no subsequent biases. Thus the factor $a_\pi(x) \in \{-1, 1\}$ passes through all hidden layers and the linear readout. This proves the asserted identity in law. The induced transformation of projective sign vectors is invertible, which proves that T_π is a bijection. The equality of target-event probabilities follows by changing variables on the cube; the rate equality follows either from these exact finite-width identities or by the same transformation in the variational functional.

For the final assertion it suffices to swap coordinate zero with an input coordinate i. If a parity has even degree and does not involve i, the transformation adds i. If a parity has odd degree and does involve i, it removes i. In the other two cases it leaves the parity unchanged. Ordinary input permutations identify all subsets of a given size. □

For uniform input sampling, transforming each example by T_π and multiplying its label by a_π also identifies the corresponding posterior learning problems, including Gibbs error and any fixed posterior-confidence threshold. This exact symmetry does not identify all parity degrees, and does not by itself give a sample-complexity lower bound.

39 A necessary depth for bounded absolute rate

The common limiting rate does not mean that a bounded rate is attained at every depth. Gaussian entropy transport gives a complementary constraint. This is a statement about representation rates, not a sample-complexity lower bound.

For $y = (y_0, \dots, y_d) \in \{-1, 1\}^{d+1}$ define the odd extension

$$\widetilde{f}(y) = y_0 f(y_1/y_0, \dots, y_d/y_0),$$

and let $k = k_{\mathrm{aug}}(f)$ be its number of relevant coordinates. Here a coordinate is relevant when some edge parallel to that coordinate has opposite endpoint labels. Since $\widetilde{f}$ is odd, $1 \le k \le d+1$. The count is unchanged by permutations of the augmented coordinates. In particular, constants and dictators both have $k = 1$. A parity of original degree j has $k = j$ when j is odd and $k = j+1$ when j is even.

Theorem 39.1 (Entropy transport and depth)**.** *Let $J = J_L(f)$ and $h = L - 2$. Put*

$$B_h(J) = \begin{cases} (1 + \sqrt{2J/h})^{2h}, & h \ge 1, \\ 1, & h = 0. \end{cases}$$

Then

$$k_{\mathrm{aug}}(f) \le 64 J B_h(J), \tag{40}$$

$$d \le 8J\big(1 + \sqrt{2J/k_{\mathrm{aug}}(f)}\big)^2 B_h(J). \tag{41}$$

In particular $B_h(J) \le \exp(2\sqrt{2hJ})$. If $J_L(f) \le K$ and $d > 8K(1 + \sqrt{2K})^2$, then

$$L \ge 2 + \frac{1}{8K}\left[\log \frac{d}{8K(1 + \sqrt{2K})^2}\right]^2.$$

Thus every target, including a constant, requires depth $\Omega((\log d)^2)$ to have a uniformly bounded critical rate as $d \to \infty$.

We use the Gaussian transport inequality in its correlated form. If $\eta \ll \mathcal{N}(0, Q)$ has entropy D, there is a coupling (Z, G) with these two marginals such that

$$\mathbb{E}\|Q^{+1/2}(Z - G)\|^2 \le 2D. \tag{42}$$

Both variables lie in range Q. This is the standard Gaussian quadratic transport inequality [5], applied after whitening on that range; singularity causes no change. In particular one common coupling works for all input coordinates, rather than separate coordinatewise couplings.

Proof of Theorem 39.1. Take any admissible variational path of total cost $A = r + \sum_{s=1}^{L-1} D_s < \infty$, where r is its readout cost. Extend every feature oddly to the augmented cube as above. This is an injective linear transformation of the feature vector and preserves relative entropy. The Gaussian reference and the tanh map commute with this extension. Finite entropy ensures that every feature coordinate is in $(-1, 1)$, so coordinatewise atanh lifts each controlled feature law to its preactivation law with exactly the same entropy.

For a later hidden layer with input second moment Q, apply (42). For any two augmented inputs y, y', put $b = e_y - e_{y'}$. Cauchy–Schwarz on range Q and Minkowski give

$$\|b^T Z\|_{L^2} \le \|b^T G\|_{L^2} + \|b^T(Z - G)\|_{L^2} \le (1 + \sqrt{2D_s})\sqrt{b^T Q b}.$$

Since tanh is 1-Lipschitz, the L^2 distance of the two output features is at most this bound. Hence each later layer multiplies pairwise feature distances by at most $1 + \sqrt{2D_s}$. Write

$$P = \prod_{s=2}^{L-1} (1 + \sqrt{2D_s}),$$

with the empty product equal to one.

At the first layer, the field $y \mapsto W \cdot y$ identifies W uniquely. Its controlled law has entropy D_1 relative to $\mathcal{N}(0, I_{d+1}/d)$. The Gaussian transport coupling therefore satisfies $\mathbb{E}\|W - G\|^2 \le 2D_1/d$. If I is the set of the k relevant augmented coordinates, projection and Minkowski give

$$\sum_{i \in I} \mathbb{E}W_i^2 \le \frac{1}{d}\left(\sqrt{k} + \sqrt{2D_1}\right)^2, \qquad \mathbb{E}\|W\|^2 \le \frac{1}{d}\left(\sqrt{d+1} + \sqrt{2D_1}\right)^2. \tag{43}$$

A first-layer edge in direction i has feature distance at most $2\sqrt{\mathbb{E}W_i^2}$.

The readout admits a representation $z(y) = \mathbb{E}[a(H)H(y)]$ with $\mathbb{E}a(H)^2 = 2r$: one may take $a(H) = z^T Q^+ H$. Each relevant direction has a bichromatic edge, and the output margin makes the output difference on that edge at least one. Consequently

$$1 \le 2\sqrt{2r}\, P \sqrt{\mathbb{E}W_i^2} \qquad (i \in I).$$

Summing the squared inequalities and applying the first estimate in (43) proves the sharper budget form

$$d \le 8rP^2\left(1 + \sqrt{2D_1/k}\right)^2. \tag{44}$$

Using the full trace instead gives

$$k \le \frac{8rP^2}{d}\left(\sqrt{d+1} + \sqrt{2D_1}\right)^2.$$

If $A < d$, its right side is at most $64AP^2$. If $A \geq d$, the same conclusion follows from $k \leq d+1 \leq 2A$.

The function $u \mapsto \log(1+\sqrt{2u})$ is concave on $[0,\infty)$. For $h \geq 1$, Jensen's inequality and $\sum_{s=2}^{L-1} D_s \leq A$ give

$$P^2 \leq (1+\sqrt{2A/h})^{2h} \leq e^{2\sqrt{2hA}}.$$

The same assertion with $P = 1$ holds for $h = 0$. Substituting $r, D_1 \leq A$ in the preceding bounds and taking $A \downarrow J$ proves both inequalities. The final depth bound follows by taking logarithms of (41) and using $k \geq 1$. □

Remark 39.2 (What the bound does and does not control)**.** *For fixed rate budget K, (40) also requires depth $\Omega((\log k)^2)$ when $k \to \infty$. Any Boolean predictor that has uniform-distribution error strictly less than $1/2$ on a parity of j specified coordinates must depend on each of those coordinates. Thus the same relevant-coordinate restriction applies to every such predictor. Neither statement compares posterior masses of functions consistent with a training sample. A bound on pairwise feature distances also does not by itself bound the number of possible correlated feature laws, so it cannot be read as a learning theorem. The ambient-dimension restriction above is compatible with Theorem 3.1, whose sufficient depth is polynomial in d.*

40 A target-relative separation at every fixed depth

The transport bound also gives a comparison beyond depth two. The comparison is asymptotic in the input dimension and does not order all targets at every finite dimension or at arbitrary growing depth.

Theorem 40.1 (Fixed-depth critical separation)**.** *Let $L \geq 2$, $d \geq 1$, and let k be the number of relevant coordinates of the odd augmented extension of a Boolean target f. Then*

$$J_L(f) \geq \frac{1}{64} \min\{d^{1/(L-1)}, (dk)^{1/L}\}. \tag{45}$$

For the constant target, and hence for a signed dictator by the projective coordinate symmetry,

$$\frac{1}{64} d^{1/L} \leq J_L(\mathbf{1}) \leq 5000L\, d^{1/L}. \tag{46}$$

Consequently

$$\frac{J_L(f)}{J_L(\mathbf{1})} \geq \frac{1}{320000L} \min\{d^{1/[L(L-1)]}, k^{1/L}\}.$$

For every fixed L, a sequence of targets with $k \to \infty$ therefore has a rate ratio tending to infinity as $d \to \infty$. In particular, every growing-degree parity eventually has strictly larger rate than a constant at each fixed depth. For full parity this strict comparison also holds whenever $L = o(\sqrt{\log d / \log\log d})$ and $d \to \infty$.

Proof. Put $J = J_L(f)$ and $h = L-2$. Every admissible path has readout cost at least $1/8$: a single output has magnitude at least $1/2$, whereas each final feature has squared magnitude at most one. Thus $J \geq 1/8$. The transport estimate gives

$$d \leq 8JB_h(J)(1+\sqrt{2J/k})^2, \qquad B_0(J) = 1, \quad B_h(J) = (1+\sqrt{2J/h})^{2h} \ (h \geq 1).$$

For $h \geq 1$, $2 + 4J/h \leq 20J$, so $B_h(J) \leq (20J)^h$. The same bound holds for $h = 0$. Also $(1 + \sqrt{2J/k})^2 \leq 2 + 4J/k$. Consequently

$$d \leq 16 \cdot 20^h J^{h+1} + \frac{32 \cdot 20^h}{k} J^{h+2}.$$

At least one summand is at least $d/2$. This proves

$$J \geq \min\left\{ \left(\frac{d}{32 \cdot 20^h} \right)^{1/(h+1)}, \left(\frac{dk}{64 \cdot 20^h} \right)^{1/(h+2)} \right\},$$

which implies (45) with the stated universal constant.

For completeness we construct the constant-target upper bound directly in the exact variational functional. Write $c = 1/100$. For $0 \leq q \leq 2$ and $0 \leq \mu \leq 1$, put

$$\psi_q(\mu) = \mathbb{E} \tanh(\mu + \sqrt{q} Z), \qquad Z \sim N(0,1).$$

The function is continuous and strictly increasing, vanishes at zero, and satisfies $\psi_q(\mu) \geq c\mu$. Indeed its derivative is at least $\Pr(|Z| \leq 1) \operatorname{sech}^2(1 + \sqrt{2}) > 1/100$ throughout this parameter range. Thus every $m \in [0, c]$ is realized by some $\mu \in [0, 1]$ with $\mu \leq m/c$.

Prescribe the common hidden-feature means

$$m_s = c\, d^{-(L-s)/(2L)}, \qquad 1 \leq s \leq L - 1.$$

At the first layer shift only the Gaussian bias by μ_1, chosen so that $\psi_{(d+1)/d}(\mu_1) = m_1$. Its relative entropy is $d\mu_1^2/2 \leq d^{1/L}/2$. The feature law is invariant under input sign translations, has mean $m_1 \mathbf{1}$, and hence its second moment satisfies $Q_1 \succeq m_1^2 \mathbf{1}\mathbf{1}^T$.

Inductively, suppose the preceding feature law is translation invariant and has mean $m_{s-1}\mathbf{1}$. Its covariance in the Gaussian transition is its second moment Q_{s-1}, whose diagonal is a common value $q_{s-1} \leq 1$. Replace the centered Gaussian preactivation by $N(\mu_s \mathbf{1}, Q_{s-1})$, choosing $\psi_{q_{s-1}}(\mu_s) = m_s$. The mean direction belongs to the range of Q_{s-1}, and the rank-one second-moment bound gives the entropy estimate

$$\frac{\mu_s^2}{2} \mathbf{1}^T Q_{s-1}^{+} \mathbf{1} \leq \frac{\mu_s^2}{2m_{s-1}^2} \leq \frac{1}{2c^2} d^{1/L}.$$

Coordinatewise tanh is injective, so its pushforward has the same relative entropy from the transition reference. Translation invariance and the prescribed common mean are preserved. This proves the induction, including singular second-moment matrices.

Finally choose the output $z = \mathbf{1}/2$. Since $Q_{L-1} \succeq m_{L-1}^2 \mathbf{1}\mathbf{1}^T$, its readout cost is at most

$$\frac{1}{8} \mathbf{1}^T Q_{L-1}^{+} \mathbf{1} \leq \frac{1}{8m_{L-1}^2} = 1250 d^{1/L}.$$

Summing the first-layer cost, the $L - 2$ later hidden costs, and the readout cost proves the upper bound in (46). Its lower bound is (45) with $k = 1$. The ratio estimate follows immediately. For full parity $k \geq d$, the logarithm of its ratio lower bound is $\log d/[L(L-1)] - \log(320000L)$, which tends to infinity in the stated growing-depth regime. □

This theorem compares unconditioned target rates. It does not compare the best training-consistent competitors or prove a posterior sample complexity lower bound. The estimates deliberately leave the regime of large growing depth open; universal rate flattening is compatible with them.

41 The absolute-rate scale at every depth

The necessary depth from entropy transport and the constant-target constructions fit into a single rate scale. This identifies the order of the smallest absolute rate, up to universal multiplicative constants; it does not assert that constants are exact minimizers at every depth.

Theorem 41.1 (A uniform depth–dimension rate profile)**.** *Define*

$$\Phi_L(d) = 1 + L\big(d^{1/(2L)} - 1\big)^2, \qquad d \ge 1, \quad L \ge 2.$$

There are universal constants $0 < c < C < \infty$ such that every Boolean function satisfies

$$J_L(f) \ge c\Phi_L(d),$$

whereas constants and dictators satisfy

$$J_L(f) \le C\Phi_L(d).$$

Thus the minimum critical absolute rate over all Boolean targets, and the rate of a constant target, are both comparable to $\Phi_L(d)$ uniformly in dimension and depth. For $L \ge \log(d+1)$, this scale is comparable to $1 + [\log(d+1)]^2/L$. These statements concern prior rates, rather than sample complexity or the ordering of every pair of targets.

Proof. Take an admissible path of cost $A = r + \sum_{s=1}^{L-1} D_s$. The exact projected-trace budget (44), with $k \ge 1$, gives

$$d \le 8r \prod_{s=1}^{L-1} (1 + \sqrt{2D_s})^2 \le 4(1 + \sqrt{2r})^2 \prod_{s=1}^{L-1} (1 + \sqrt{2D_s})^2.$$

Concavity of $u \mapsto \log(1 + \sqrt{2u})$ and Jensen's inequality on these L costs imply

$$d \le 4(1 + \sqrt{2A/L})^{2L}.$$

Taking $A \downarrow J_L(f)$ yields the explicit lower bound

$$J_L(f) \ge \frac{L}{2} \left[(d/4)^{1/(2L)} - 1 \right]_+^2. \tag{47}$$

Also $J_L(f) \ge 1/8$ by the single-input readout bound.

For $d \ge 16$, write $u = \log d/(2L)$ and $v = \log 4/(2L)$. Then $u \ge 2v$ and $e^v \le \sqrt{2}$, so

$$e^u - 1 = e^v(e^{u-v} - 1) + (e^v - 1) \le (\sqrt{2} + 1)(e^{u-v} - 1).$$

Combining this with (47) and the readout bound proves $J_L(f) \ge c\Phi_L(d)$ for a universal $c > 0$. For $1 \le d < 16$, the function $L \mapsto L(d^{1/(2L)} - 1)^2$ is nonincreasing on $(0, \infty)$: putting $t = \log d/(2L)$ reduces this to monotonicity of $(e^t - 1)^2/t$. Its derivative is positive for $t > 0$, since $2te^t - (e^t - 1) > 0$. Hence $\Phi_L(d) \le 3$ for $L \ge 2$, and the readout bound proves the same lower comparison.

For the upper comparison, the constant-target seed satisfies $S_{\mathbf{1}} \le C_0 \log(d+1)$. Theorem 17.1 therefore gives, whenever $L \ge C_1 \log(d+1)$,

$$J_L(\mathbf{1}) \le V_* + C_2 \frac{[\log(d+1)]^2 + [\log(2L)]^2}{L} \le C_3 \left(1 + \frac{[\log(d+1)]^2}{L} \right).$$

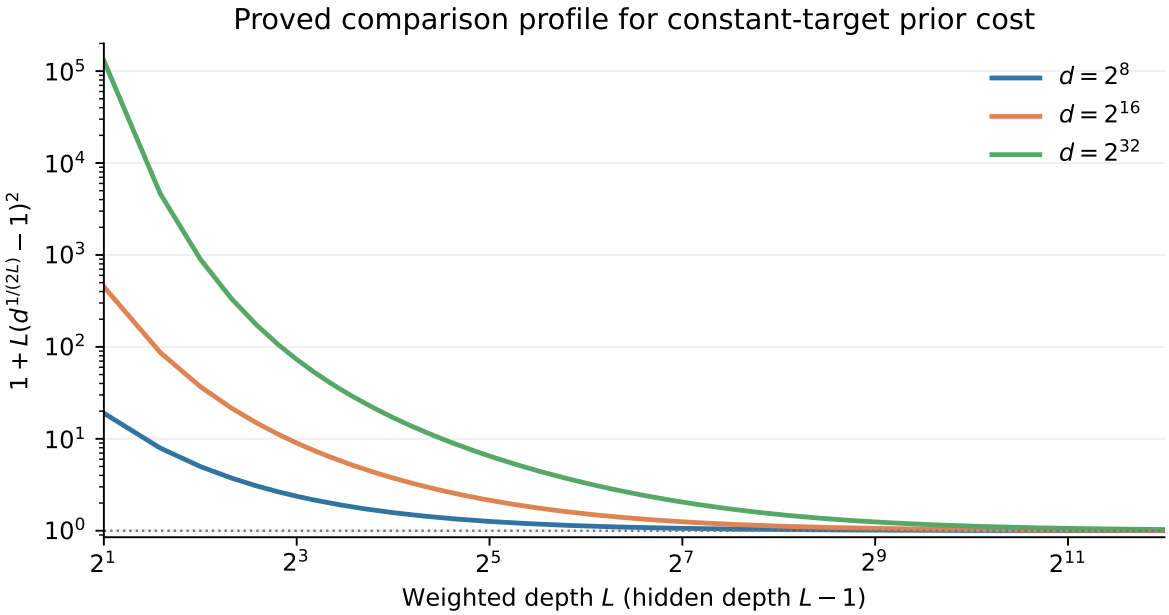


Figure 1: The comparison function in the all-depth constant-target theorem. The actual critical rate lies between universal constant multiples of this function; the curves are not numerical estimates of that rate. The shallow-depth growth and the transition to bounded cost are parts of one quantitative statement. Depth counts weighted layers.

Here $[\log(2L)]^2/L$ is universally bounded for $L \geq 2$. For $d \geq 2$, $\log(d+1) \leq 2\log d$ and $e^t - 1 \geq t$ show that this last expression is at most $C_4\Phi_L(d)$. For $d = 1$ it is bounded universally and $\Phi_L(1) = 1$.

In the remaining range $L < C_1 \log(d+1)$, use (46): $J_L(\mathbf{1}) \leq 5000Ld^{1/L}$. For $d \geq 2$, the parameter $t = \log d/(2L)$ obeys $t \geq 1/(4C_1)$, so

$$\frac{e^{2t}}{(e^t-1)^2} = (1-e^{-t})^{-2} \leq (1-e^{-1/(4C_1)})^{-2}.$$

Thus this upper bound is at most a universal multiple of $L(d^{1/(2L)} - 1)^2$. When $d = 1$ the remaining range has universally bounded L, and the same conclusion with $\Phi_L(1) = 1$ follows. The dictator case follows from projective symmetry.

Finally, when $L \geq \log(d+1)$, $t = \log d/(2L) \in [0, 1/2]$, so $t \leq e^t - 1 \leq e^{1/2}t$; replacing $\log d$ by $\log(d+1)$ changes $1 + (\log d)^2/L$ by at most universal factors. This proves the final scale comparison. □

42 A strict target-relative ordering at depth two

The common limiting rate does not imply equality at finite depth. At depth two one can determine a strict comparison between simple targets and every nontrivial parity.

Theorem 42.1 (Constants have lower critical rate than nontrivial parities)**.** *At critical scaling and depth $L = 2$, let $\mathbf{1}$ denote the constant-positive Boolean function. If f is nonconstant and antipodally even, meaning $f(-x) = f(x)$, then*

$$J_2(f) > J_2(\mathbf{1}).$$

Consequently every parity of degree at least two has strictly larger critical rate than a constant or a dictator. The negative constant and signed dictators have the same rate as $\mathbf{1}$.

Write

$$X_f(W) = \frac{1}{m} \sum_{x \in \{-1,1\}^d} f(x) \tanh(W \cdot \bar{x}), \qquad m = 2^d.$$

The proof uses the following exact formula for character targets.

Lemma 42.2 (Depth-two character rate)**.** *If f is a Boolean character, including the constant character, then*

$$J_2(f) = \max_{t \geq 0} \left\{ \frac{t}{2} - \log \mathbb{E}_W \exp\left(\frac{t^2 X_f(W)^2}{2} \right) \right\}.$$

The maximizing parameter for the constant character exists and is strictly positive. For an arbitrary Boolean f, the right-hand side is a lower bound for $J_2(f)$.

Proof. At depth two, the output vector has the distribution

$$z_N = \frac{1}{N} \sum_{i=1}^{N} Z_i h(W_i), \qquad h(W)(x) = \tanh(W \cdot \bar{x}),$$

where $Z_i \sim \mathcal{N}(0,1)$ are independent of the first-layer weights. Its logarithmic moment generating function is

$$\Lambda(\theta) = \log \mathbb{E}_W \exp\{(\theta \cdot h(W))^2/2\}.$$

It is finite for every θ. The depth-two output rate is the convex Legendre transform $I(z) = \sup_\theta(\theta \cdot z - \Lambda(\theta))$. This also follows directly from the variational functional: among joint laws of (h, Z) having a given marginal ν and $\mathbb{E}[Zh] = z$, the minimum relative entropy in the Gaussian Z coordinate is $z^T Q(\nu)^+ z/2$. Minimizing the remaining marginal entropy gives precisely the entropy formula for I.

For completeness the rate is finite and continuous on the whole output space. The first feature covariance K is positive definite. Jensen's inequality gives $\Lambda(\theta) \geq \mathbb{E}(\theta \cdot h)^2/2 \geq \lambda_{\min}(K)\|\theta\|_2^2/2$. Thus its dual is finite; the Gaussian scalar also makes every output direction available. Equivalently, the finite-dimensional entropy duality here has no boundary obstruction. Strict or non-strict output margins consequently give the same infimum.

The law of the output vector is invariant under input translations $x \mapsto sx$ and under global output negation. For a character f, the transformation $z(x) \mapsto f(s)z(sx)$ preserves both this law and the constraints $f(x)z(x) \geq 1/2$. Averaging over all s and using convexity replaces any feasible z by af, where $a = m^{-1}\sum_x f(x)z(x) \geq 1/2$, without increasing I. Convexity and $I(0) = 0$ then reduce the optimum to $z = f/2$. The same group averaging in the dual restricts θ to tf/m, giving the displayed formula. Symmetry in t permits $t \geq 0$. For general f, substituting this dual vector directly gives the asserted lower bound, because $\theta \cdot z \geq t/2$ on the target-margin event.

For the constant character, $X_{\mathbf{1}}$ is nonzero with positive probability. Therefore for some $a, p > 0$, $\Pr\{|X_{\mathbf{1}}| \geq a\} \geq p$, and the objective tends to $-\infty$ as $t \to \infty$. Its derivative at zero is $1/2$. It has a finite, strictly positive maximizer. □

Proof of Theorem 42.1. Write $W \cdot \bar{x} = b + w \cdot x$. For every real u,

$$\tanh(b+u) + \tanh(b-u) = \frac{\sinh(2b)}{\cosh(b+u)\cosh(b-u)}.$$

For $b \neq 0$ every such antipodal pair sum is nonzero and has the same sign as b. If f is even and nonconstant, it takes both signs on the set of antipodal pairs. Strict triangle inequality therefore gives, pointwise for every finite w and $b \neq 0$,

$$|X_f(b, w)| < |X_{\mathbf{1}}(b, w)|.$$

The Gaussian bias is nonzero almost surely, so for every $t > 0$,

$$\mathbb{E}e^{t^2 X_f^2/2} < \mathbb{E}e^{t^2 X_{\mathbf{1}}^2/2}.$$

Evaluate the lower bound in the preceding lemma at a positive maximizer for the constant character. This proves the strict rate inequality.

Every positive even-degree parity is antipodally even and nonconstant. A parity of odd degree at least three has the same rate as a parity of the preceding even degree by the projective coordinate symmetry. That symmetry also identifies constants and dictators, and global output negation identifies their negatives. □

The strict gap implies an exponentially small zero-data prior ratio between such a target and a constant as width tends to infinity at this fixed depth. It does not give a sample lower bound: a few oppositely labeled observations already eliminate the constant competitor. The exact ordering at every finite dimension and arbitrary depth remains open; the preceding comparison covers growing-relevance targets at fixed depth and sufficiently large dimension. Neither result completely orders training-consistent extensions. The later sample lower bound uses a separate comparison with an explicit interpolating function.

The pointwise comparison used in the proof is specific to a first-layer feature. For example, on two input bits put

$$h_1 = \tanh(x_1 + x_2 - 1), \quad h_2 = \tanh(x_1 + x_2 + 1), \quad h_3 = \tanh(1), \qquad g = \tanh(h_1 - h_2 + h_3).$$

This uses only first-layer biases. The feature g is positive when $x_1 x_2 = 1$ and negative when $x_1 x_2 = -1$: its two values are $\tanh(2\tanh 1 - \tanh 3) > 0$ and $-\tanh(\tanh 1) < 0$. Consequently its parity coefficient has greater magnitude than its constant coefficient. Thus the preceding pointwise domination fails already for a second-hidden-layer feature. This example does not reverse the full prior rate comparison; it identifies a limitation of this proof method.

43 A polynomial sample lower bound at each fixed critical depth

A low-rate interpolating halfspace gives a sample lower bound for the actual critical posterior. The bound is polynomial in the parity and ambient dimensions, and the width limit is taken before those dimensions grow.

Lemma 43.1 (Amplifying a first-feature mean through its second moment)**.** *Let μ_1 be the uncontrolled first-feature law. Suppose a feature law ν satisfies $D(\nu\|\mu_1) \le D$, where $D \ge 1$, and its mean h satisfies $g(x)h(x) \ge a$ at every input, with $0 < a \le 1$ and g Boolean. Set $R = D/a^2$ and $q = R^{1/L}$. If $q \le D$, then*

$$J_L(g) \le 5000 L R^{1/L}.$$

Proof. Put $\theta = q/D \le 1$. The first controlled law $(1-\theta)\mu_1 + \theta\nu$ has entropy at most $\theta D = q$ by convexity, and its second moment dominates $\theta h h^T$. Thus it has a tracked rank-one vector $b_1 = \sqrt{\theta} h$ with

$$g(x) b_1(x) \ge \sqrt{\theta} a = q^{-(L-1)/2}.$$

This is a second-moment component; it need not equal the first controlled law's mean.

For $L = 2$, choose the readout $z = h/(2a)$. Its cost is at most $1/(8\theta a^2) = q/8$, proving the assertion in this case. Suppose $L \geq 3$ and put $c = 1/100$. For every $0 \leq v \leq 1$ and $u \geq 0$, the elementary estimate used in the constant-target construction gives

$$\mathbb{E}\tanh(u + \sqrt{v}Z) \geq c\min\{u, 1\}.$$

At the second hidden layer shift the reference Gaussian by $\sqrt{q}\,b_1$. Its mean-shift entropy is at most $q/2$, because $Q_1 \succeq b_1 b_1^T$. After tanh, its actual mean b_2 satisfies $g(x)b_2(x) \geq cq^{-(L-2)/2}$. Its second moment dominates $b_2 b_2^T$.

At each hidden layer $s = 3, \ldots, L-1$, shift the reference Gaussian by $(\sqrt{q}/c)b_{s-1}$. The entropy is at most $q/(2c^2)$, and the new feature mean satisfies

$$g(x)b_s(x) \geq cq^{-(L-s)/2}.$$

The induction uses only that every reference diagonal is at most one, not that the diagonals or feature magnitudes are equal. If some shift coordinate exceeds one in magnitude, the displayed minimum estimate still gives the required lower bound. All shift vectors lie in the corresponding covariance range; the formulas therefore also apply to singular covariances. Tanh pushforward cannot increase the entropy.

Finally take $z = b_{L-1}/(2cq^{-1/2})$. It has the required global margin and Gaussian readout cost at most $q/(8c^2)$. Summing all costs gives the claimed bound. □

Lemma 43.2 (A low-rate majority interpolant for parity samples)**.** *Let the labels be a parity on a nonempty coordinate set S, and draw $n \geq 1$ independent uniform cube inputs. Fix one $i_0 \in S$ and an odd number $m \leq d-1$ of other coordinate indices I. There is a sample-dependent Boolean majority function*

$$g(x) = \operatorname{sign}\left(\sum_{j\in I} s_j x_j\right), \qquad s_j \in \{-1, 1\},$$

which fits every training label with probability at least $1 - n\exp[-m/(6n)]$. For every sample, this function admits a first-feature law ν with

$$D(\nu\|\mu_1) \leq dm, \qquad g(x)\mathbb{E}_\nu H(x) \geq 1/100 \quad \textit{for every input } x.$$

Consequently, whenever $dm \geq 10000$,

$$J_L(g) \leq 5000L(10000dm)^{1/L} \qquad (L \geq 2).$$

Proof. Write Y_i for the parity label of training row i, and choose

$$s_j = \operatorname{sign}\left(\sum_{i=1}^n Y_i X_{ij}\right),$$

with either fixed convention for a zero sum. Conditional on the entire label vector Y, all coordinate columns other than i_0 are independent uniform sign vectors: the missing parity coordinate is uniquely determined by these columns and Y. Thus $A_{ij} = Y_i X_{ij}$, for $j \in I$, are independent uniform sign columns conditionally on Y.

If $T = \sum_{i=1}^n \varepsilon_i$ is a sum of independent uniform signs, exchangeability gives

$$\mathbb{E}\left[\varepsilon_i \operatorname{sign}(T)\right] = \frac{\mathbb{E}|T|}{n} \geq \frac{1}{\sqrt{3n}}.$$

Indeed $\mathbb{E}T^2 = n$ and $\mathbb{E}T^4 \le 3n^2$, while Holder's inequality gives $\mathbb{E}T^2 \le (\mathbb{E}|T|)^{2/3}(\mathbb{E}T^4)^{1/3}$. For each training row, the variables $s_j A_{ij}$ are therefore independent across j, bounded by one in absolute value, and have mean at least $1/\sqrt{3n}$. Hoeffding's inequality bounds the probability that their sum is nonpositive by $e^{-m/(6n)}$. A union bound proves the training-fit probability. Dependence between different training rows is irrelevant to that union bound.

Since m is odd, the affine form defining g is a nonzero integer at every cube input, so its global signed margin is at least one. Shift the first-row Gaussian weight mean to have coefficient s_j on I and zero elsewhere, including its bias. This costs exactly $dm/2$ in relative entropy. At each input its residual Gaussian variance is $(d+1)/d \le 2$, and its feature mean is

$$\mathbb{E}\tanh\left(\sum_{j\in I} s_j x_j + \sqrt{(d+1)/d}\, Z\right).$$

The elementary mean-response bound from the constant-target construction gives signed mean at least $1/100$ everywhere. Data processing therefore proves the first-feature-law assertion. Apply Lemma 43.1 with $D = dm$ and $a = 1/100$. Its hypothesis $(10000D)^{1/L} \le D$ holds whenever $D \ge 10000$ and $L \ge 2$. □

Theorem 43.3 (Critical parity sample lower bound)**.** *Fix $L \ge 2$ and $0 < \eta < 1$. Let the target be parity on $k \le d$ specified input coordinates, and put*

$$M = \min\{k, d^{1/(L-1)}\}.$$

Suppose $M \to \infty$ and $1 \le n \le M^{1-\eta}$. Draw n independent uniform cube inputs labeled by this parity. For all sufficiently large M, with training-sample probability at least $1 - 1/(2M)$, the globally margin-conditioned posterior at $\gamma = 1$ has Gibbs error tending to $1/2$ as $N \to \infty$ at this fixed d, L, n. If $p_N(x)$ is its posterior probability of the correct label, then on the same event

$$\liminf_{N\to\infty} \Pr_x\{p_N(x) < 2/3\} \ge 1/4.$$

For full parity, this gives $n \le d^{(1-\eta)/(L-1)}$. If $d \ge k^{L-1}$, it gives $n \le k^{1-\eta}$.

Proof. Let m be the smallest odd integer at least $12n\log(2M(n+1))$. Since $n \le M^{1-\eta}$, one has $m = O(n\log(M+1)) = o(M) \le o(d)$, so $m \le d-1$ for large M. Lemma 43.2 supplies, with probability at least $1 - 1/(2M)$, a training-consistent halfspace and a first-feature law of entropy at most $D = dm$ with mean margin $a = 1/100$.

Apply Lemma 43.1. Its hypothesis $(D/a^2)^{1/L} \le D$ is equivalent to $D^{L-1} \ge 10000$ and holds for large M. The resulting training-consistent competitor has rate at most

$$5000L(10000dm)^{1/L}.$$

Any Boolean function depending on every one of the k parity coordinates has at least k relevant augmented coordinates. By Theorem 40.1, its critical rate is at least

$$\frac{1}{64}\min\{d^{1/(L-1)}, (dk)^{1/L}\} = \frac{1}{64}(dM)^{1/L}.$$

The ratio of the preceding competitor upper bound to this lower bound is at most

$$CL\, M^{-\eta/L}[\log(M+1)]^{1/L},$$

which tends to zero. Thus there is a positive rate gap between a training-consistent competitor and every function retaining all parity coordinates.

The exact finite-domain posterior-rate principle now implies that posterior mass on the latter functions tends to zero in the width limit. A function omitting any one parity coordinate has uniform parity error exactly $1/2$, even if it depends on irrelevant coordinates. This proves the Gibbs limit. Finally $\mathbb{E}_x p_N(x) \to 1/2$ and Markov's inequality give $\Pr_x\{p_N(x) < 2/3\} \geq 1 - (3/2)\mathbb{E}_x p_N(x)$. □

The theorem concerns the actual posterior, but takes width to infinity at fixed dimension, depth, and sample size. It gives no finite-width threshold uniform in growing dimension and no corresponding lower bound for posterior-majority prediction at confidence threshold $1/2$.

Corollary 43.4 (Slowly growing learner depth)**.** *Fix $0 < \eta < 1$. For full parity, the same conclusion holds at sample sizes $n \leq d^{(1-\eta)/(L-1)}$ along integer depths $L = L(d) \geq 2$ satisfying*

$$L = o\left(\sqrt{\frac{\log d}{\log\log d}}\right).$$

For each d, the width limit is taken with its corresponding $L(d)$ and sample size fixed.

Proof. Here $M = d^{1/(L-1)} \to \infty$. The halfspace construction has $m \leq Cn\log(M+1) = o(M) \leq o(d)$ uniformly in $L \geq 2$ at the specified sample sizes. Its first-feature entropy bound and amplification are also uniform. The rate ratio is therefore at most

$$CL\, d^{-\eta/[L(L-1)]}[\log(d+1)]^{1/L}$$

with universal C. Its logarithm tends to minus infinity under the stated depth assumption. The mixture hypothesis and the training success probability also hold uniformly, so the same rate-gap argument applies. □

44 Parity remains hard at nearly logarithmic learner depth

The comparison can be extended to a larger growing-depth range by retaining the leading constant in both the interpolating competitor's rate and the transport lower bound. The first improvement comes from tracking the freely cooling feature variance during mean amplification. It avoids charging a fixed loss at every hidden layer.

Lemma 44.1 (Mean amplification with its leading constant)**.** *Fix $a \in (0, 1]$. Suppose a first-feature law ν has entropy at most $D \geq 1$ relative to the unmodified first-feature law μ, and its mean h satisfies $g(x)h(x) \geq a$ everywhere, for a Boolean g. Put $q = (D/a^2)^{1/L}$. There are constants K, P, C, depending only on a, such that, if $L \geq 2$ and $q \geq KL^P$, then*

$$J_L(g) \leq \frac{q}{2}\{L + C\log L\}.$$

The first law and the subsequent covariance diagonals need not be constant over inputs.

Proof. Let $F(v) = \mathbb{E}\tanh^2(\sqrt{v}Z)$ and $u_s = F^s(2)$. The function F is nondecreasing and one-Lipschitz. Monotonicity is pointwise in v, and Gaussian differentiation gives $F' \leq 1$ because $(\tanh^2)'' \leq 2$. On $[0, 2]$, one has $v - F(v) \geq cv^2$ with universal $c > 0$: the ratio tends to two at zero and is strictly positive away from zero. It follows that

$$u_1 < 1, \qquad u_s \leq \frac{C_0}{s+1}, \qquad \sum_{s=1}^{L} u_s \leq C_1\log(L+1).$$

For a centered Gaussian G of variance v, the same second-derivative bound and symmetry give

$$\mathbb{E}\tanh^2(G+t) \le F(v) + t^2.$$

For $t \ge 0$, integration of $\mathbb{E}\operatorname{sech}^2(G+s) \ge 1 - v - s^2$ gives the convenient weaker estimate

$$\mathbb{E}\tanh(G+t) \ge t(1 - v - t^2).$$

Set $\theta = q/(2D) = 1/(2a^2 q^{L-1})$. Use $(1-\theta)\mu + \theta\nu$ at the first hidden layer. Its entropy is at most $q/2$, and its second moment dominates $b_1 b_1^T$ with $b_1 = \sqrt{\theta}h$. Its maximum diagonal is at most $u_1 + \theta$. For $L = 2$, a readout proportional to b_1 costs at most $q/4$, proving the assertion. Assume henceforth $L \ge 3$.

Let $\kappa = 1 - u_1 > 0$, $\epsilon = \kappa/(8L^2)$, and

$$c_s = 1 - u_{s-1} - \epsilon \qquad (2 \le s \le L-1).$$

These constants are uniformly positive. The preceding bounds on u_s imply, for universal constants A_0, C_2,

$$\prod_{s=2}^{L-1} c_s^{-1} \le L^{A_0}, \qquad \sum_{s=2}^{L-1} c_s^{-2} \le L - 2 + C_2 \log L.$$

Indeed, $-\log c_s$ and $c_s^{-2} - 1$ are both at most a universal multiple of $u_{s-1} + \epsilon$.

At hidden layer s, shift the reference Gaussian by

$$m_s(x) = \frac{\sqrt{q}}{c_s} b_{s-1}(x),$$

and let b_s be its tanh feature mean. For $s = 2$, b_1 denotes the tracked component just constructed; thereafter b_{s-1} is the actual preceding mean. In either case its outer product is dominated by the preceding second moment. Thus the shift lies in the covariance range and costs at most $q/(2c_s^2)$, including singular references.

The centered-Gaussian mean $\mathbb{E}\tanh(G+t)$ is odd, has the sign of t, and has absolute value at most $|t|$. Therefore, without yet using a variance bound,

$$\max_x |m_s(x)| \le \frac{L^{A_0}}{\sqrt{2}a}\, q^{-(L-s)/2}.$$

For $q \ge 2$, summing the resulting geometric series gives

$$\theta + \sum_{s=2}^{L-1} \max_x |m_s(x)|^2 \le C_a L^{2A_0}/q.$$

Choose K, P so that $q \ge KL^P$ makes this at most ϵ; the same choice ensures $\theta \le 1$. Since F is one-Lipschitz, the shifted-square estimate gives inductively

$$\max_x Q_s(x,x) \le u_s + \theta + \sum_{t=2}^{s} \max_x |m_t(x)|^2.$$

In particular the variance error before layer s, together with the squared current shift, is at most ϵ. The shifted-mean estimate consequently yields

$$g(x) b_s(x) \ge c_s g(x) m_s(x) = \sqrt{q}\, g(x) b_{s-1}(x).$$

The last hidden mean has global signed margin at least

$$\sqrt{\theta} a q^{(L-2)/2} = \frac{1}{\sqrt{2q}}.$$

A readout proportional to this mean has output margin $1/2$ and costs at most $q/4$. Adding all costs gives

$$\frac{q}{2} + \frac{q}{2} \sum_{s=2}^{L-1} c_s^{-2} + \frac{q}{4} \le \frac{q}{2}\{L + C \log L\},$$

as required. □

Lemma 44.2 (Joint optimization of the transport budget)**.** *If a Boolean function depends on all d original input coordinates, then any admissible critical path of cost A satisfies, for every $t > 0$,*

$$d \le 8(2 + 1/t)^{L-2} \left(\frac{A + (L-2)t}{L-1} \right)^{L-1} \exp\left(2\sqrt{2A/d} \right).$$

In particular, if $A \le (q/2)\{L + C \log L\}$, then

$$d \le 4q^{L-1} \exp\left(C' \log L + C' L/\sqrt{q} + 2\sqrt{2A/d} \right),$$

where C' depends only on C.

Proof. Use the sharper pathwise transport inequality (44), with readout cost r, hidden costs D_s, and $k \ge d$ relevant augmented coordinates. Cauchy–Schwarz gives

$$(1 + \sqrt{2D_s})^2 \le (2 + 1/t)(D_s + t).$$

The arithmetic-geometric mean inequality gives

$$r \prod_{s=2}^{L-1} (D_s + t) \le \left(\frac{A + (L-2)t}{L-1} \right)^{L-1}.$$

The first-layer factor is at most $\exp(2\sqrt{2A/d})$. Combining these facts proves the first display, also when $L = 2$ and the products are empty. For the second choose $t = \sqrt{q}$, factor out 2^{L-2} and $(q/2)^{L-1}$, and use $\log(1 + u) \le u$. The resulting errors are bounded by $C' \log L + C' L/\sqrt{q}$. □

Theorem 44.3 (Critical parity lower bound at nearly logarithmic depth)**.** *Fix $0 < \eta < 1$. Let the target be full parity on d coordinates and let integer depths $L = L(d) \ge 2$ satisfy*

$$L = o\left(\frac{\log d}{\log \log d} \right).$$

Put $M = d^{1/(L-1)}$, and draw $1 \le n \le M^{1-\eta}$ independent uniform cube inputs, labeled by parity. For all sufficiently large d, with training probability at least $1 - 1/(2M)$, the original globally margin-conditioned posterior at $\gamma = 1$ satisfies

$$\lim_{N \to \infty} \textit{Gibbs error} = \frac{1}{2}, \qquad \liminf_{N \to \infty} \Pr_x\{p_N(x) < 2/3\} \ge \frac{1}{4}.$$

For each d, the width limit fixes d, L, n first. Moreover, there is a universal $c > 0$ for which the same conclusion holds for every

$$2 \le L \le c\eta \frac{\log d}{\log \log d},$$

once d is sufficiently large depending on η.

Proof. Let m be the smallest odd integer at least $12n\log(2M(n+1))$. For large d,

$$m \le 50n\log M \le 50M^{1-\eta}\log M = o(M) \le o(d).$$

Lemma 43.2 supplies, with the stated training probability, a consistent Boolean function and a first-feature law of entropy $D = dm$ and global signed mean at least $a = 1/100$. Put $q = (10000dm)^{1/L}$. The depth hypothesis implies

$$\log q \ge \frac{\log d}{L} \gg \log\log d,$$

so q exceeds every fixed power of L. Lemma 44.1 gives a consistent competitor of rate at most $U = (q/2)\{L + C\log L\}$.

Suppose a Boolean function retaining all d coordinates had rate at most U. Apply Lemma 44.2 to paths whose costs approach its infimum, then use continuity and monotonicity in the budget. It follows that

$$\log\frac{d}{q^{L-1}} \le \log 4 + C'\log L + C'L/\sqrt{q} + 2\sqrt{2U/d}.$$

The last two terms tend to zero. For $L \ge 3$, use $m \le d$, $q \le (10000d^2)^{1/L}$, and $L = O(\log d)$ to see $U/d \to 0$. For $L = 2$, use $m \le 50d^{1-\eta}\log d$ instead. On the other hand, the sample bound and $q^L = 10000dm$ give

$$\log\frac{d}{q^{L-1}} \ge \frac{\eta\log d}{L} - \log(500000\log M).$$

Here the first term dominates $\log\log d$, whereas the other logarithmic terms are $O(\log\log d)$. This is a contradiction. There is therefore a strict rate gap between the consistent competitor and every Boolean function retaining all d input coordinates.

The exact finite-domain posterior rate principle makes posterior mass on the latter class tend to zero. Every remaining function omits a parity coordinate and has uniform parity error exactly $1/2$. This proves the Gibbs limit, and Markov's inequality gives the displayed confidence-error bound from $\mathbb{E}_x p_N(x) \to 1/2$.

For the final depth range, choose the universal c small enough that $\log q \ge \log d/L$ guarantees the fixed power of L required by the amplification lemma, and that $\eta\log d/L$ exceeds $(C'+2)\log\log d$ by a fixed factor. This is possible independently of $\eta \le 1$. The remaining estimates hold for all sufficiently large d depending on η, completing the proof. □

This theorem concerns the actual posterior, including mixtures of minimizers. It does not reach the order-$\log^2 d$ depths at which controlled parity seeds attain a bounded absolute rate, and it does not by itself give an exponential-in-teacher-depth lower bound.

45 The exact first-order scalar entrance cost

Theorem 45.1 (Small-variance scalar expansion)**.** *There are universal constants $q_0, C > 0$ such that*

$$\bigl|V(q) - (V_* - 2q)\bigr| \le Cq^2 \qquad (0 < q \le q_0).$$

In particular, $\lim_{q\downarrow 0}(V_* - V(q))/q = 2$.

Proof. We first construct a nearly optimal small-state bridge. At state q, replace the reference Gaussian by $\mathcal{N}(0, (1+4q)q)$. Since

$$F(u) = u - 2u^2 + O(u^3),$$

its new state q' and entropy cost D satisfy

$$q' = q + 2q^2 + O(q^3), \qquad D = \tfrac{1}{2}[4q - \log(1+4q)] = 4q^2 + O(q^3).$$

For sufficiently small q, therefore,

$$q' - q \geq q^2, \qquad D \leq (2 + Cq)(q' - q).$$

Starting at any $0 < a < b$ in this small interval, take full steps while they remain below b. The states reach the final interval in finitely many steps, since every increment is at least a^2. The full-step costs sum to at most $2(b-a) + Cb^2$. On the last step reduce the variance inflation continuously to hit b exactly; its entropy is at most Cb^2. Thus, after changing C, there is a bridge from a to b of cost

$$2(b-a) + Cb^2. \tag{48}$$

Appending a nearly optimal path from q and letting $a \downarrow 0$ proves

$$V_* - V(q) \leq 2q + Cq^2.$$

For the reverse inequality take q still smaller, and start a near-optimal scalar path at $0 < a < q$. Its cost may be assumed less than two, by the uniform upper bound for V. It must cross level q: otherwise the readout alone costs more than $1/(8q) > 2$. Let b be its first state at or above q, and a_- the preceding state. Then $a_- < q$. The crossing satisfies $b < 8q$. Indeed the preactivation second moment is at least b, because $\tanh^2 z \leq z^2$. At a ratio $r = b/a_- \geq 8$, Gaussian entropy duality gives a single-step cost at least

$$\tfrac{1}{2}(r - 1 - \log r) \geq \tfrac{1}{2}(8 - 1 - \log 8) > 2,$$

contrary to the total-cost bound.

The scalar exponential potential with coefficient two gives prefix entropy at least $2(b-a)$. The remaining path costs at least $V(b)$. Apply the bridge (48) from q to b, which stays in the small-state interval since $b < 8q$:

$$V(b) \geq V(q) - 2(b-q) - Cb^2.$$

The entire path therefore costs at least

$$V(q) + 2q - 2a - 64Cq^2.$$

Taking the infimum over near-optimal paths and then $a \downarrow 0$ gives $V_* - V(q) \geq 2q - C''q^2$, proving the theorem. □

The coefficient two describes the scalar entrance to the common rare-event trajectory. It does not identify the function-dependent cost of bringing a particular target direction to that scalar regime. Thus it is an ingredient for a possible target-relative expansion, not a characterization of the posterior's learning behavior.

A An exact variational principle at the critical exponent

For completeness we reproduce the finite-domain variational foundation. Here $\phi = \tanh$. For arbitrary fixed depth, the critical prior admits a direct finite-domain large-deviation description. This result requires neither a propagation-of-chaos assumption nor differentiability of the activation.

It identifies the functional used in the critical-rate capacity constructions with the actual Gaussian probability rate. Covariance-process large deviations for Gaussian networks have been studied in a more general functional setting by Andreis, Bassetti, and Hirsch [3]. No novelty claim is made for the general large-deviation framework; the finite-domain proof below fixes the precise normalizations and strict-margin convention used here.

Fix $\gamma = 1$, enumerate $\mathcal{X} = \{x^1, \dots, x^m\}$, where $m = 2^d$, and put $r = L - 1$. All limits in this section hold with d and L fixed. Write $E = [-1, 1]^m$ and let $\mathcal{P}(E)$ carry its weak topology. For $\nu \in \mathcal{P}(E)$ define

$$K(\nu) = \int_E hh^{\mathsf{T}}\, d\nu(h), \qquad Q_0(a, b) = \frac{\bar{x}^a \cdot \bar{x}^b}{d}.$$

For any positive semidefinite $m \times m$ matrix Q, let

$$\mathsf{T}_Q(u) = (\phi((Q^{1/2}u)_a))_{a=1}^m, \qquad \mathsf{M}_Q = (\mathsf{T}_Q)_{\#}\mathcal{N}(0, I_m).$$

Here $T_{\#}\rho$ is the distribution of $T(U)$ when U has distribution ρ, and $Q^{1/2}$ is the positive semidefinite square root. Relative entropy is

$$D(\nu\|\rho) = \begin{cases} \displaystyle\int \log \frac{d\nu}{d\rho}\, d\nu, & \nu \ll \rho, \\ +\infty, & \text{otherwise.} \end{cases}$$

Finally define the Gaussian energy

$$\mathcal{G}_Q(z) = \begin{cases} \frac{1}{2} z^{\mathsf{T}} Q^{+} z, & z \in \operatorname{range}(Q), \\ +\infty, & z \notin \operatorname{range}(Q), \end{cases}$$

where Q^{+} is the Moore–Penrose inverse.

Theorem A.1 (Critical finite-domain large deviations)**.** *Let $Z_N = (z_w(x^a))_{a=1}^m$ under the unconditioned prior $\mu_{N,1}$. The sequence Z_N satisfies a large-deviation principle with speed N and good rate function*

$$I(z) = \inf_{\nu_1, \dots, \nu_r \in \mathcal{P}(E)} \left\{ D(\nu_1 \| \mathsf{M}_{Q_0}) + \sum_{s=2}^{r} D(\nu_s \| \mathsf{M}_{K(\nu_{s-1})}) + \mathcal{G}_{K(\nu_r)}(z) \right\}. \tag{49}$$

In particular, for every open $O \subseteq \mathbb{R}^m$ and closed $F \subseteq \mathbb{R}^m$,

$$-\inf_{z \in O} I(z) \le \liminf_{N\to\infty} \frac{1}{N} \log \mathbb{P}(Z_N \in O), \qquad \limsup_{N\to\infty} \frac{1}{N} \log \mathbb{P}(Z_N \in F) \le -\inf_{z \in F} I(z).$$

The infimum in (49) is attained whenever it is finite. No covariance matrix in the statement is required to be nonsingular.

Proof. We give an exact representation of the finite-width law and then apply Sanov's theorem and continuous contraction [4].

Let $\Gamma_m = \mathcal{N}(0, I_m)$. Independently for each $s = 1, \dots, r$, let $U_{s,1}, \dots, U_{s,N}$ have law Γ_m, and set

$$\eta_{s,N} = \frac{1}{N} \sum_{i=1}^{N} \delta_{U_{s,i}}.$$

Also let $V_N = G/\sqrt{N}$, where $G \sim \Gamma_m$ is independent of these empirical measures. Recursively define

$$\nu_{1,N} = (\mathsf{T}_{Q_0})_{\#}\eta_{1,N}, \qquad \nu_{s,N} = (\mathsf{T}_{K(\nu_{s-1,N})})_{\#}\eta_{s,N} \quad (2 \le s \le r).$$

These are exactly the joint laws of the network's empirical hidden-activation vectors on $\mathcal{X}$. Indeed, first-layer preactivation vectors are independent Gaussians with covariance Q_0. Conditional on the preceding layer, each subsequent row of weights gives an independent Gaussian vector with covariance

$$\frac{1}{N}\sum_{i=1}^{N}\left(h_i^{s-1}(x^a)h_i^{s-1}(x^b)\right)_{a,b} = K(\nu_{s-1,N}).$$

The conditional distribution depends on the previous layer only through its empirical measure. Finally, readout weights have variance N^{-2}, so

$$Z_N \overset{\text{law}}{=} K(\nu_{r,N})^{1/2}V_N. \tag{50}$$

This representation is exact for every N.

Sanov's theorem on $\mathcal{P}(\mathbb{R}^m)$ gives the good rate $D(\eta\|\Gamma_m)$ for each $\eta_{s,N}$. The elementary Gaussian large-deviation principle gives rate $\|v\|_2^2/2$ for V_N. Independence and the finite-product principle therefore give the good rate

$$\sum_{s=1}^{r} D(\eta_s\|\Gamma_m) + \frac{1}{2}\|v\|_2^2$$

for $(\eta_{1,N}, \dots, \eta_{r,N}, V_N)$.

The recursive map from these variables to $(\nu_1, \dots, \nu_r, z)$ is continuous. To verify the only potentially delicate point, suppose $Q_j \to Q$ and $\eta_j \Rightarrow \eta$ weakly. For every continuous $b : E \to \mathbb{R}$, the functions $b \circ \mathsf{T}_{Q_j}$ converge uniformly on each compact subset of $\mathbb{R}^m$ to $b \circ \mathsf{T}_Q$, since matrix square roots are continuous and ϕ is continuous. Their absolute values are uniformly bounded. Tightness of a weakly convergent sequence of probability measures then gives

$$\int b \circ \mathsf{T}_{Q_j}\, d\eta_j \longrightarrow \int b \circ \mathsf{T}_Q\, d\eta.$$

Also $\nu \mapsto K(\nu)$ is continuous because every coordinate $h_a h_b$ is continuous and bounded on E. This proves continuity of the whole recursive map, including (50).

Continuous contraction now gives the rate by minimizing over its preimages. For a measurable map T and a reference probability ρ, the entropy contraction identity is

$$\inf_{\eta : T_{\#}\eta = \nu} D(\eta\|\rho) = D(\nu\|T_{\#}\rho). \tag{51}$$

The lower bound is data processing. When the right side is finite and $q = d\nu/d(T_{\#}\rho)$, the measure $d\eta = q(T(u))\, d\rho(u)$ attains equality. If the right side is infinite, data processing already proves the identity in the extended real sense. Apply (51) successively with $T = \mathsf{T}_{Q_0}$ and $T = \mathsf{T}_{K(\nu_{s-1})}$. The last minimization is

$$\inf_{v : Q^{1/2}v = z} \frac{1}{2}\|v\|_2^2 = \mathcal{G}_Q(z),$$

including singular Q. This proves (49).

The contraction theorem supplies a good rate and exponential tightness. For completeness, the displayed minimization is also directly well behaved: relative entropy is jointly lower semicontinuous, the map $\nu \mapsto \mathsf{M}_{K(\nu)}$ is weakly continuous, and

$$\mathcal{G}_Q(z) = \sup_{t \in \mathbb{R}^m} \left\{ t^{\mathsf{T}} z - \frac{1}{2} t^{\mathsf{T}} Q t \right\}$$

is jointly lower semicontinuous in (Q, z). Since $\mathcal{P}(E)^r$ is compact, every finite infimum in (49) is attained. Moreover $K(\nu) \preceq mI_m$, so $I(z) \geq \|z\|_2^2/(2m)$. □

Corollary A.2 (Full-margin masses and posterior rates)**.** *For a Boolean function f, regarded as a vector in $\{-1, 1\}^m$, set*

$$J(f) = \inf_{z:\ f_a z_a \geq 1/2\ \forall a} I(z).$$

Then the strict-margin event used throughout this paper has the exact rate

$$\lim_{N\to\infty} -\frac{1}{N} \log \mu_{N,1}(A_{N,f}) = J(f). \tag{52}$$

For a fixed noiseless training set S, let $\mathcal{B}_S$ be the Boolean functions agreeing with its labels. If $J_S = \min_{g\in\mathcal{B}_S} J(g) < \infty$, the actual globally conditioned posterior satisfies, for every $f \in \mathcal{B}_S$,

$$\lim_{N\to\infty} -\frac{1}{N} \log \Pi_{N,1}(f \mid S) = J(f) - J_S.$$

In particular, posterior mass concentrates on the minimizers of J within $\mathcal{B}_S$. These logarithmic asymptotics do not determine how mass is divided between tied minimizers.

Proof. For $c \geq 1$, retaining any feature laws in (49) and scaling only the output gives

$$I(cz) \leq c^2 I(z),$$

because the entropy terms are nonnegative and the Gaussian energy scales quadratically. If $f_a z_a \geq 1/2$ for all a, then cz has strict margin for every $c > 1$. Letting $c \downarrow 1$ shows that the infimum of I over the open strict-margin orthant equals its infimum over the closed orthant. The lower and upper large-deviation bounds therefore agree, proving (52).

Conditioning the prior on the global margin event and subsequently on the labels gives exactly

$$\Pi_{N,1}(f \mid S) = \frac{\mu_{N,1}(A_{N,f})}{\sum_{g\in\mathcal{B}_S} \mu_{N,1}(A_{N,g})}.$$

For fixed d this is a finite sum, whose exponential rate is the smallest $J(g)$ among its terms. This proves the ratio formula and concentration claim. □

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